ZOO 405 ECOLOGY

FINAL TERM PPT (051- 101)

MERGED BY BAQIR HUSSAIN

BS ZOOLOGY 03336361244

5TH SEMESTER

Module# 51

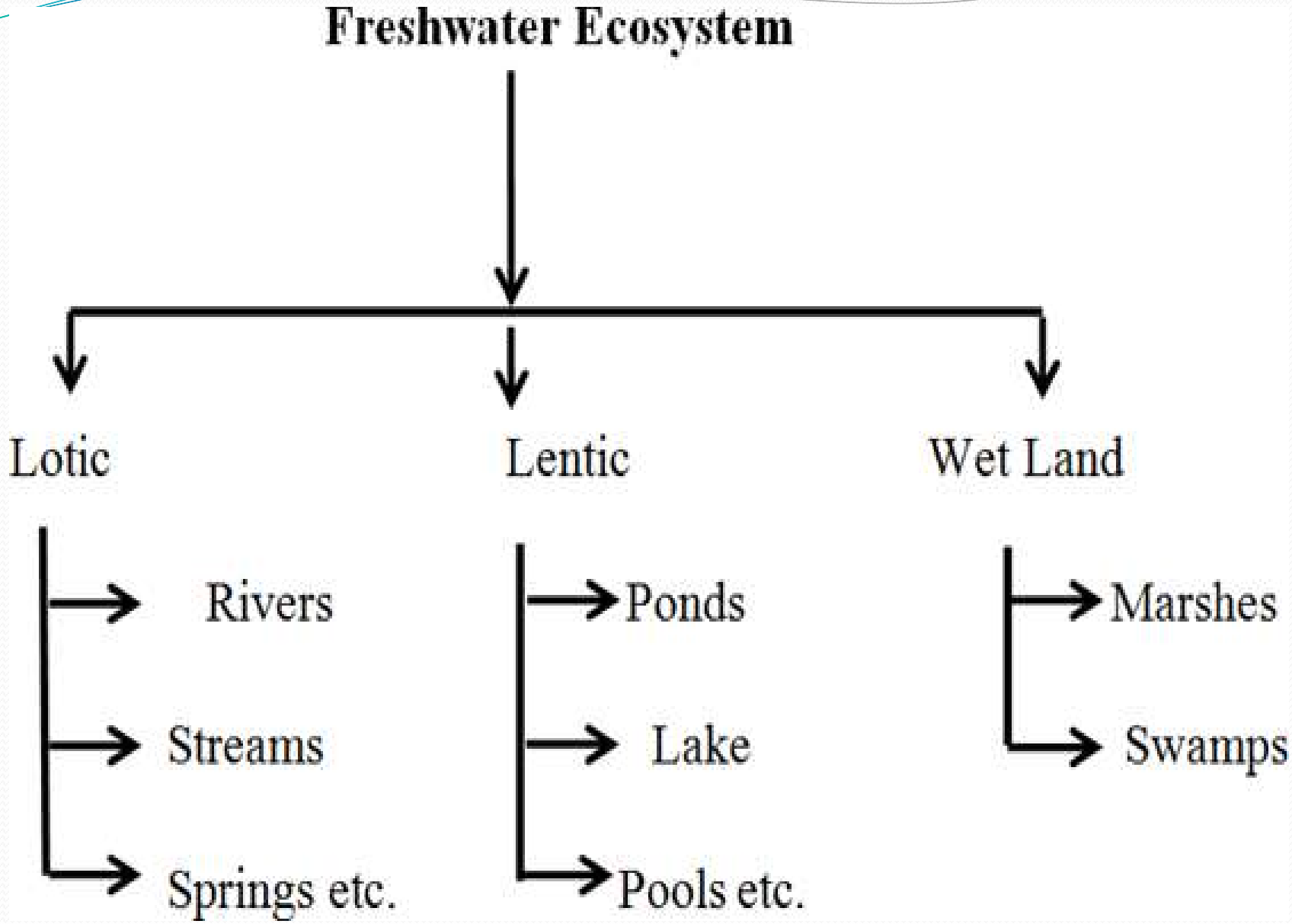
Freshwater Habitat

- Any waterbody having a low salt concentration usually less than 1% is called a freshwater body. Animals as well as plants in these freshwaters are adapted to the low salt concentrations and are unable to survive high concentration of salt present in oceans.

- The study of relationship between living organisms and the fresh water environment constitutes the freshwater ecology. The study of physical, chemical, geological and biological aspects of freshwaters is called **limnology**.

Types of Freshwater Habitat

- Freshwater habitats are studied under three major categories for convenience:
  - Standing water or **Lentic** (*lenis,* calm) Habitat: e.g., Lake, pond, swamps etc.
  - Running water or **Lotic** (*lotus,* washed) Habitat: e.g., spring, streams, rivers etc.
  - **Wet Lands**: Marshes and swamps.


Freshwater Ecosystem
Lotic
Lentic
Wet Land
Rivers
Streams
Springs etc.
Ponds
Lake
Pools etc.
Marshes
Swamps

Fresh Water Lake (Lentic Habitat)



River (Lotic Habitat)



- Freshwater habitats occupy a relatively small portion of earth's surface as compared to marine and terrestrial habitats. However, they play a far more important role for Humans because of the following reasons:

1. They are the cheapest and the most convenient source of water for industrial as well as domestic needs.

2. They are the —bottle-neck of the hydrological cycle.

3. They are a convenient and cheapest sinks for waste disposal systems.

Module# 052

Limiting Factors of Freshwater Ecosystems

Temperature

The unique properties of water maintain the temperature fluctuation in the water bodies. The variation is small, decreasing gradually from the surface to the bottom of the water body. Some of these thermal properties are:

- High specific heat: To change the temperature of water, a relatively very large amount of heat is required compared to other substance of same mass. One gram-calorie (gcal) of heat is required to raise one milliliter (ml) of water by one degree centigrade (between 150C to 160C).
- High latent heat of fusion: 80 calories are required to change one gram of ice into water without changing its temperature.

- **Highest latent heat of evaporation: 536** calories per gram are absorbed during evaporation which occurs more or less continually from vegetation, water, and ice surfaces. Major portion of the solar energy dissipates in the evaporation of water from the ecosystems of the world. This flow of energy moderates the climate which makes life possible on earth.
- Highest density at 40 C helps in expanding water from both above and below the surface and making it lighter. This property prevents the water bodies like lake from freezing.

Transparency

- Transparency is an important parameter to measure the penetration of light. Penetration of light is limited by suspended materials, restricting the photosynthetic zone wherever aquatic habitats have appreciable depth.
- Turbidity is caused either by clay and split particles or by the excretory products of the living organisms. Turbidity caused due to suspended clay and split particles is considered as an important limiting factor.
- On the other hand turbidity caused by living organism is the measurement of productivity of the water body.

- Turbidity has a direct impact on the penetration of light. Turbid water restricts the amount of light reaching the bed.
- Transparency can be measured with an instrument called **Secchi disk**.
- In the most widely studied Wisconsin lakes, Secchi disk transparency represents the zone of light penetration down to about 5% of the solar radiation reaching the surface. The 5% level marks the lowest limit of major photosynthetic zone.

Turbid and Clear freshwater Lake in Maryland



Use of Secchi Disk to measure the transparency of lake



Module# 53

Limiting Factors of Freshwater Ecosystems

Current

It plays important role in estimating the distribution of salts, vital gases and small organisms. It serves as an important limiting factor for the lotic systems especially for streams.

Concentration of respiratory gases

- Oxygen and the carbon dioxide concentration are often limiting in freshwater environment. Dissolved oxygen content (DO) and Biological oxygen demand (BOD) are important determinants for measuring the biological and physical health of a waterbody.

- These are one of the most intensively studied physical factors. Dissolved oxygen reflects the productivity. Biochemical Oxygen Demand or Biological Oxygen Demand (BOD), measures the amount of dissolved oxygen (DO) which is utilized by the aerobic microorganisms while decomposing the organic matter in waterbody, at certain temperature and over a specific period of time.

- BOD affects the amount of dissolved oxygen (DO) in streams and rivers. The oxygen consumption rate is affected by many variables: viz. temperature, pH, the presence of certain kinds of microorganisms, and the type of organic and inorganic material in the water.
- The greater the BOD, the more rapidly oxygen is depleted in the stream. Less BOD indicates less availability of oxygen to higher forms of aquatic life. A high BOD mean low dissolved oxygen (DO). As a result of such conditions aquatic organisms become stressed, suffocate, and die.

Isolation due to land barriers

- Distribution and number of organisms in freshwater habitats is also influenced by the ability of the organisms to disperse in their surroundings. Isolation due to the presence of land barriers does not allow free movement of the organism to become established in places otherwise favorable.
- Example: Fishes of the streams, which are separated only by a few miles from land but isolated by water, may have their niches occupied by different species. On the other hand, most small organisms-algae, crustacea, protozoa and bacteria, etc. have a wide range of dispersal. Thus, same kinds of water flea are observed (Daphnia, for example) in a pond in the United States as in England.


Freshwater
Ecosystems

Module# 54

Lotic Ecosystems

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The Lotic Aquatic System

- Lotic aquatic systems are the fresh water bodies with flowing waters.
- Water in lotic system is in a state of constant motion.
- Rivers and streams are the most common examples of such systems.
- The basic function of lotic water bodies is assumed to carry the excess of rain water back to the sea.
- It is estimated that the amount water that these lotic bodies carry to the oceans amounts to approximately 25 cm of rains that can be distributed evenly on the land surface.

Lotic Systems (Running Water)







- **Rivers, streams, creeks**

Lotic Ecosystem

- The ecosystem of a river, stream, or spring.
- Lotic refers to flowing water.
- Included in the lotic environment are the biotic interactions (plants, animals and micro-organisms) as well as the abiotic interactions (physical and chemical).



Lotic Ecosystem

- A natural system provides a variety of habitats for many species of aquatic plants and animals.
- The instream habitats provide areas for feeding, resting, and reproduction and generally support a great diversity of organisms.



Examples include:

- Rivers
- Streams
- Brooks
- Creeks
- Human made channels



Typically, the lotic aquatic systems have the following properties:

- The flow of water in continuous and unidirectional in a lotic system. Water mostly flows in narrow channels that are less deep than the lakes. Water from diverse sources can pool into a lotic body. The bottoms of such water bodies continuously receive materials from the biomass of surface water.
- The volume of water changes rapidly which results in change in the velocity of water currents. There is also a wide range of fluctuation in the water level in a lotic system.
- Water in these systems act as effective agent of transportation, transfer, and dilution. They continuously erode materials along their channels and deposit them to other places.

- There is no vertical stratification and is a thorough mixing of contents.
- The physico-chemical statistics of water quality is in a state of perpetual change. The environment of a lotic aquatic system is relatively unstable as compared to the lentic system.
- There is a rich supply of oxygen which is derived from air above and also the photosynthetic outputs of the autotrophs. Oxygen is evenly distributed across the water body and oxygen depletion is very rare in unpolluted lotic systems.
- Turbidity limits the light penetration to the deeper zones. Very less amount of carbon dioxide present in these waters. Such conditions limit productivity. Therefore, algae bloom and growth of other organisms is rarely observed in moving waters.

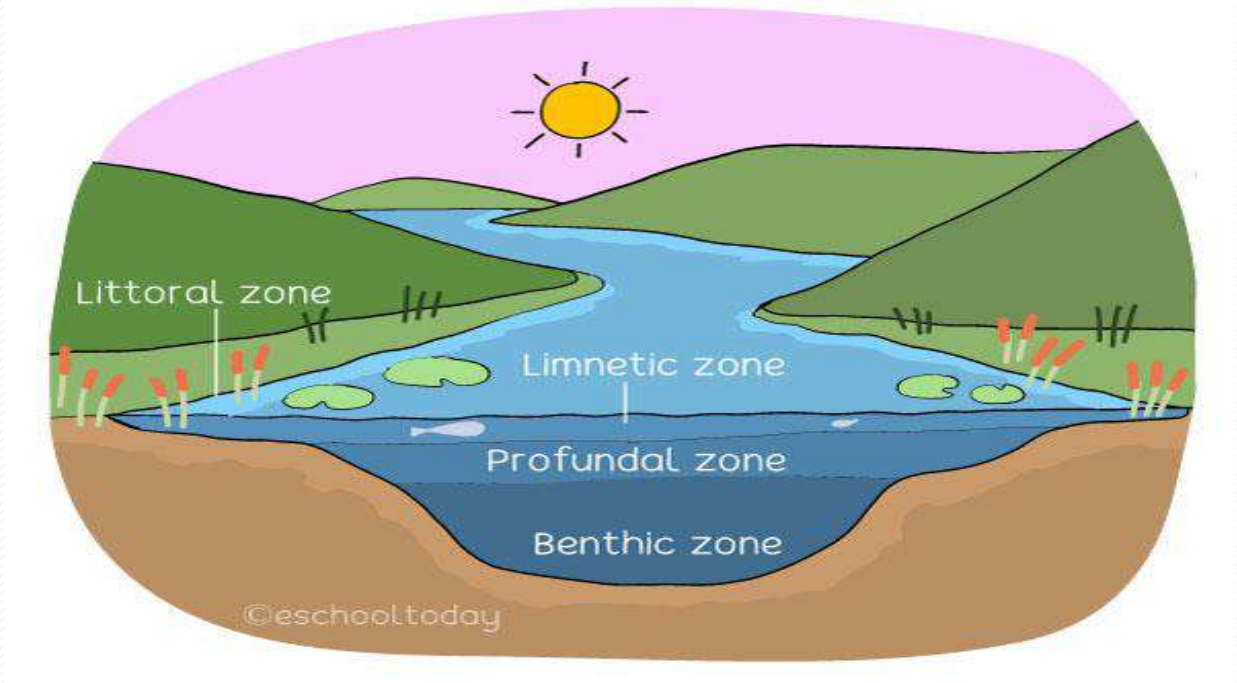
Module# 55

lentic ecosystem

lentic ecosystem

- A **lentic ecosystem** entails a body of standing water, ranging from ditches, seeps, ponds, seasonal pools and lakes.
- Deeper waters, such as lakes, may have layers of ecosystems, influenced by light.
- Ponds, due to their having more light penetration, are able to support a diverse range of water plants.

- Animals and plants in a lentic system behave and adapt differently because there are various zones of every lentic system with different abiotic conditions.
- The diagram below illustrates the zones of a lentic system.



Littoral Zone:

- This zone forms the banks or boundary of the water. There is usually warmer and shallow water, with lots of plants such as reeds and cattails. Many of the plants are rooted in the muddy floor and their leaves form a good hideout for insects and fish. There is a lot of fish, frogs, reptiles, and insects in this zone.

Limnetic Zone:

- It is the zone after the Littoral zone, extending a few meters into the water body. Fish swim freely here, with some plants floating on the water. There is enough sunlight here, and algae and plankton usually thrive here.

Profundal Zone:

- This zone extends beyond the Limnetic zone and is a deeper region of the water. Here, there is usually very little or no sunlight. It is cooler. Living things in this region depend on drifting organic matter for food.

Benthic Zone:

- This is the deepest region of the water body and farthest from the banks. There is no sunlight in this region, and therefore very little or no photosynthesis occurs here. Organisms that live here include worms, bacteria, decomposers, and other scavengers.


Lotic
Lentic

Lentic Ecosystem

- Includes lakes, ponds, dammed rivers, and other static freshwater habitats





Aquatic Ecosystems - Lentic

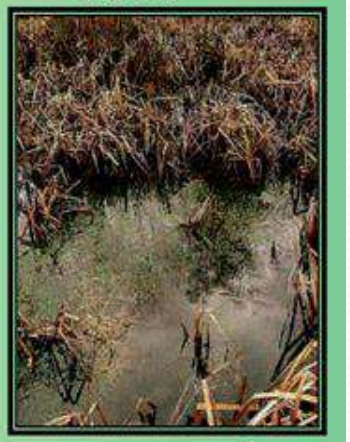
Lake





Bog

Pond





Swamp



Marsh

Module# 56

Wetlands Ecosystem

Wetlands Ecosystem

- Wetlands are some of the most valuable ecosystems on Earth. They act like giant sponges or reservoirs.
- During heavy rains, wetlands absorb excess water, limiting the effects of flooding. Wetlands also protect coastal areas from storm surges that can wash away fragile beaches and coastal communities.
- Saltwater swamps and tidal salt marshes help secure coastal soil and sand.

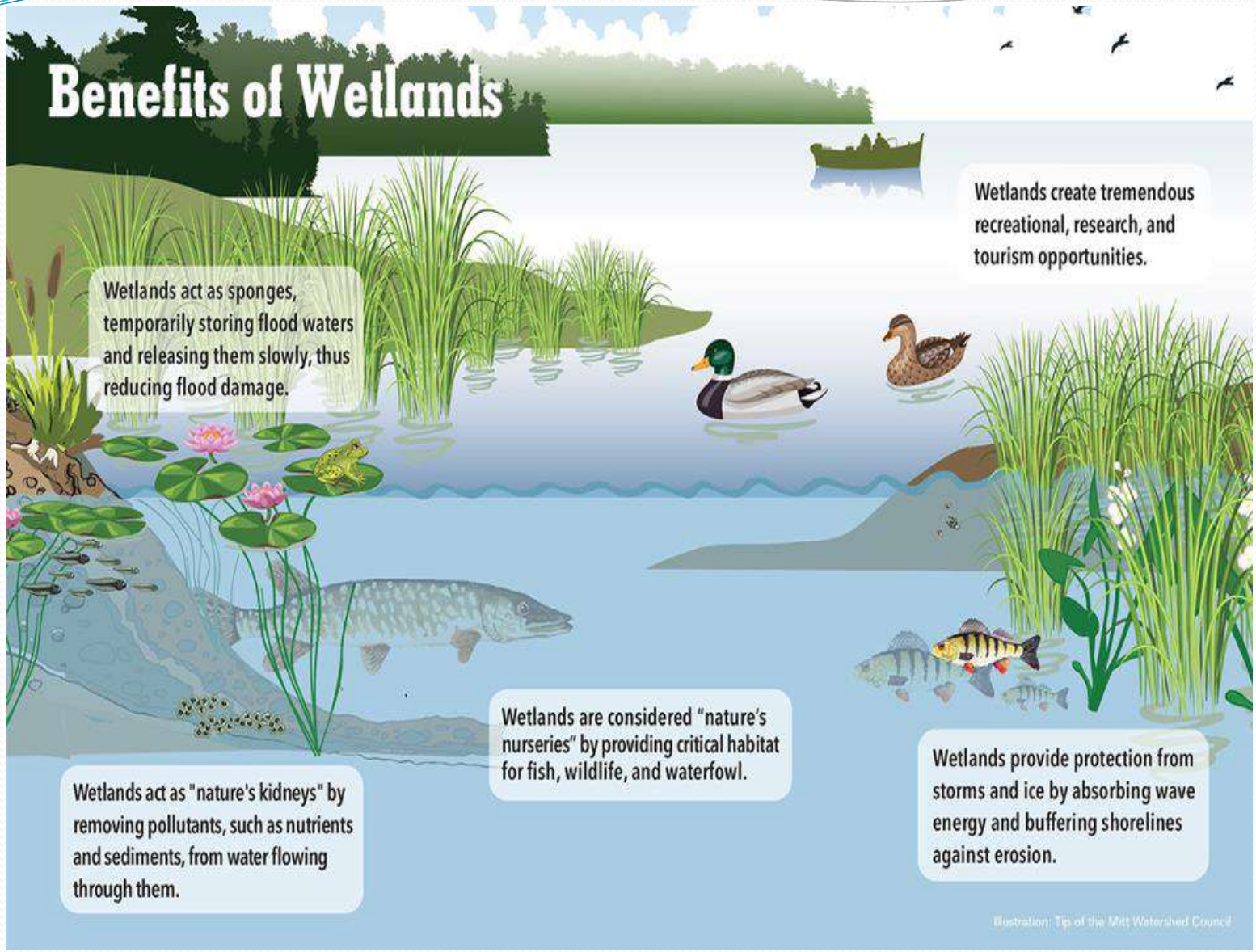



Wetlands
tamarack tree
nesting osprey
cedar tree
black-toed woodpecker
yellow spatterdock
viceroy butterfly
dragonfly
red-winged blackbird
jack-in-the-pulpit
smoky shrew
pickerelweed
sundew
blueberries
blue-winged teal
mink
spring peeper
cattails
pitcher plant
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- Wetland ecosystems also act as water-treatment facilities.
- The plants, fungi, and algae of a wetland filter wastes and purify water.
- Nitrates and other runoff chemicals often wash into wetlands from urban areas and farms.
- Organisms there absorb the harmful chemicals. Pollutants not absorbed by plants slowly sink to the bottom, where they are buried in sand and other sediment.

- Wetlands, especially marshes and swamps, are home to a wide variety of plant and animal life.
- Some animals, such as shrimp, live in tidal marshes. Many marine fishes, such as striped bass, enter coastal wetlands to spawn.
- Bass swim from the ocean and into salt marshes to lay their eggs. When the eggs hatch, the young bass find plenty of food and some protection in the grasses or tree roots.
- Oysters live in huge reefs in salt marshes. All of these wetlands are home to economically valuable fisheries.


Benefits of Wetlands
Wetlands create tremendous recreational, research, and tourism opportunities.
Wetlands act as sponges, temporarily storing flood waters and releasing them slowly, thus reducing flood damage.
Wetlands are considered "nature's nurseries" by providing critical habitat for fish, wildlife, and waterfowl.
Wetlands act as "nature's kidneys" by removing pollutants, such as nutrients and sediments, from water flowing through them.
Wetlands provide protection from storms and ice by absorbing wave energy and buffering shorelines against erosion.
Illustration: Tip of the Mitt Watershed Council

Benefits of wetland ecosystems



Controls Flood



Provides Clean Source Of Water



Improves Air Quality



Helps sustain healthy livelihood



Filters out toxins & adds in nutrition



Habitats to hundreds of species, including the endangered ones


What makes a freshwater wetland?
Rivers and Streams
Bogs
Ponds
Vernal Pools
Wet Meadows
Marshes
Swamps

Module# 57 Population Ecology

- Population ecology is the study of populations in relation to the environment

- Includes environmental influences on population density and distribution, age structure, and variations in population size

Definition of a Population

- A population is the total number of individuals of a given biological species found in one place at one time.

- In practice, ecologists often deal with density—numbers per unit area for land organisms and numbers per unit volume in aquatic systems—or even weight rather than raw numbers.



Population Ecology





- Population ecology is the study of populations in relation to environment
  - Including environmental influences on population density and distribution, age structure, and variations in population size





Module# 58

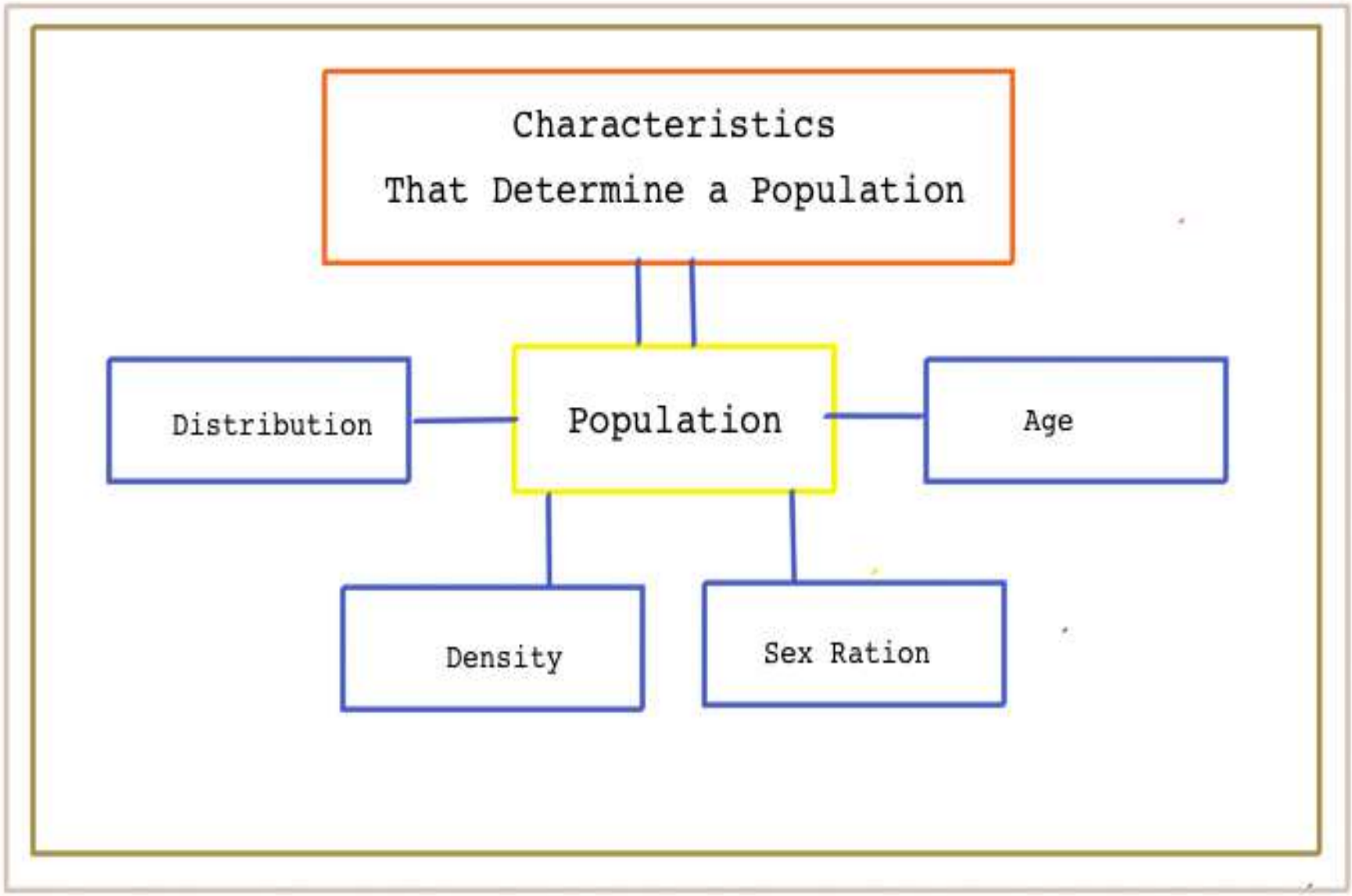
Characteristics of Population Ecology

- All species occur in groups called populations. There are certain characteristics that all populations have, such as population density, spatial distribution, and growth rate. These characteristics are used to classify all populations of organisms, including bacteria, animals, and plants.

Characteristics of a Population

- **Population** - individuals inhabiting the same area at the same time

- **Population Dynamics:** Population change due to
  - **Population Size** - number of individuals
  - **Population Density** - population size in a certain space at a given time
  - **Population Dispersion** - spatial pattern in habitat
  - **Age Structure** - proportion of individuals in each age group in population


Characteristics
That Determine a Population
Distribution
Population
Age
Density
Sex Ration

Population Ecology: Population Characteristics

Population Characteristics

1. **Population Density:**
   - The number of organisms per unit area
2. **Spatial Distribution:**
   - Dispersion: The pattern of spacing a population within an area
     - 3 main types of dispersion
       - Clumped
       - Uniform
       - Random
   - The primary cause of dispersion is resource availability

Module#59 Population density

Population density one characteristic of a population is its population density, which is the number of organisms per unit area.

For example, the population density of cattle egrets, shown with the water buffalo in Figure, is greater near the buffalo than farther away. Near the water buffalo, there might be three birds per square meter. Fifty meters from the water buffalo, the density of birds might be zero



Spatial distribution

- Another characteristic of a population is called dispersion—the pattern of spacing of a population within an area. Figure shows the three main types of dispersion—uniform, clumped groups, and random. Black bears are typically dispersed in a uniform arrangement. American bison are dispersed in clumped groups or herds. White-tailed deer are dispersed randomly with unpredictable spacing. One of the primary factors in the pattern of dispersion for all organisms is the availability of resources such as foo

Population density describes how many individual organisms live in a given area. Dispersion describes how the individuals are spaced within that area. Population range describes a species' distribution.

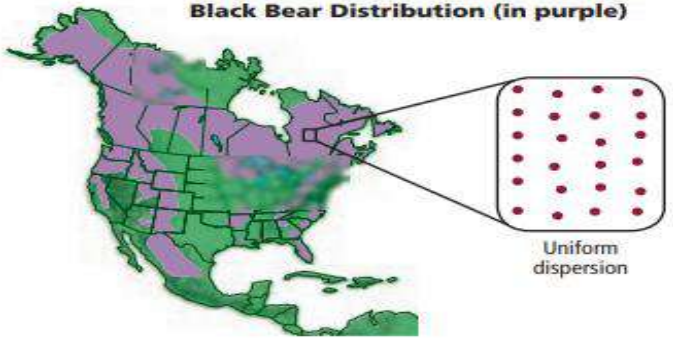
Black Bear



Density: one bear per several hundred square kilometers

Dispersion: American black bear males usually are dispersed uniformly within territories as large as several hundred square kilometers. Females have smaller territories that overlap those of males.

Black Bear Distribution (in purple)



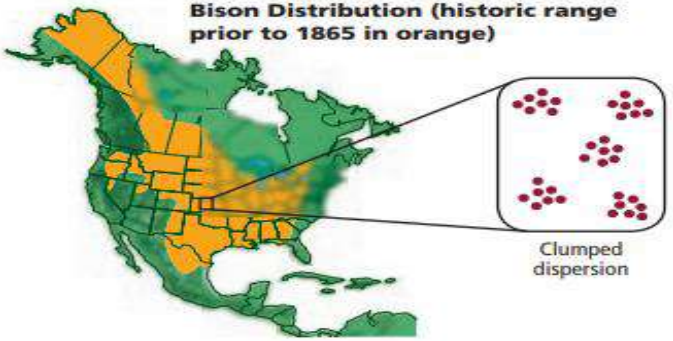
American Bison



Density: four bison/km^2 in Northern Yellowstone in 2000

Dispersion: American bison are found in clumped groups called herds.

Bison Distribution (historic range prior to 1865 in orange)



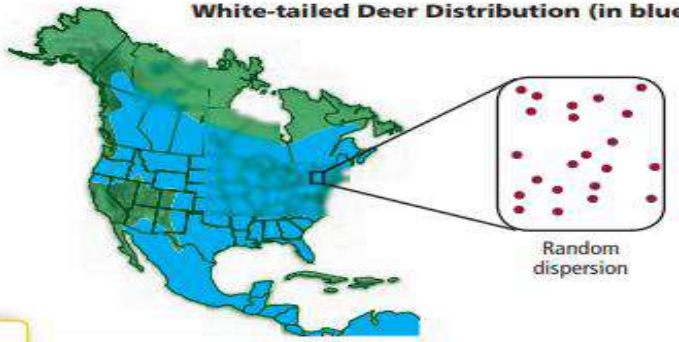
White-tailed Deer



Density: 10 deer/km^2 in some areas of the northeastern United States

Dispersion: White-tailed deer are dispersed randomly throughout appropriate habitats.

White-tailed Deer Distribution (in blue)



- Uniform dispersion refers to organisms that live in a specific territory. One example would be penguins. Penguins live in territories, and within those territories the birds space themselves out relatively uniformly.
- Random dispersion refers to the spread of individuals such as wind-dispersed seeds, which fall randomly after traveling.

- Clustered or clumped dispersion refers to a straight drop of seeds to the ground, rather than being carried, or to groups of animals living together, such as herds or schools. Schools of fish exhibit this manner of dispersion.

Module# 60 Population ranges

Population ranges No population, not even the human population, occupies all habitats in the biosphere. Some species, such as the iiwi (EE ee wee) shown in Figure, have a very limited population range. This songbird is found only on some of the islands of Hawaii.



The iiwi lives only on some of the Hawaiian islands. The peregrine falcon is found worldwide.

- Other species, such as the peregrine falcon shown in Figure, have a vast distribution. Peregrine falcons are found on all continents except Antarctica.

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Peregrine falcon

- A species might not be able to expand its population range because it cannot survive the abiotic conditions found in the expanded region.
- A change in temperature range, humidity level, annual rainfall, or sunlight might make a new geographic area uninhabitable for the species.
- In addition, biotic factors, such as predators, competitors, and parasites, present threats that might make the new location difficult for survival

Module# 61 Population Dynamics and Regulation

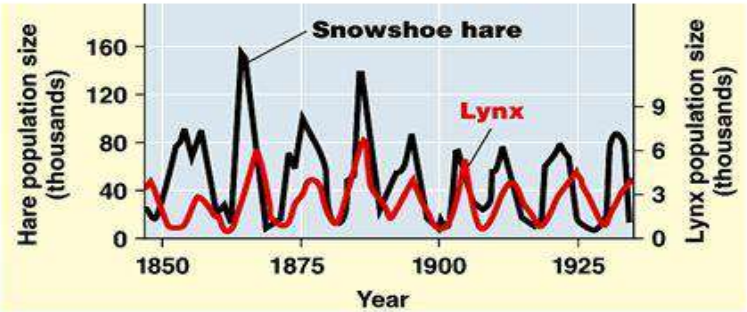
- The logistic model of population growth, while valid in many natural populations and a useful model, is a simplification of real-world population dynamics. Implicit in the model is that the carrying capacity of the environment does not change, which is not the case.
- The carrying capacity varies annually: for example, some summers are hot and dry whereas others are cold and wet. In many areas, the carrying capacity during the winter is much lower than it is during the summer.

- Also, natural events such as earthquakes, volcanoes, and fires can alter an environment and hence its carrying capacity. Additionally, populations do not usually exist in isolation.
- They engage in interspecific competition: that is, they share the environment with other species, competing with them for the same resources. These factors are also important to understanding how a specific population will grow. Nature regulates population growth in a variety of ways..

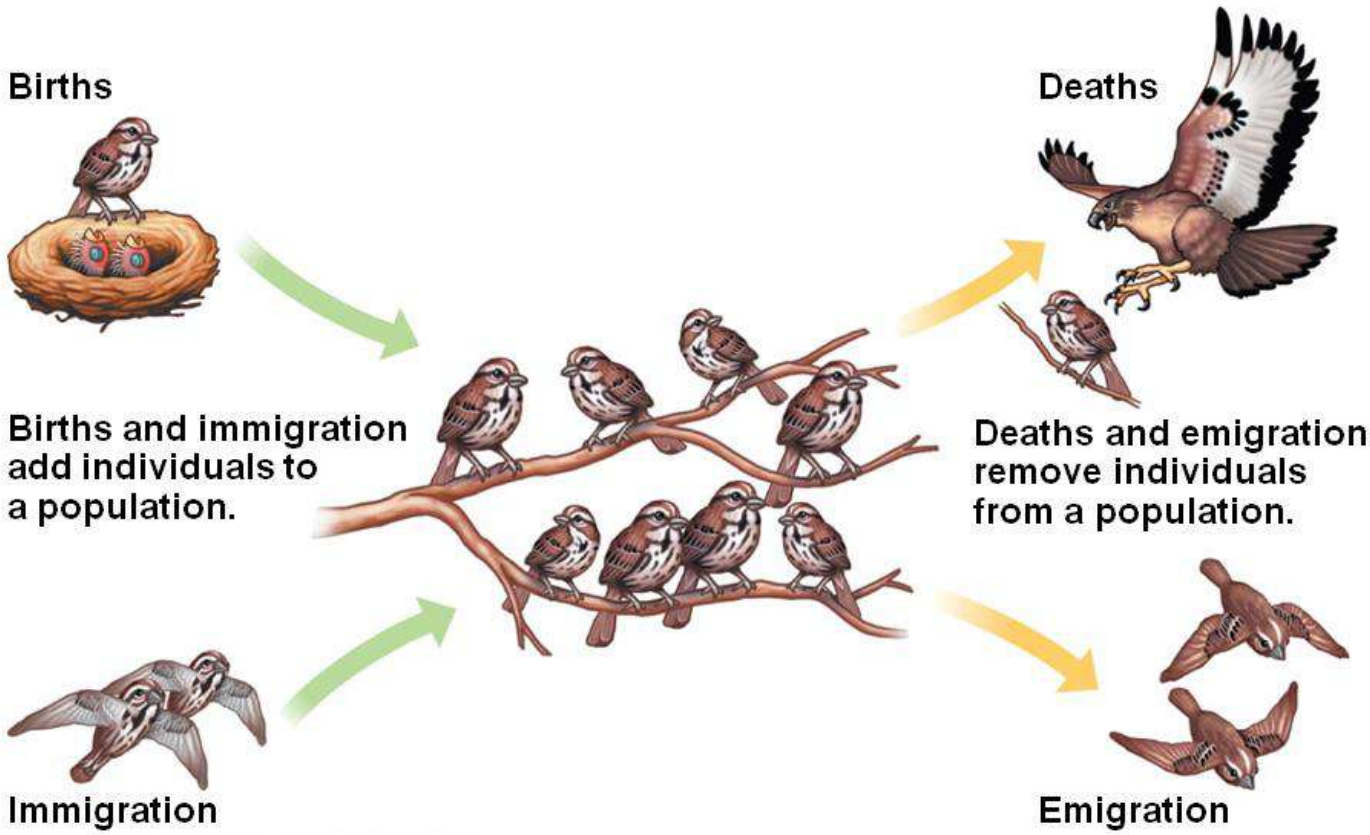
- These are grouped into **density-dependent factors**, in which the density of the population at a given time affects growth rate and mortality, and **density-independent factors**, which influence mortality in a population regardless of population density.
- Note that in the former, the effect of the factor on the population depends on the density of the population at onset. Conservation biologists want to understand both types because this helps them manage populations and prevent extinction or overpopulation

2 types of Population Regulations

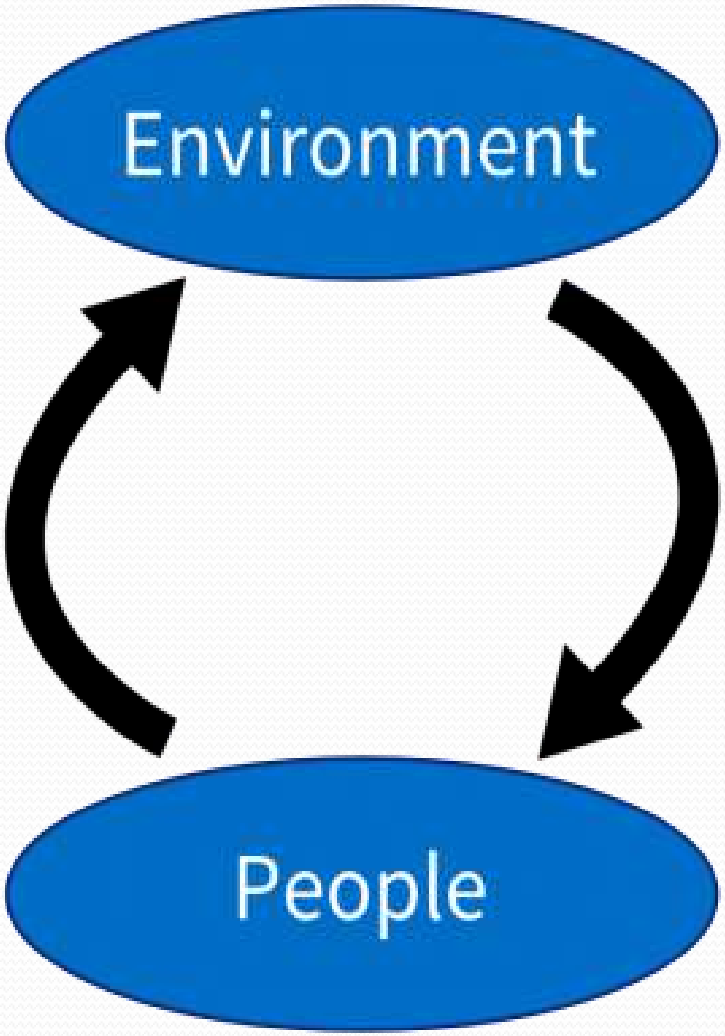
1. **Density Dependent** – death is a result of over population
   - Competition among members of a growing population for limited resources, like food or territory.
   - Health of organisms
   - Predation
   - Physiological factors (reproduction, growth, hormone changes)
2. **Density Independent-** death is a result regardless of the population size
   - Weather conditions
   - Acidity
   - Salinity
   - Fires
   - Catastrophies



Population dynamics



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The environment affects people:

Climate
Water availability
Shelter availability
Soil for plant growth

People affect the environment:

Burning fossil fuels
Creating urbans areas
Cutting forests for wood
Flushing waste into rivers

Module# 62 Density-dependent Regulation

- Most density-dependent factors are biological in nature (biotic), and include predation, inter- and intraspecific competition, accumulation of waste, and diseases such as those caused by parasites.

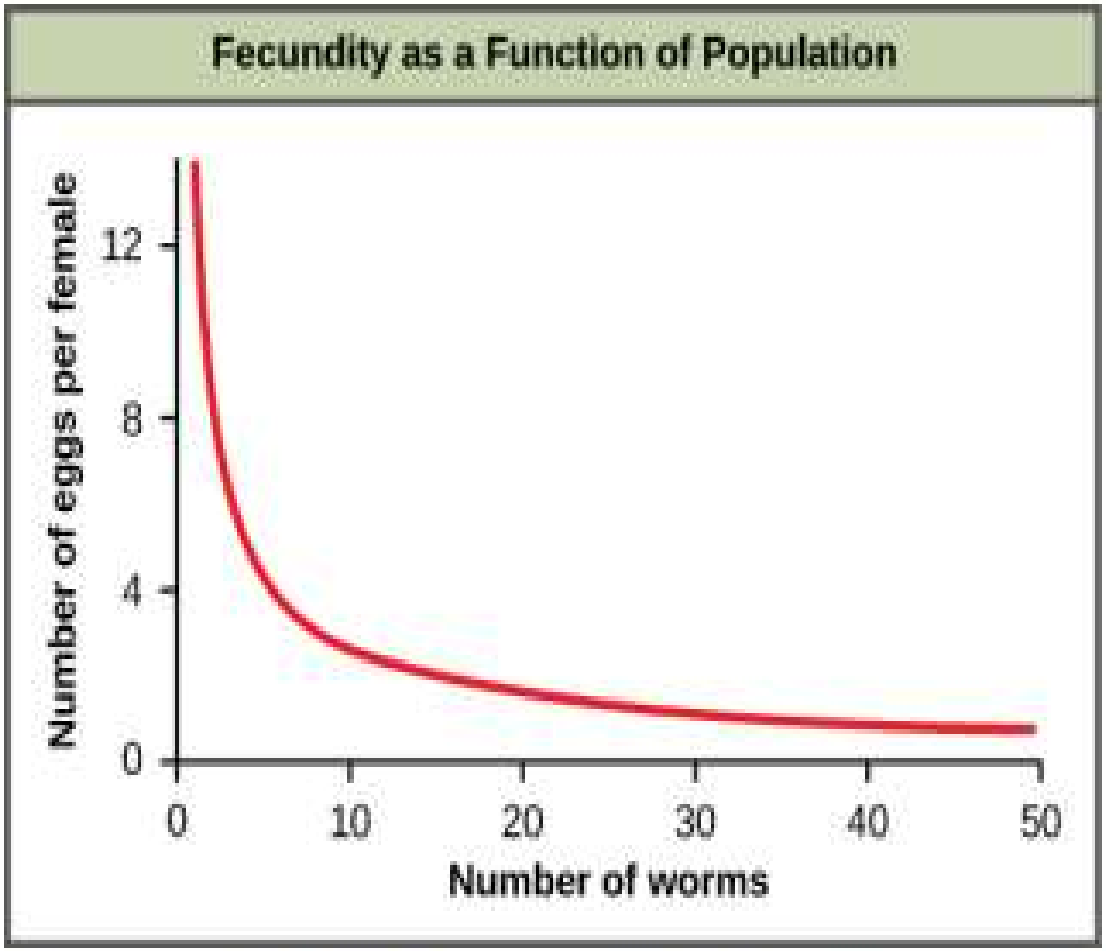
- Usually, the denser a population is, the greater its mortality rate.

- For example, during intra- and interspecific competition, the reproductive rates of the individuals will usually be lower, reducing their population's rate of growth.

- In addition, low prey density increases the mortality of its predator because it has more difficulty locating its food source.

- An example of density-dependent regulation is shown in Figure with results from a study focusing on the giant intestinal roundworm (Ascaris lumbricoides), a parasite of humans and other mammals. Denser populations of the parasite exhibited lower fecundity: they contained fewer eggs.

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In this population of roundworms, fecundity (number of eggs) decreases with population density

- One possible explanation for this is that females would be smaller in more dense populations (due to limited resources) and that smaller females would have fewer eggs. This hypothesis was tested and disproved in a 2009 study which showed that female weight had no influence.
- The actual cause of the densitydependence of fecundity in this organism is still unclear and awaiting further investigation

Module# 63

Density-independent Regulation and Interaction with Density dependent Factors

- Many factors, typically physical or chemical in nature (abiotic), influence the mortality of a population regardless of its density, including weather, natural disasters, and pollution.
- An individual deer may be killed in a forest fire regardless of how many deer happen to be in that area.
- Its chances of survival are the same whether the population density is high or low. The same holds true for cold winter weather

- In real-life situations, population regulation is very complicated and density-dependent and independent factors can interact.
- A dense population that is reduced in a density-independent manner by some environmental factor(s) will be able to recover differently than a sparse population.
- For example, a population of deer affected by a harsh winter will recover faster if there are more deer remaining to reproduce.

Factors affecting population size

Density independent

Physical (abiotic) factors

- Rainfall
- Humidity
- Temperature
- Acidity
- Salinity





Catastrophic events

- Flood
- Fire
- Drought
- Volcanic eruptions
- Earthquakes
- Tsunamis







Density dependent (biotic factors)

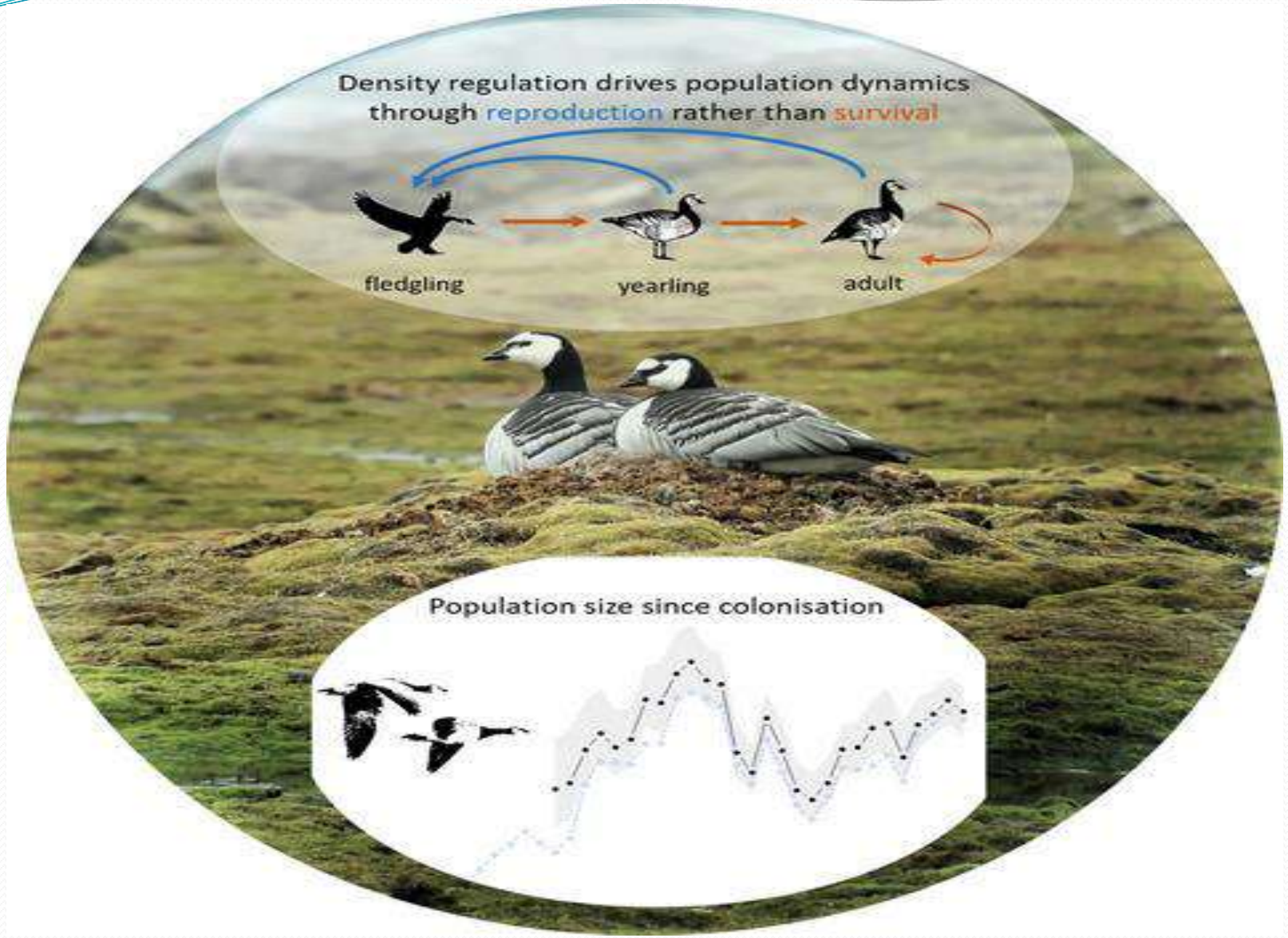
- Food supply (abundance or distribution of prey species) - this becomes more critical as population size and density increases
- Disease - this is easier to spread as population size and density increases
- Parasites - these spread more easily as population size and density increases
- Competition - this becomes more critical as population size and density increases
- Predation (abundance or distribution of predator species) - this becomes more critical as population size and density decreases







The **carrying capacity** is the number of organisms that can be supported by the environment in which the population exists.


Density regulation drives population dynamics
through reproduction rather than survival
fledgling
yearling
adult
Population size since colonisation


DENSITY-DEPENDENT FACTORS
competition for resources
Study.com

Deforestation

- Density-Independent
- Human activities
  - Deforestation; over-hunting/fishing
  - Effects are harmful on population regardless of size

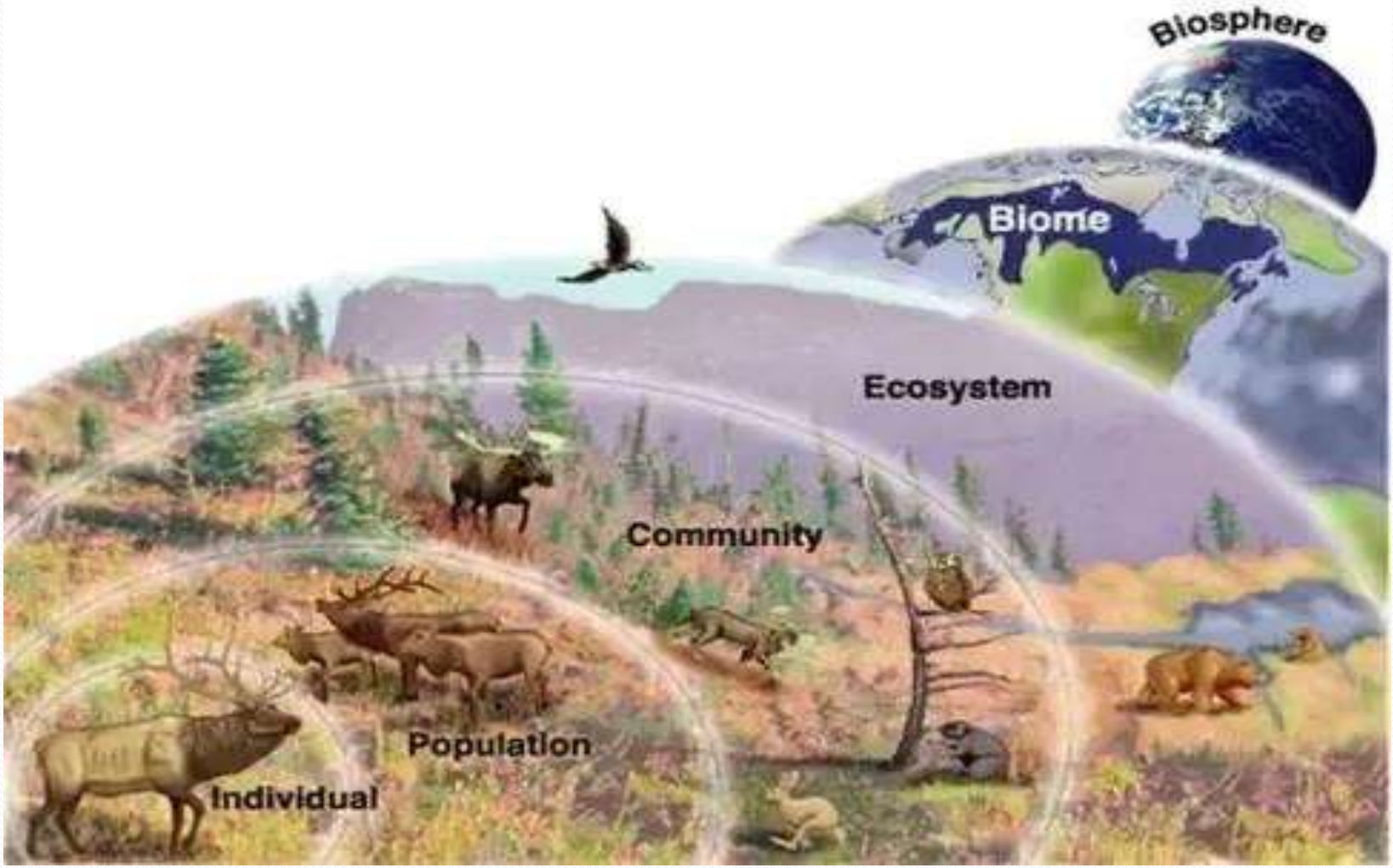


Module# 64

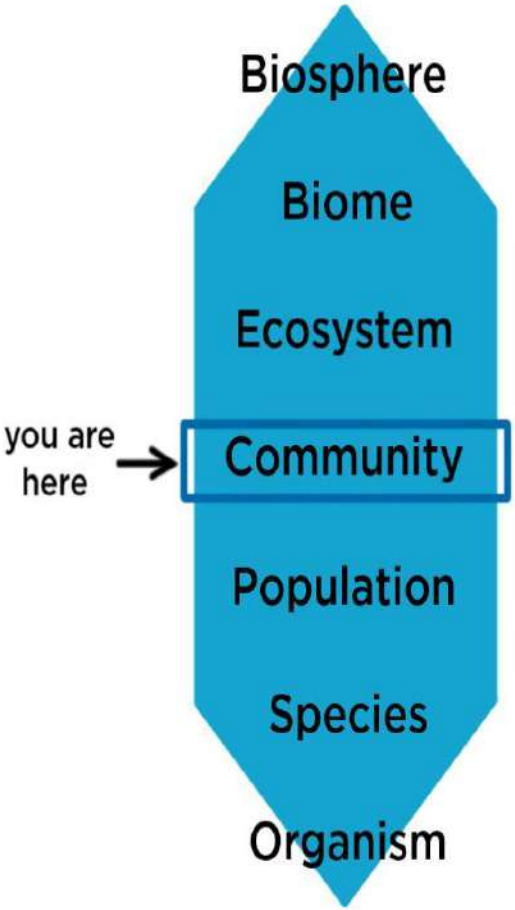
Community Ecology

- A community is defined as an assemblage of species living close enough together for potential interaction.
- Communities differ in their species richness, the number of species they contain, and the relative abundance of different species.


Community Ecology
organism
population
community
ecosystem


Biosphere
Biome
Ecosystem
Community
Population
Individual

Levels of Biological Organization: Community



What is a community?

A community is made up of all the living organisms that reside within a given area and interact with one another.

How is it different from a population?

A population is made up of members of single species, which live in a particular area. A community is made up of populations of different species, which live in an area.

How is it different from an ecosystem?

While a community consists of all the living organisms within an area, an ecosystem also includes the nonliving components within the area, like water and sunlight.

13 **FIGURE 1–21 LEVELS OF ORGANIZATION**

Biosphere	The part of Earth that contains all ecosystems	
Ecosystem	Community and its nonliving surroundings	
Community	Populations that live together in a defined area	
Population	Group of organisms of one type that live in the same area	 Bison herd
Organism	Individual living thing	 Bison
Groups of Cells	Tissues, organs, and organ systems	 Nervous tissue Brain Nervous system
Cells	Smallest functional unit of life	 Nerve cell
Molecules	Groups of atoms; smallest unit of most chemical compounds	 Water DNA

Module# 65
Community's interactions

A community's interactions include

Competition,
Predation,
Herbivory
Symbiosis and disease

- There are a number of possible **interspecific interactions** that link the species of a community.

- Interspecific interactions can be symbolized by the positive (+) or negative (–) effects of the interaction on the individual populations.

- **Interspecific competition** can occur when species compete for a specific limiting resource.

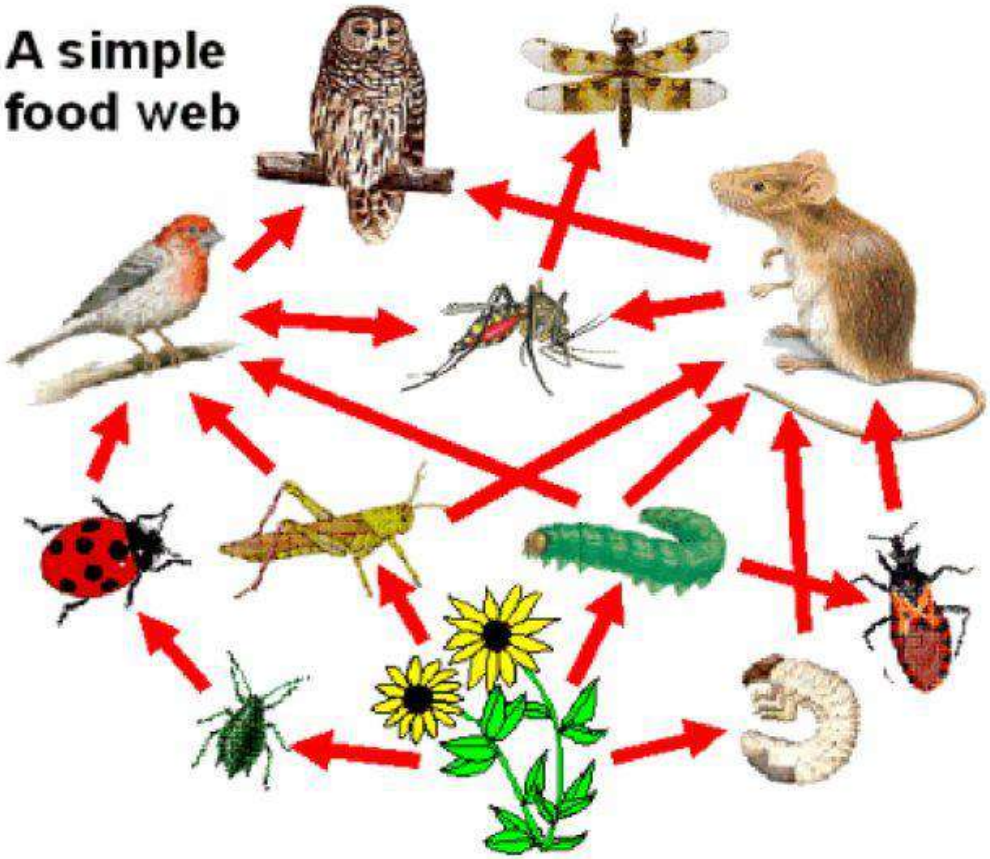
- When two species compete for a resource, the result is detrimental to one or both species.

- Strong competition can lead to the local elimination of one of the two competing species, a process called competitive exclusion.

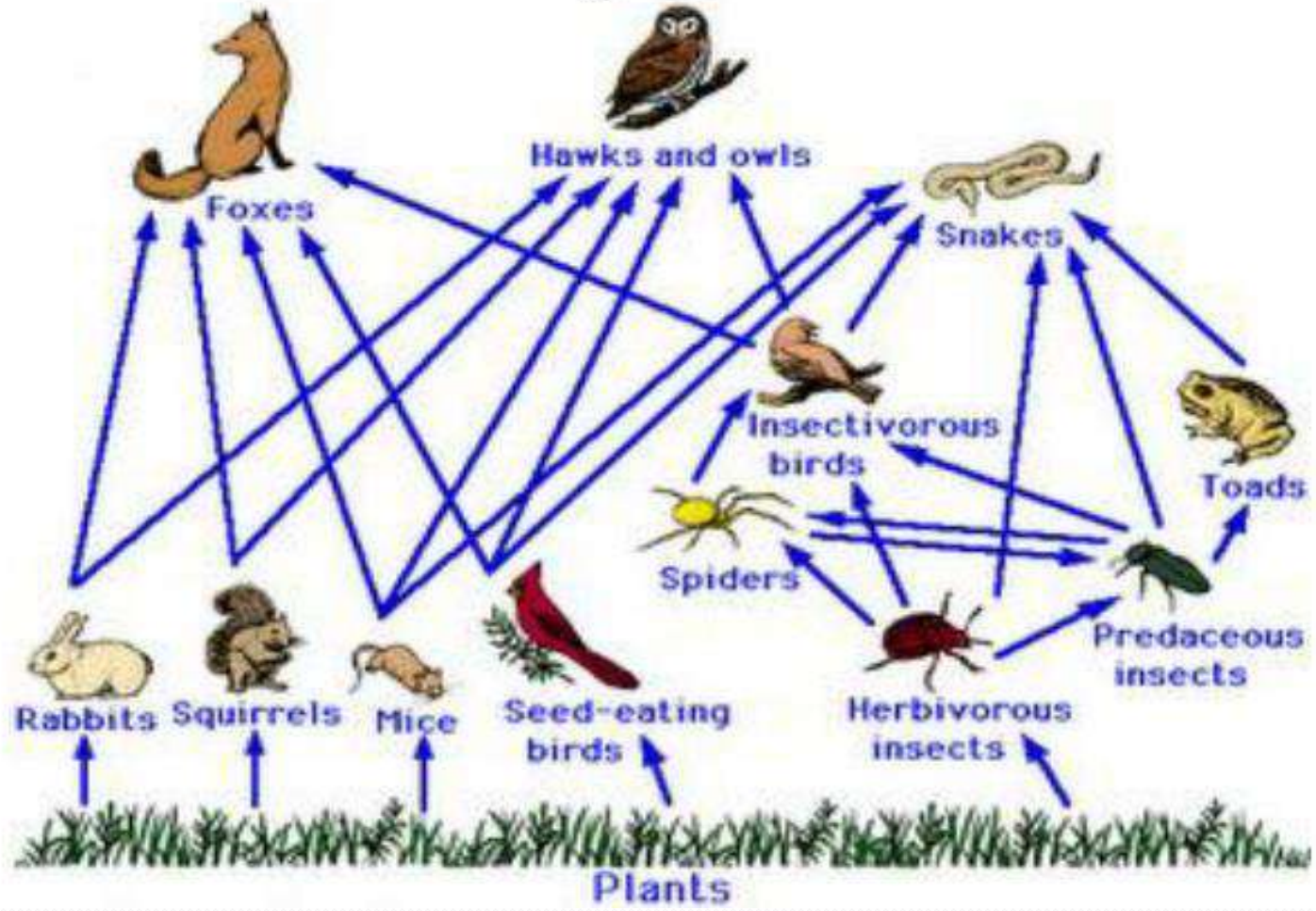
- The competitive exclusion principle states that two species with similar needs for the same limiting resources cannot coexist in the same place.

Community Interactions

- **Community**: Many different species interacting in the same environment.
- Three types of interactions:
  - Competition
  - Predation
  - Symbiosis



Community Interactions



Module# 66 Competition

- Competition is a relationship between organisms in which one is harmed when both are trying to use the same resource related to growth, reproduction, or survivability.
- Competition stems from the fact that resources are limited. There are simply not enough of some resources for all individuals to have equal access and supply.
- Competition can occur between organisms of the same species, or between members of different species.

- Competition between species can either lead to the extinction of one of the species, or a decline in both of the species.
- However, this process can often be interrupted by environmental disturbances or evolution, which can change the rules of the game.
- Competition is often involved when species are limited in their range, often by direct competition from other organisms.

Competition

- Different species living in the same environment, or habitat, may require the same resources. When the resources are limited, competition occurs among the species.
- Competition- is the struggle between different species for the same limited resources. The more similar the needs of the species, the more intense the competition.

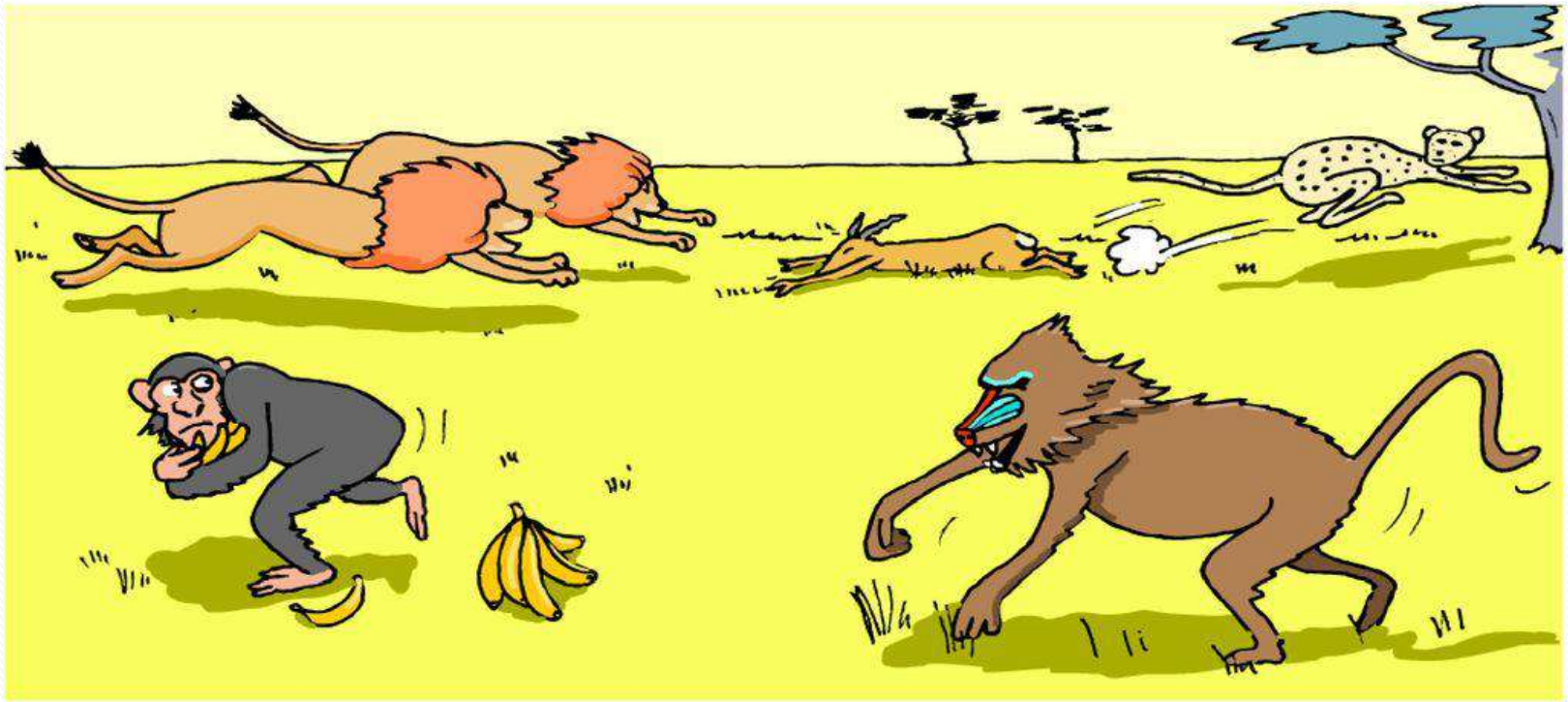


Competition: Common use of Scarce Resources

- In Africa, lions and hyenas both compete for prey. Fierce rivalry between both species sometimes leads to standoffs and battles.
- Most competition among species does not involve direct contact.
- Species most often interact by their effects on the abundance of a resource.



INTERspecific Competition: Competition **BETWEEN** different species of organisms



INTRAspecific Competition: Competition **WITHIN** the same species of organism



Module# 67 Predation

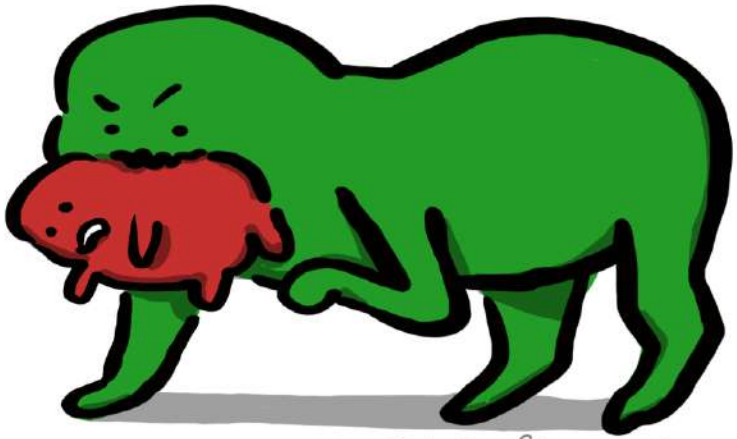
- **Predation** is a biological interaction where one organism, the **predator**, kills and eats another organism, its **prey**.
- A *predator* is an organism that consumes all or part of the body of another—living or recently killed—organism, which is its *prey*. "Living or recently killed" distinguishes predators from decomposers, such as fungi and bacteria that break down the leftover remains of organisms that have died.

- If we see a lion eating a zebra, we can feel comfortable in saying that the lion is a predator. In the broad definition, however, the zebra is too A predator's prey can be an animal, but it can also be a plant or fungus.
- Nor does a predator necessarily have to kill its prey. Instead, as in a grazing cow or a bloodsucking mosquito, it may simply take a portion of the prey's body and leave it alive.

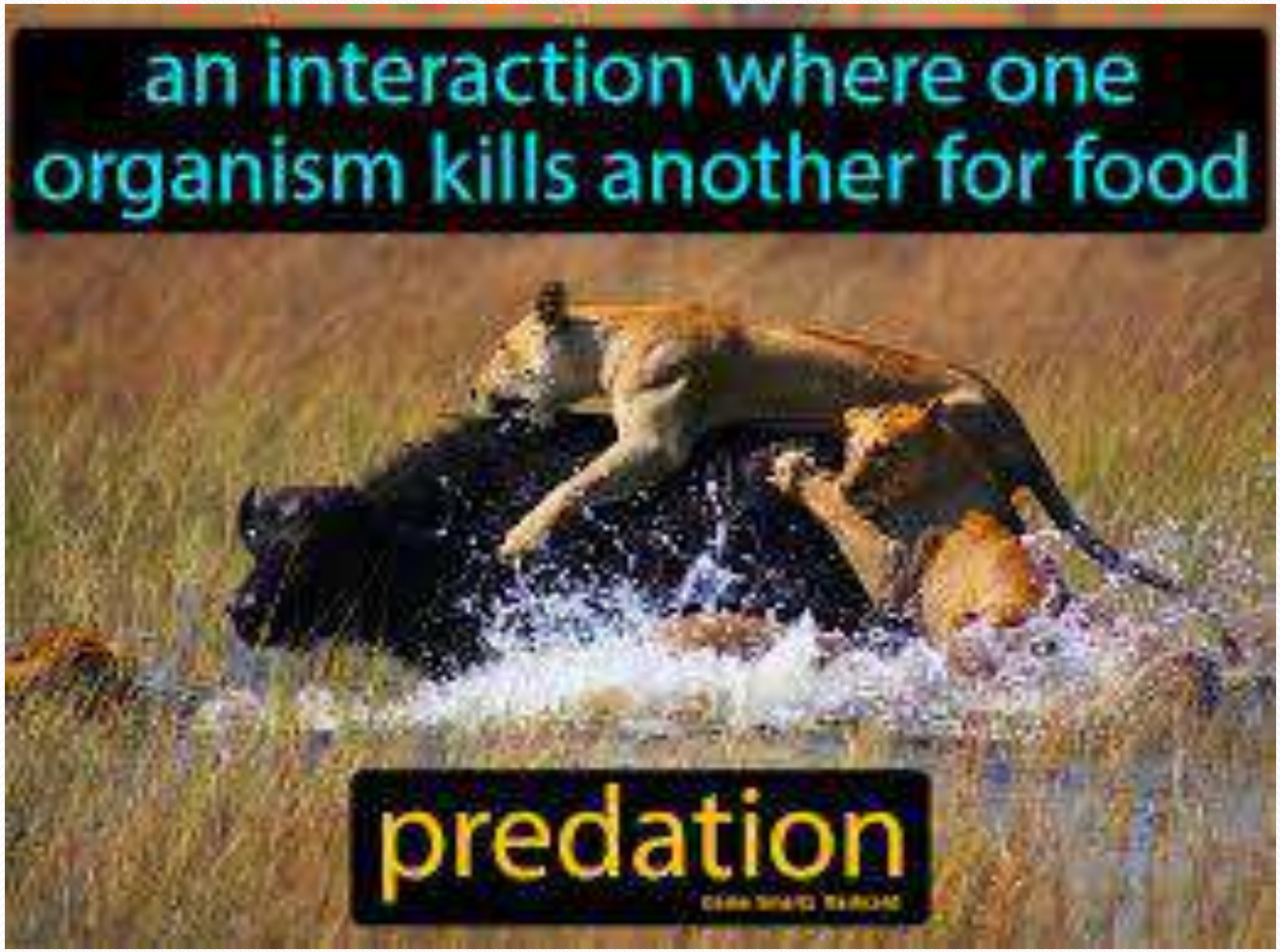
Predation

Predation is a (+/–) relationship between the **predator** and the **prey** species. The predator survives by killing and eating prey. Predators are always in a higher **trophic level** than their prey.

(+/–)



Predators are at least secondary consumers or above, while prey can be primary consumers (herbivores).


an interaction where one organism kills another for food
predation

Ecological Relationships

- Predation – the act of one organism consuming another organism for food.





- Example: cat (predator) mouse (prey)

Module# 68 Herbivory

Herbivory

- Herbivory is the consumption of plant material by animals, and herbivores are animals adapted to eat plants.
- As in predator-prey interactions, this interaction drives adaptations in both the herbivore and the plant species it eats.
- For example, to reduce the damage done by herbivores, plants have evolved defenses, including thorns and chemicals.

- To maximize nutrient intake, many herbivores have evolved adaptations that allow them to determine which plants contain fewer defensive compounds and more high-quality nutrients.
- Some insects, such as butterflies, have chemical sensors on their feet that allow them to taste the plant before they consume any part of it.
- Mammalian herbivores often use their keen sense of smell to detect bitter compounds, and they preferentially eat younger leaves that contain fewer chemicals.

What is herbivory?

- Herbivory is the eating or consumption of plants (primary producers) or plant material by animals known as primary consumers
  - Heterotrophs
  - Autotrophs



- Herbivory, like parasitism & predation, is a +,- interaction
- Herbivory is important because plants are the ultimate source of energy for food webs in most environments.

Herbivores

Herbivores are animals that only eat plants.

These can be flowers, fruit, nuts, grass or even wood.











Module# 69

Symbiotic Relationships

- Symbiosis is a close relationship between two species in which at least one species benefits.
- For the other species, the relationship may be positive, negative, or neutral. There are three basic types of symbiosis:
- mutualism
- commensalism
- parasitism.

Mutualism

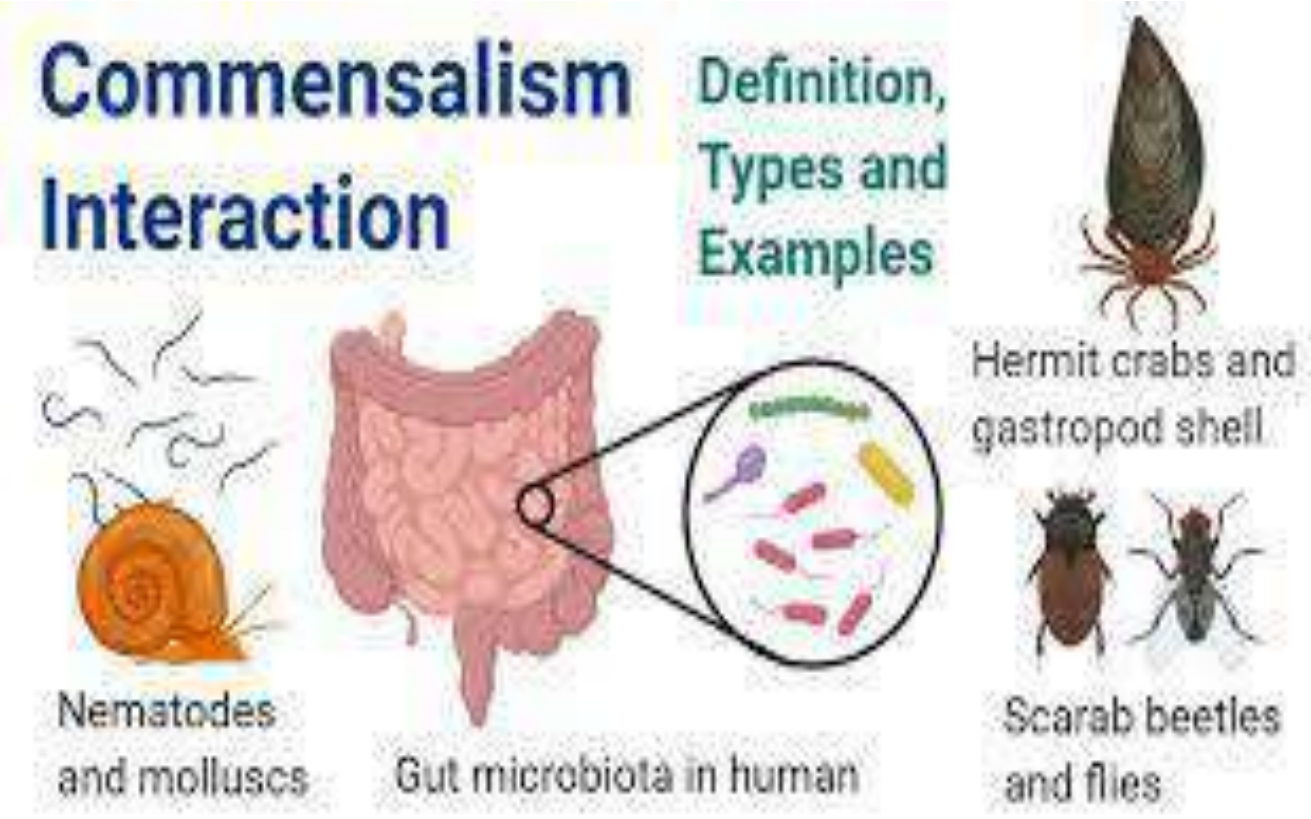
- **Mutualism** is a symbiotic relationship in which both species benefit. An example of mutualism involves goby fish and shrimp . The nearly blind shrimp and the fish spend most of their time together.
- The shrimp maintains a burrow in the sand in which both the fish and shrimp live. When a predator comes near, the fish touches the shrimp with its tail as a warning. Then, both fish and shrimp retreat to the burrow until the predator is gone.
- From their relationship, the shrimp gets a warning of approaching danger. The fish gets a safe retreat and a place to lay its eggs.

The multicolored shrimp in the front and the green goby fish behind it have a mutualistic relationship.



Commensalism

- Commensalism is a symbiotic relationship in which one species benefits while the other species is not affected.
- One species typically uses the other for a purpose other than food.
- For example, mites attach themselves to larger flying insects to get a "free ride." Hermit crabs use the shells of dead snails for homes.

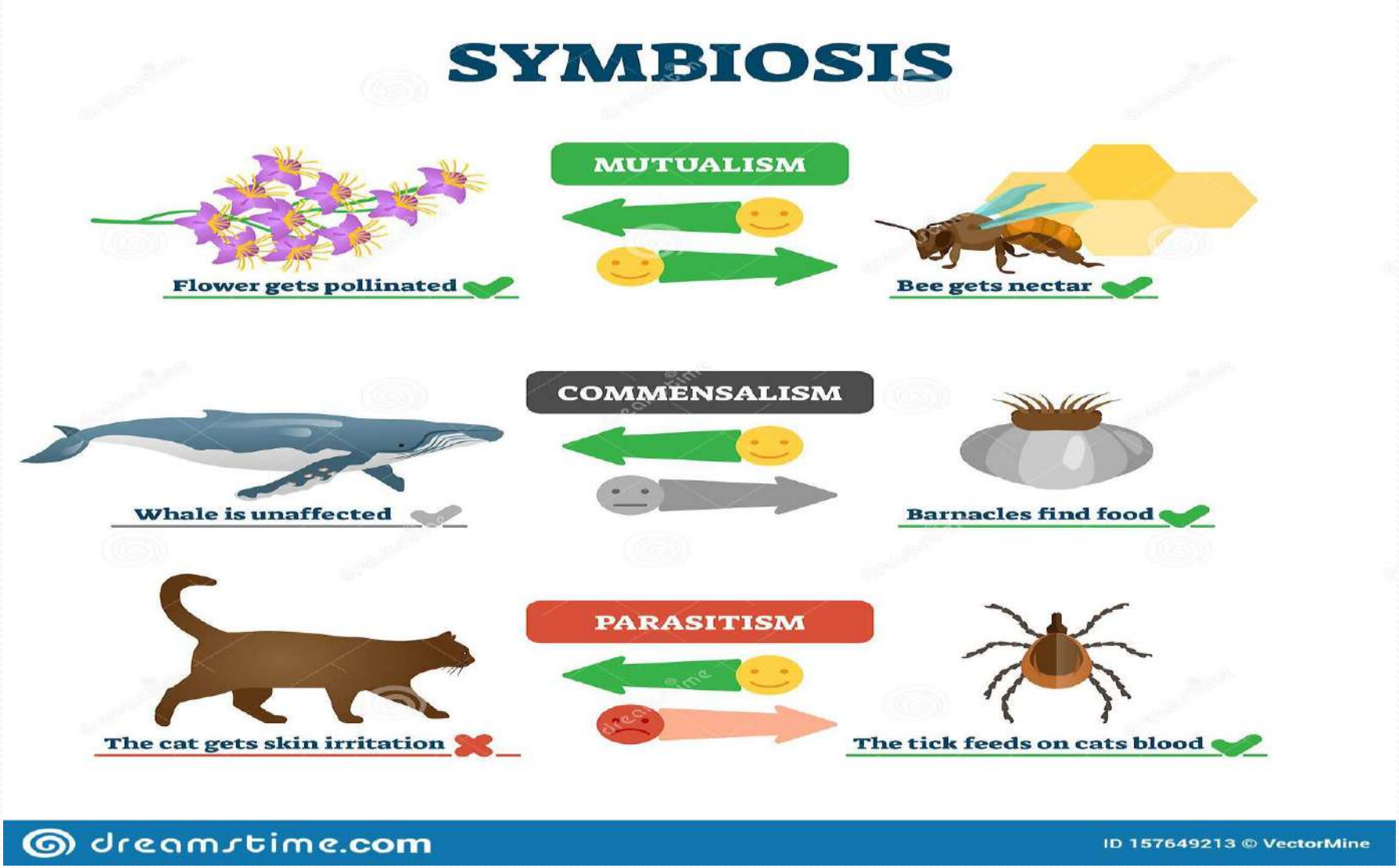

Commensalism Interaction
Definition, Types and Examples
Hermit crabs and gastropod shell
Nematodes and molluscs
Gut microbiota in human
Scarab beetles and flies

Parasitism

- Parasitism is a symbiotic relationship in which one species (the parasite) benefits while the other species (the host) is harmed.
- Many species of animals are parasites, at least during some stage of their life.
- Most species are also hosts to one or more parasites.

Parasitism is the relationship between two organisms wherein one organism, the parasite, thrives at the cost of the other, the host.




SYMBIOSIS
MUTUALISM
Flower gets pollinated
Bee gets nectar
COMMENSALISM
Whale is unaffected
Barnacles find food
PARASITISM
The cat gets skin irritation
The tick feeds on cats blood
dreamstime.com
ID 157649213 © VectorMine

Module# 70 Ecotone

- An ecotone is an area that acts as a boundary or a transition between two ecosystems.
- A common example could be an area of marshland between a river and its riverbank.
- Ecotones are of great environmental importance. Because the area is a transition between two ecosystems or biomes, it is natural that it contains a large variety of species of fauna and flora as the area is influenced by both the bordering ecosystems.

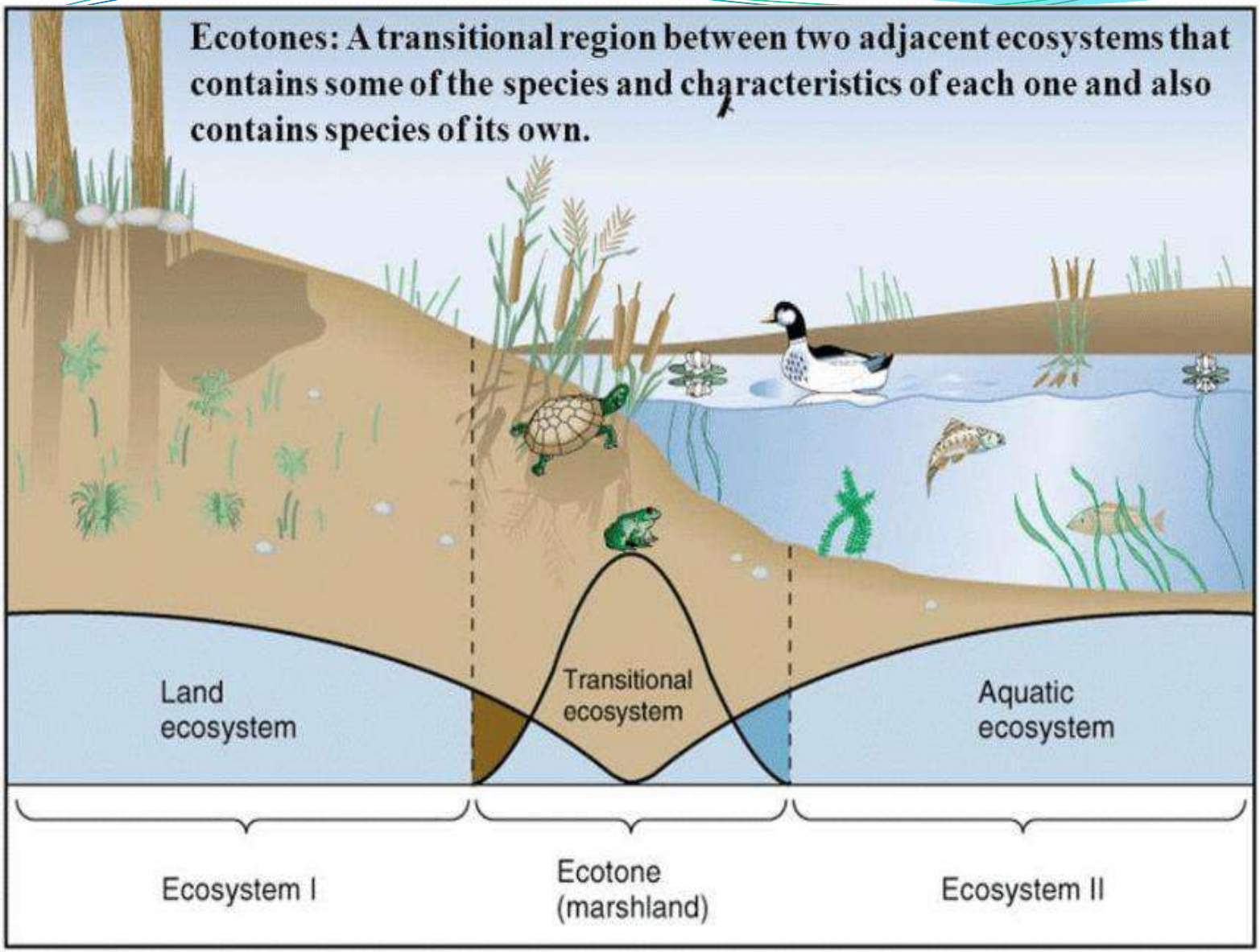
- Examples of ecotones include marshlands (between dry and wet ecosystems), mangrove forests (between terrestrial and marine ecosystems), grasslands (between desert and forest), and estuaries (between saltwater and freshwater).
- Mountain ranges can also create ecotones due to the changes in the climatic conditions on the slopes.

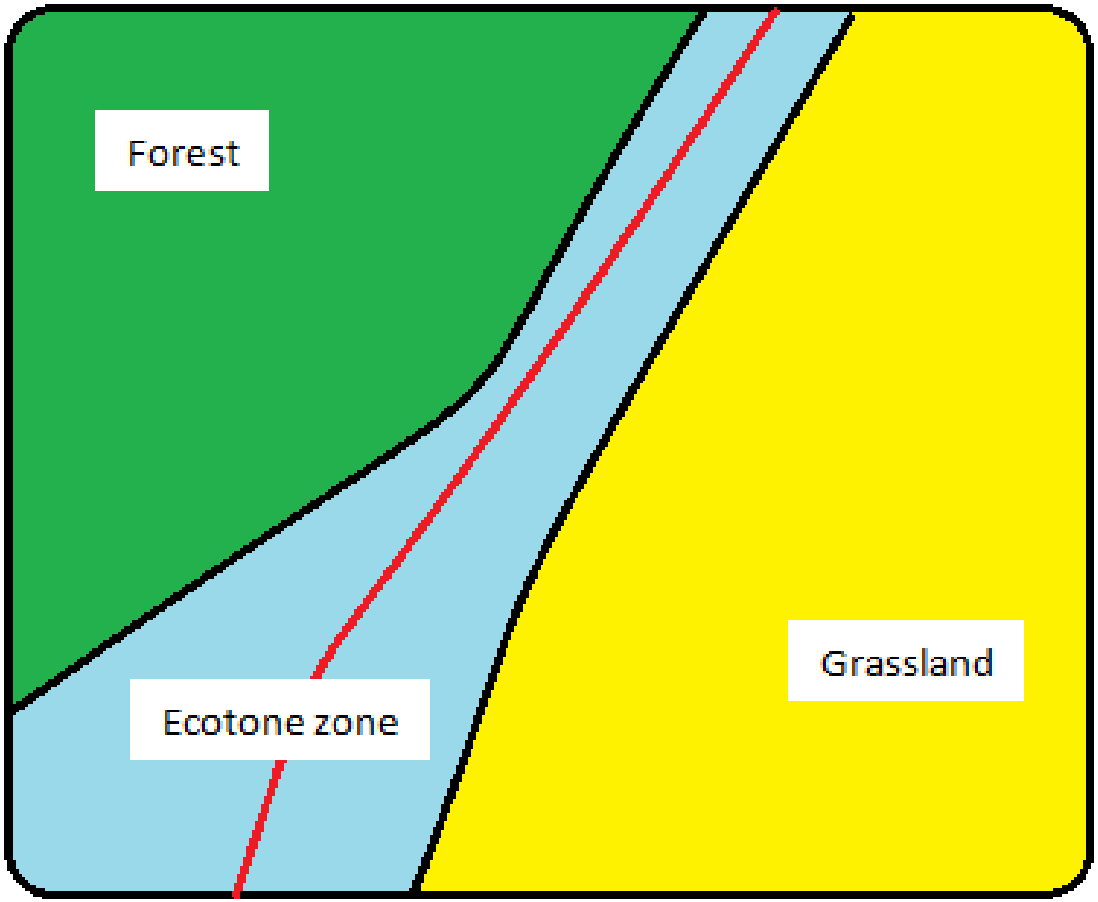
Characteristics of Ecotones

- It may be wide or narrow.
- It is a zone of tension (as it has conditions intermediate to the bordering ecosystems).
- It could contain species that are entirely different from those found in the bordering systems.
- Ecotones can be natural or man-made. For example, the ecotone between an agricultural field and a forest is a man-made one

Importance of Ecotone

- They have a greater variety of organisms.
- They also offer a good nesting place for animals coming in search of a nesting place or food.
- They serve as a bridge of gene flow from one population to another because of the larger genetic diversity present.
- They can act as buffer zones offering protection to the bordering ecosystems from possible damage. For example, a wetland can absorb pollutants and prevent them from seeping into the river.
- Ecotones are also a sensitive indicator of global climate change. A shifting of boundaries between ecosystems is thought to be due to climate change. So, scientists and environmentalists are studying ecotones with greater interest now.


Ecotones: A transitional region between two adjacent ecosystems that contains some of the species and characteristics of each one and also contains species of its own.
Land ecosystem
Transitional ecosystem
Aquatic ecosystem
Ecosystem I
Ecotone (marshland)
Ecosystem II


Forest
Ecotone zone
Grassland

Module# 071 Applied Ecology

- Applied ecology is a sub-field within ecology that considers the application of the science of ecology to real-world (usually management) questions.
- It is also described as a scientific field that focuses on the application of concepts, theories, models, or methods of fundamental ecology to environmental problems

- Applied ecology is an integrated treatment of the ecological, social, and biotechnological aspects of natural resource conservation and management.
- Applied ecology typically focuses on geomorphology, soils, and plant communities as the underpinnings for vegetation and wildlife (both game and non-game) management.

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- Applied ecology includes all disciplines that are related to human activities so that it does not only cover agriculture, forestry and fisheries but also global change.
- It has two study categories. The first involves the outputs or those fields that address the use and management of the environment, particularly for its ecosystem services and exploitable resources.
- The second are the inputs or those that are concerned with the management strategies or human influences on the ecosystem or biodiversity

Applied uses ecological approaches to address problems of specific parts of the environment



a small area of wetland regeneration in south-eastern Australia, recently constructed.



Module# 072 Approaches and Applications of Applied Ecology

- Applied ecologists often use one or more of the following approaches, namely, observation, experimentation, and modeling.
- For example, a wildlife preservation project could involve: observational studies of the wildlife ecology; experiments to understand causal relationships; and the application of modeling to determine the information beyond the scope of experimentation.

- The ecological approach used in applied ecology could include inputs from management strategies such as conservation biology, restoration ecology, global change, ecotoxicology, biomonitoring, biodiversity, environmental policies, and economics, among others.
- Restoration ecology is a particularly prominent strategy in the discipline since it applies the principles of restoring and repairing damaged ecological systems to their original state.

- Like those used in ecological theory, many areas of the discipline employ approaches that are based on simple statistical and analytic models (e.g. spatial models) as well as those with mathematical properties (e.g.matrix models).
- There is also the digital computer simulation modeling, which is designed to solve statistical ecology problems and to achieve bioeconomic goals such as the forecasting and the evaluation of consequences for specific activities.[10]

- Applied ecology also requires human interest, particularly the exercise of judgments of relative values and goals.

Applications

- Applied ecology can be applied to the economic development process.
- The discipline, for example, can be integrated into the national economic planning to comprehensively address environmental concerns since these problems are intersectoral and interdisciplinary in nature.

Aspects of applied ecology include:

- Agro-ecosystem management
- Biodiversity conservation
- Biotechnology
- Conservation biology
- Disturbance management
- Ecosystem restoration
- Environmental engineering
- Environmental technology
- Habitat management

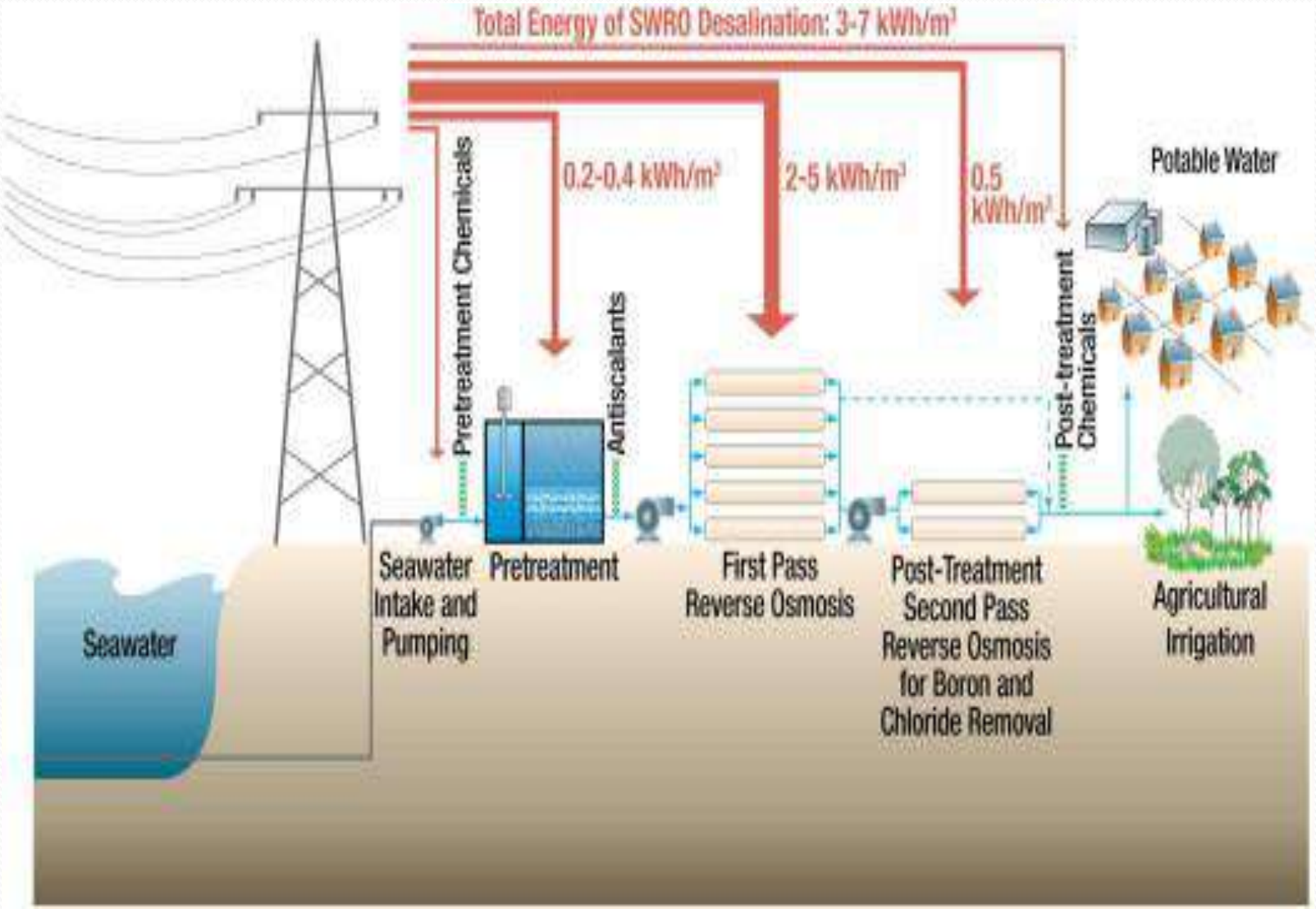
Module# 73 Agricultural Desalination

- Among the options for augmenting freshwater resources is the desalination of salty groundwater, brackish drainage water and seawater.
- Distilling drinking-water from seawater has been studied over many centuries by Mediterranean and Near East civilizations.
- Large-scale solar ponding to serve as domestic drinking-water was practiced more than 100 years ago in Egypt.

- However, progress on modern desalination was made during the 1960s and plants have been developed since the 1970s, starting with some countries of the Persian Gulf because of their ready availability of energy and relevant scarcity of freshwater resources.
- Intensive research for large-scale commercial desalinating technologies began in the United States of America in the early 1960s.

- The first desalination technology to be developed was thermal distillation. In this process, saline water is distilled into steam, which in turn is condensed into pure water.
- Later, membrane processes, such as electro-dialysis (ED) and reverse osmosis (RO), were developed.
- In an ED process, salts are separated from water by means of an electric load application.

- In RO, pressure is applied to the intake water to force it to flow through a semi-permeable membrane that prevents most of the salts from passing through.
- The higher the salt concentration of the intake water is, the higher the pressure that is required.
- Solar energy may also be used to produce water vapour, which is then condensed on a cooler surface to form the desalinated water.
- However, this process produces only a small amount of water and is generally ignored as a technological solution.


Total Energy of SWRO Desalination: 3-7 kWh/m³
0.2-0.4 kWh/m³
2-5 kWh/m³
0.5 kWh/m³
Pretreatment Chemicals
Antiscalants
Post-treatment Chemicals
Potable Water
Seawater
Seawater Intake and Pumping
Pretreatment
First Pass Reverse Osmosis
Post-Treatment Second Pass Reverse Osmosis for Boron and Chloride Removal
Agricultural Irrigation

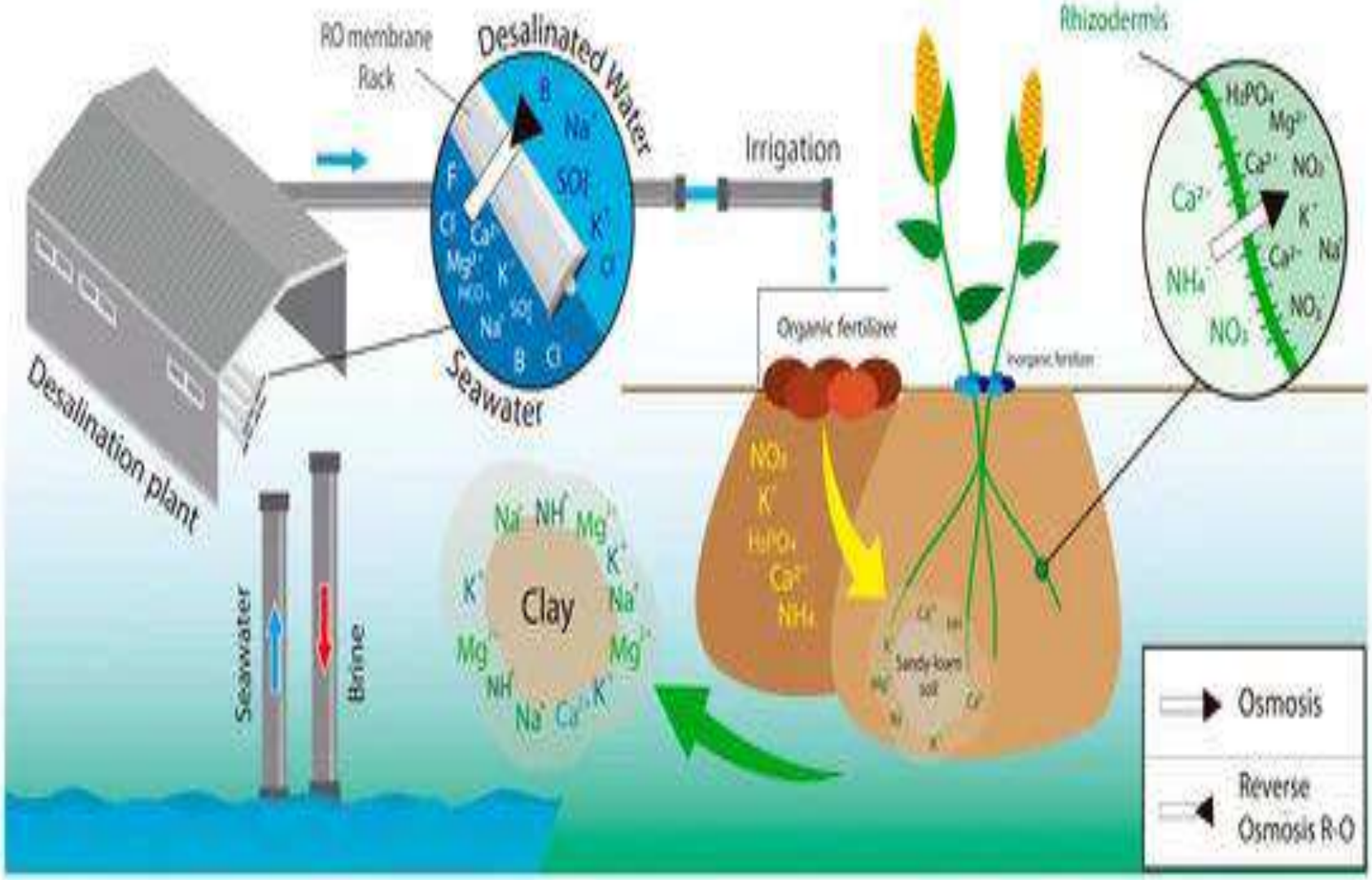
Module# 074 USE OF DESALINATED WATER IN AGRICULTURE

- In the past, the high cost of desalinating and the energy required have been major constraints on large-scale production of freshwater from brackish waters and seawater.
- However, desalinated water is becoming more competitive for urban uses because desalinating costs are declining and the costs of surface water and groundwater are increasing.
- In spite of this development, the costs of desalinated water are still too high for the full use of this resource in irrigated agriculture, with the exception of intensive horticulture for high-value cash crops, such as vegetables and flowers (mainly in greenhouses), grown in coastal areas (where safe disposal is easier than in inland areas).

- For agricultural uses, RO is the preferred desalination technology because of the cost reductions driven by improvements in membranes in recent years.
- Spain provides a significant example of the application of desalinated water in irrigation.
- Spain has more than 300 treatment plants (about 40 percent of the total number of existing plants) and 22.4 percent of the total desalinated water is used for agriculture.

- Most of these plants process brackish water (only 10 percent of the total desalinated water for agriculture originates from seawater).
- In this country, small and medium-sized brackish-water desalination plants, with a capacity of less than 1 000 m3 /d (11.6 litres/ s), are common because they adapt better to individual farmer requirements and to the existing hydraulic structures.

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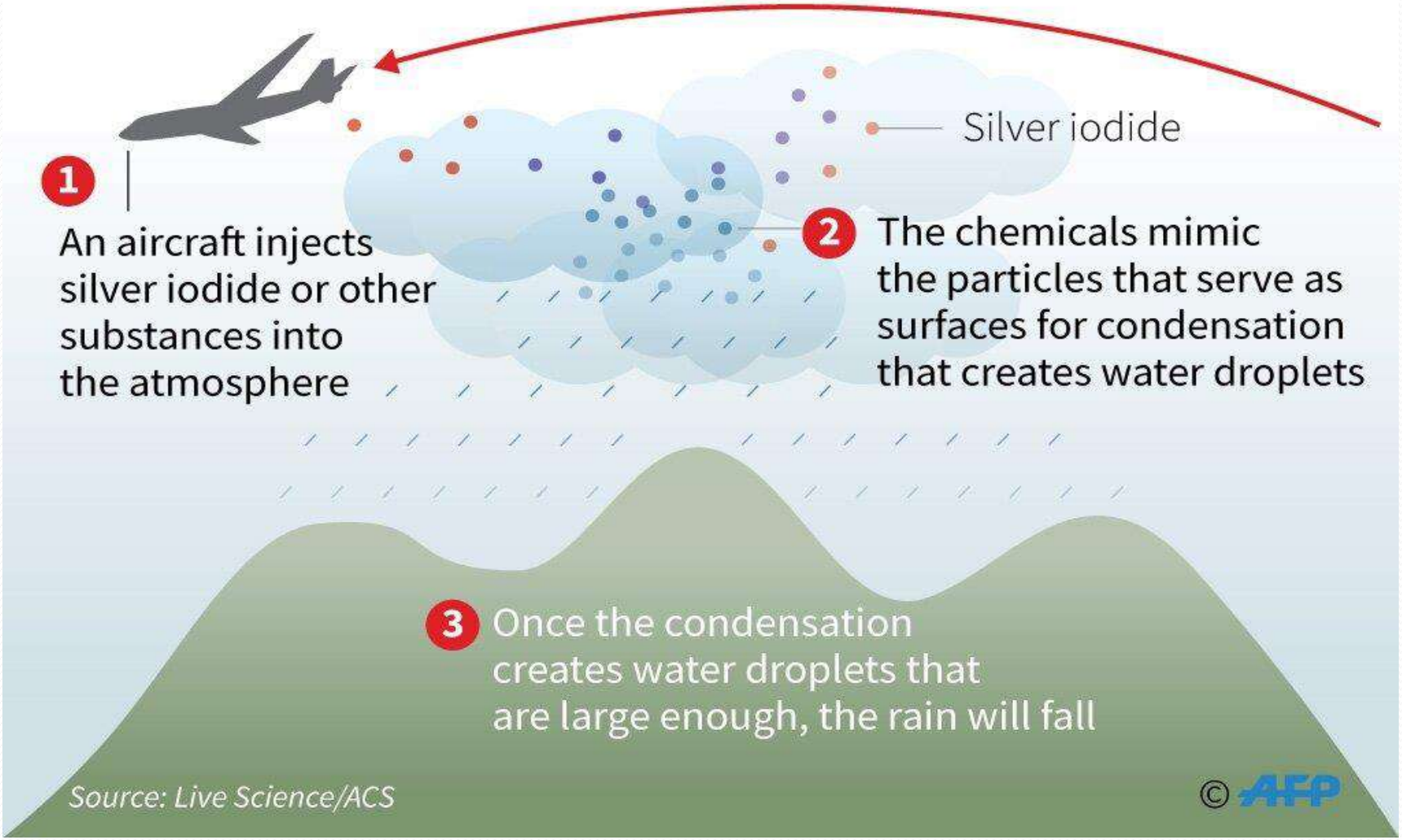

RO membrane
Rack
Desalinated Water
Irrigation
Rhizodermis
Organic fertilizer
Desalination plant
Seawater
Brine
Clay
Sandy-loam soil
Osmosis
Reverse Osmosis R-O

Module# 075 Weather modification

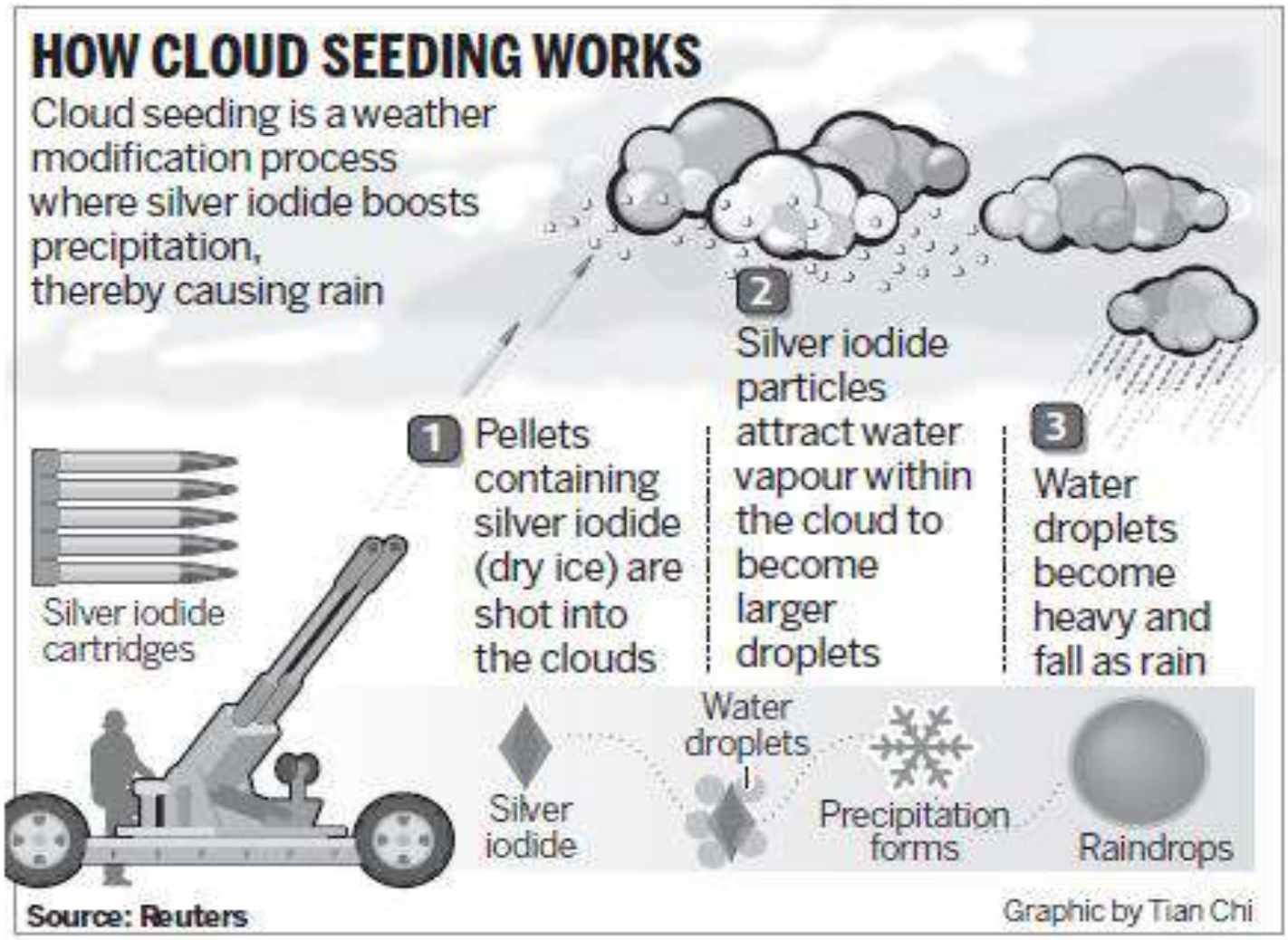
- Weather modification (also known as weather control) is the act of intentionally manipulating or altering the weather.
- The most common form of weather modification is cloud seeding, which increases rain or snow, usually for the purpose of increasing the local water supply.

Cloud seeding

Traditional method of rainmaking, in use since the 1940s



- Weather modification can also have the goal of preventing damaging weather, such as hail or hurricanes, from occurring; or of provoking damaging weather against the enemy, as a tactic of military or economic warfare like Operation Popeye, where clouds were seeded to prolong the monsoon in Vietnam.
- Weather modification in warfare has been banned by the United Nations under the Environmental Modification Convention.


HOW CLOUD SEEDING WORKS
Cloud seeding is a weather modification process where silver iodide boosts precipitation, thereby causing rain
1 Pellets containing silver iodide (dry ice) are shot into the clouds
2 Silver iodide particles attract water vapour within the cloud to become larger droplets
3 Water droplets become heavy and fall as rain
Silver iodide cartridges
Silver iodide
Water droplets
Precipitation forms
Raindrops
Source: Reuters
Graphic by Tian Chi

Module# 076 Cloud seeding

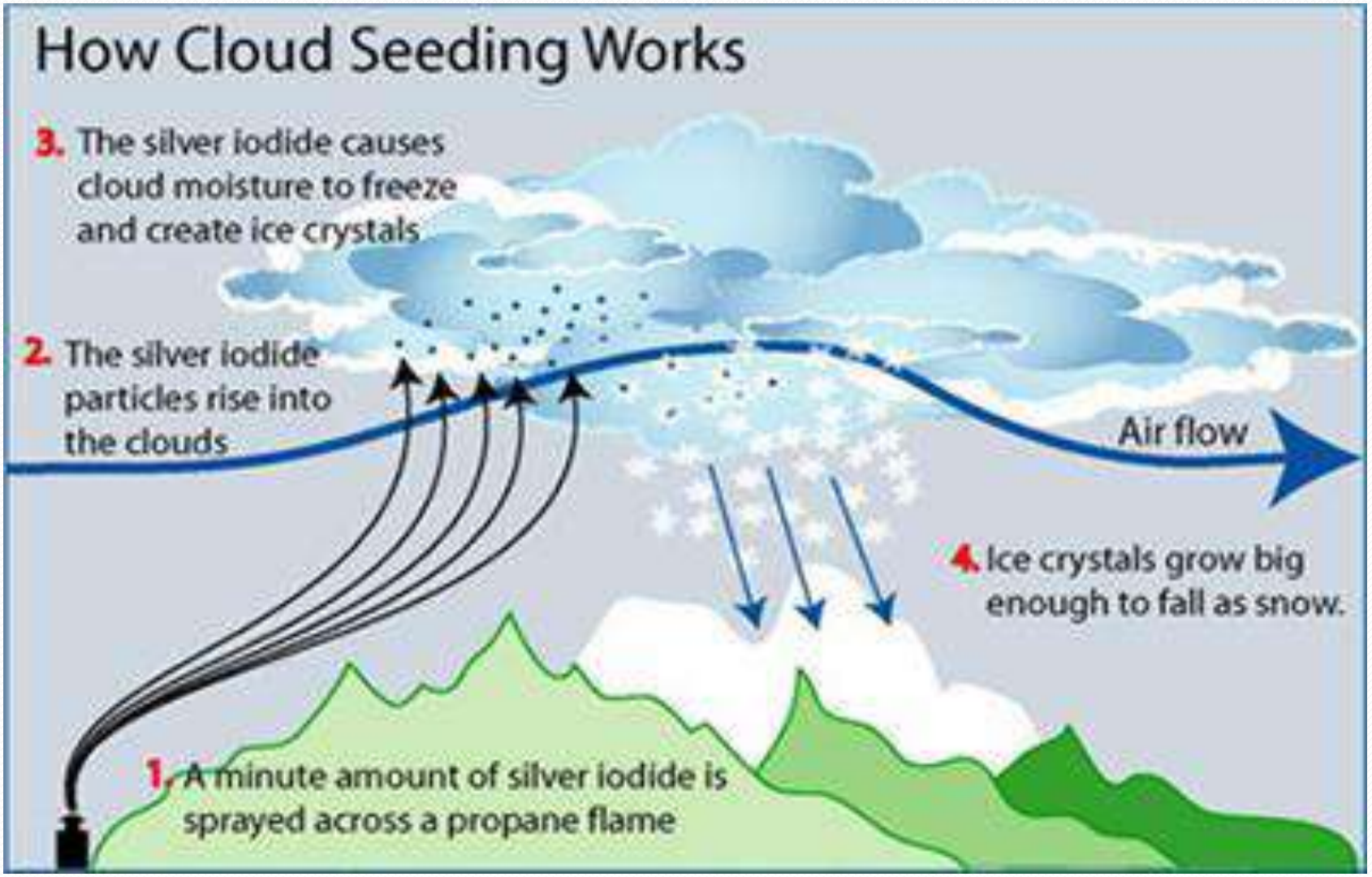
- Cloud seeding is a common technique to enhance precipitation. Cloud seeding entails spraying small particles.

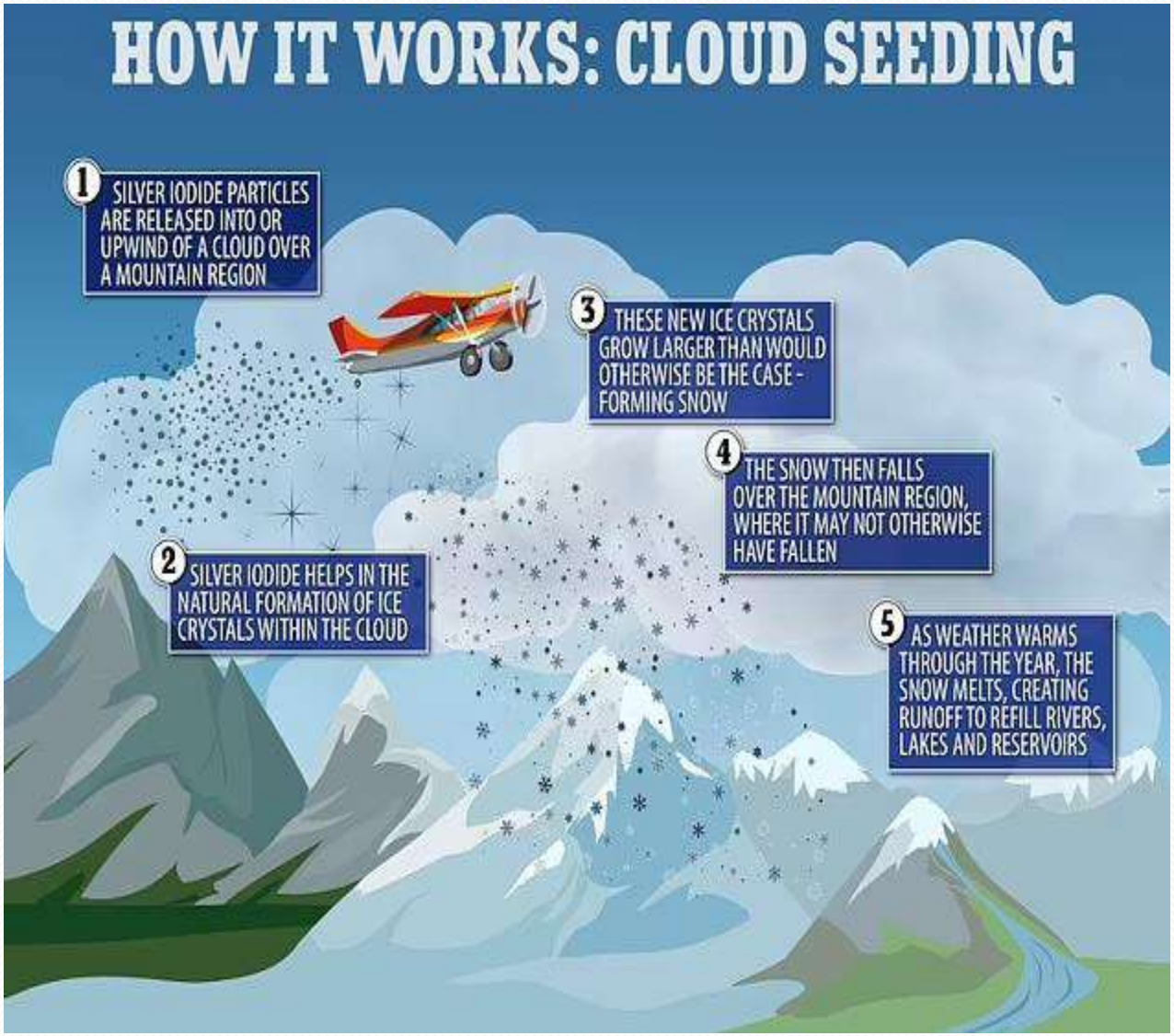
- such as silver iodide, onto clouds to attempt to affect their development, usually with the goal of increasing precipitation.

- Cloud seeding only works to the extent that there is already water vapor present in the air.

- Critics generally contend that claimed successes occur in conditions which were going to lead to rain anyway.

- It is used in a variety of drought-prone countries, including the United States, China, India, and Russia.
- In China there is a perceived dependency upon it in dry regions, and there is a strong suspicion it is used to "wash the air" in dry and heavily polluted places, such as Beijing.


How Cloud Seeding Works
3. The silver iodide causes cloud moisture to freeze and create ice crystals
2. The silver iodide particles rise into the clouds
Air flow
4. Ice crystals grow big enough to fall as snow.
1. A minute amount of silver iodide is sprayed across a propane flame


HOW IT WORKS: CLOUD SEEDING
1 SILVER IODIDE PARTICLES ARE RELEASED INTO OR UPWIND OF A CLOUD OVER A MOUNTAIN REGION
2 SILVER IODIDE HELPS IN THE NATURAL FORMATION OF ICE CRYSTALS WITHIN THE CLOUD
3 THESE NEW ICE CRYSTALS GROW LARGER THAN WOULD OTHERWISE BE THE CASE - FORMING SNOW
4 THE SNOW THEN FALLS OVER THE MOUNTAIN REGION, WHERE IT MAY NOT OTHERWISE HAVE FALLEN
5 AS WEATHER WARMS THROUGH THE YEAR, THE SNOW MELTS, CREATING RUNOFF TO REFILL RIVERS, LAKES AND RESERVOIRS

Module# 077 Rangeland

- Rangeland, also called range, any extensive area of land that is occupied by native herbaceous or shrubby vegetation which is grazed by domestic or wild herbivores.
- The vegetation of ranges may include tallgrass prairies, steppes (shortgrass prairies), desert shrublands, shrub woodlands, savannas, chaparrals, and tundras.
- Temperate and tropical forests that are used for grazing as well as timber production can also be considered rangeland. Rangelands thus occupy about 40–50 percent of the land area of Earth.

- Rangelands are distinguished from pastureland by the presence on them of native vegetation, rather than of plants established by human societies, and by their management principally through the control of the number of animals grazing on them, as opposed to the more intensive agricultural practices of seeding, irrigation, and the use of fertilizers.

- The tallgrass prairies of the North American Great Plains, Ukraine, and parts of Argentina and Hungary formerly made ideal rangelands but were too well suited to cultivated crops to be left for grazing purposes.
- Rangelands are thus more generally confined to areas of marginal or submarginal agricultural land or to areas that are entirely unsuited to permanent cultivation

Buffalo grazing on rangeland



Range management

- Range management is a professional field whose aim is to ensure a sustained yield of rangeland products while protecting and improving the basic range resources of soil, water, and plant and animal life. Besides producing forage for domestic and wild animals, a range can provide timber, minerals, natural beauty, and recreational opportunities.
- Modern range management utilizes the concept of multiple use, which requires that all the resources of a rangeland be managed simultaneously, using constant monitoring and adjustments to provide a mix of material products and intangible assets that best satisfy the needs of both landowners and the general public.

- Range management depends for its effectiveness on range science, which is a body of knowledge drawn from the botanical and zoological sciences as well as from ecology, climatology, pedology (soil science), hydrology, and so on.
- The responses of rangeland to grazing and other uses are predicted from range science's accumulated knowledge of the functioning of rangeland ecosystems, which in turn has been aided by computer-simulated mathematical models.



Rangelands of the World





Module# 078 Pollution

- Pollution is the introduction of harmful materials into the environment.
- These harmful materials are called pollutants. Pollutants can be natural, such as volcanic ash.
- They can also be created by human activity, such as trash or runoff produced by factories.
- Pollutants damage the quality of air, water, and land.

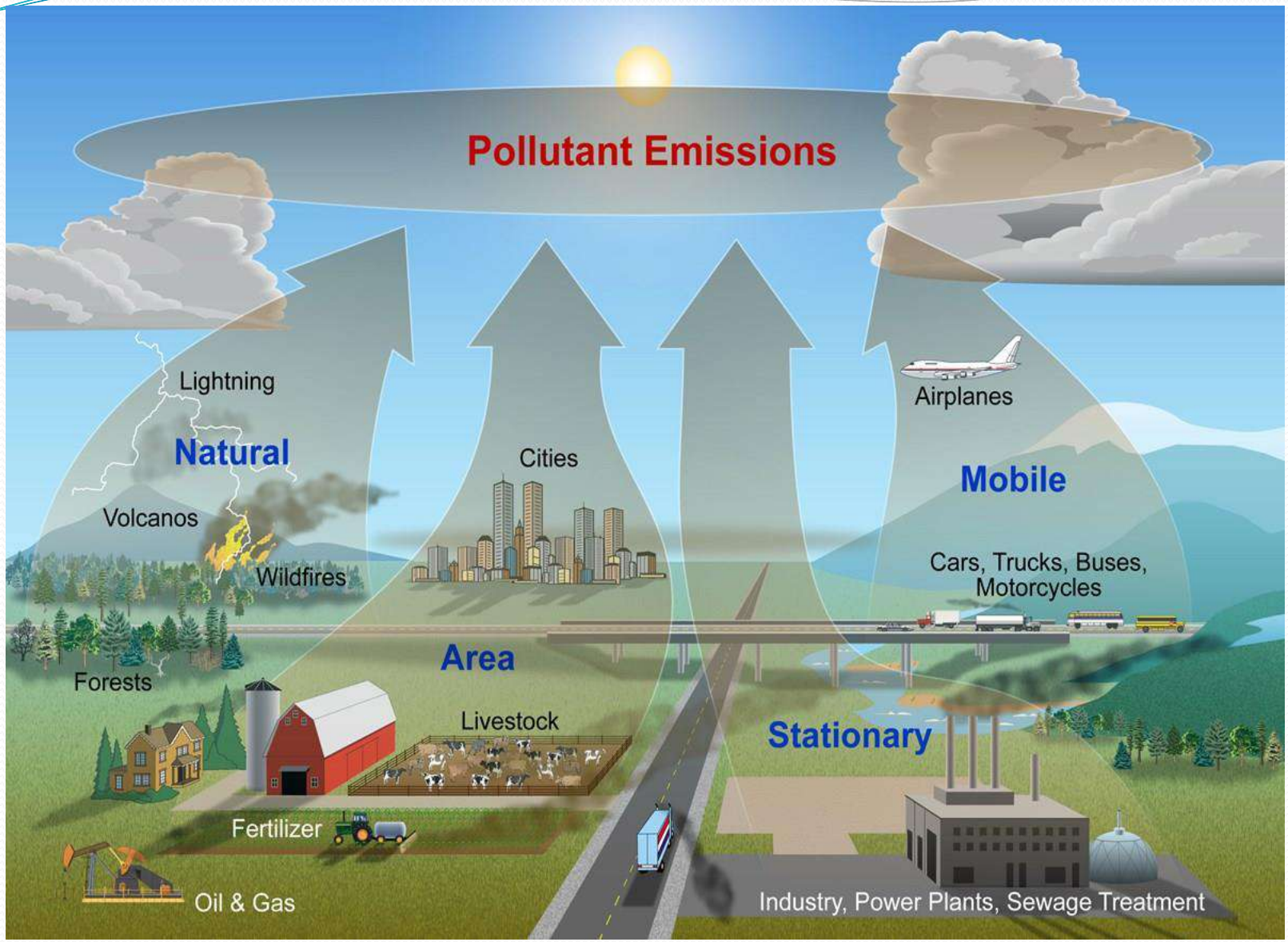
- Many things that are useful to people produce pollution.
- Cars spew pollutants from their exhaust pipes. Burning coal to create electricity pollutes the air. Industries and homes generate garbage and sewage that can pollute the land and water.
- Pesticides—chemical poisons used to kill weeds and insects—seep into waterways and harm wildlife.
- All living things—from one-celled microbes to blue whales—depend on Earth's supply of air and water. When these resources are polluted, all forms of life are threatened.

- The major types of pollution are

- Water pollution
- Air pollution
- Land pollution
- Noise pollution






Pollutant Emissions
Lightning
Natural
Volcanos
Wildfires
Forests
Cities
Airplanes
Mobile
Cars, Trucks, Buses, Motorcycles
Area
Livestock
Fertilizer
Oil & Gas
Stationary
Industry, Power Plants, Sewage Treatment

Module# 079 Water pollution

- Water pollution happens when chemicals or dangerous foreign substances are introduced to water, including chemicals, sewage, pesticides and fertilizers from agricultural runoff, or metals like lead or mercury.
- According to the Environmental Protection Agency (EPA), 44% of assessed stream miles, 64% of lakes and 30% of bay and estuarine areas are not clean enough for fishing and swimming.

- The EPA also states that the United State's most common contaminants are bacteria, mercury, phosphorus and nitrogen. These come from the most common sources of contaminates, that include agricultural runoff, air deposition, water diversions and channelization of streams.



- Water pollution isn't just a problem for the United States. According to United Nations, 783 million people do not have access to clean water and around 2.5 billion do not have access to adequate sanitation. Adequate sanitation helps to keep sewage and other contaminants from entering the water supply.

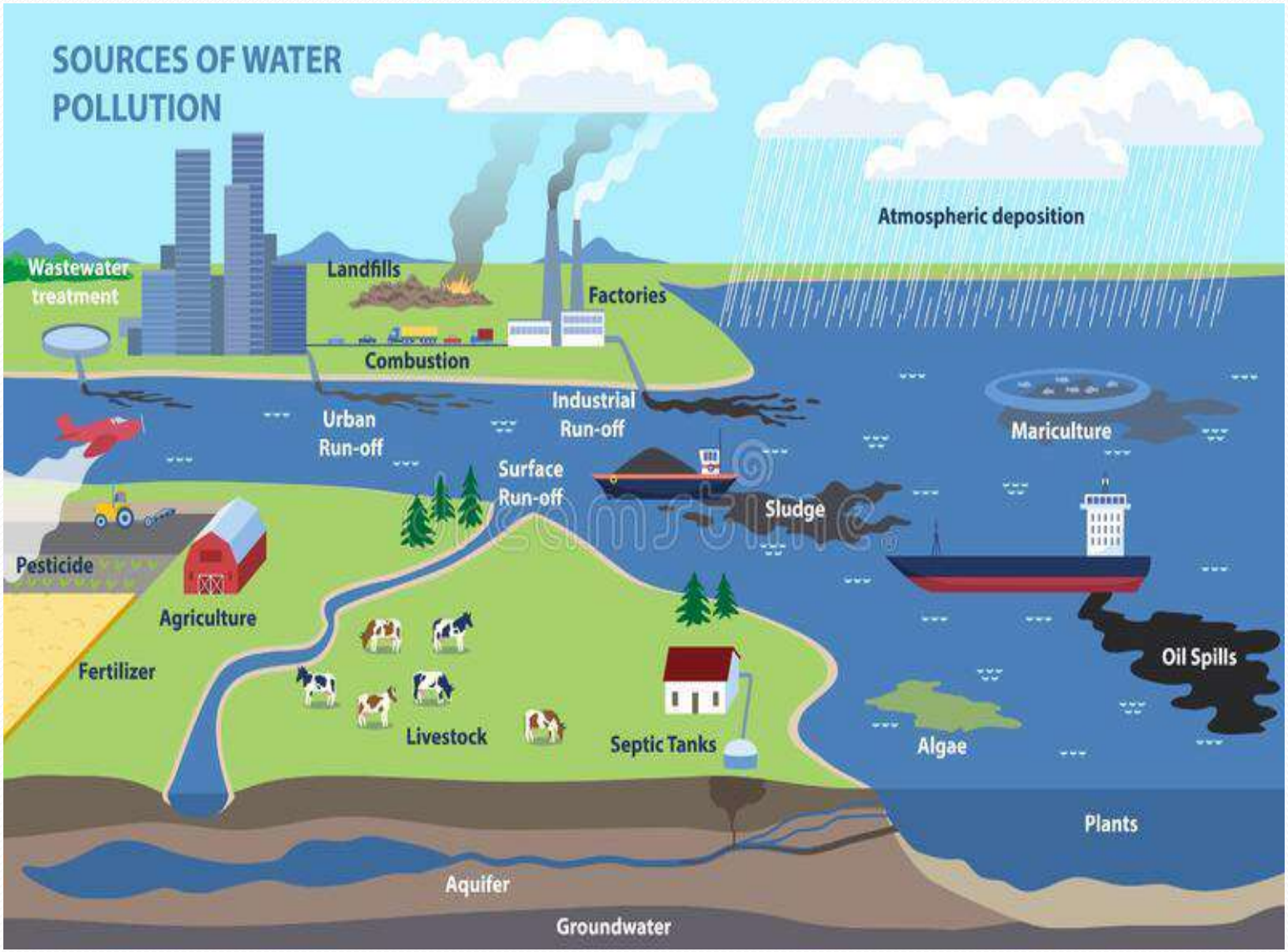




- According to National Oceanic and Atmospheric Administration (NOAA), 80% of pollution in marine environment comes from the land through sources like runoff.
- Water pollution can also severely affect marine life. For example, sewage causes pathogens to grow, while organic and inorganic compounds in water can change the composition of the precious resource.
- According to the EPA, low levels of dissolved oxygen in the water are also considered a pollutant. Dissolved is caused by the decomposition of organic materials, such as sewage introduced into the water.

Module# 080 SOURCES OF WATER POLLUTION

- Sewage leakages
- High population density
- oil spillage
- Menace of Nipa palm and water hyacinth
- Industrial waste dumped into our waters
- Pollution of ground water through drilling activities


SOURCES OF WATER POLLUTION
Atmospheric deposition
Wastewater treatment
Landfills
Factories
Combustion
Urban Run-off
Industrial Run-off
Mariculture
Surface Run-off
Sludge
Pesticide
Agriculture
Fertilizer
Oil Spills
Livestock
Septic Tanks
Algae
Plants
Aquifer
Groundwater

- Flooding during rainy season which carries waste deposits into our waters.
- Water pollution is generally induced by humans. It results from actions of humans carried on to better self. These could be treated under the various activities that man engages in, that lead to pollution. The growth of human population, industrial and agricultural practices is the major causes of pollution


SOURCES OF WATER POLLUTION
Atmospheric deposition
Landfills
Oil Spills
Mariculture
Sludge
Abandoned mines
Factories
Industrial Run-off
Combustion (Cars)
Urban Run-off
Livestock
Farms
Residential Buildings
Agriculture
Surface Run-off
Sewage Treatment Facilities
Septic Tanks
Fertilizer
Algae
Aquifer
Plants
Groundwater

Human activities

- Water pollution becomes worse as a result of overcrowding in urban areas. Agricultural, domestic and industrial wastes are the major pollutants of agnostic habitats.
- Sewage is the biggest pollutant of fresh water when discharged into them.
- Sewage is the waterborne waster of society and the discharge of untreated sewage into a river is very enormous and unhealthy.

- The striking consequence is a substantial and immediate drop in the amount of dissolved oxygen in the water.
- This happens because organic matter stimulates decomposers especially bacteria which break down suspended solids in the sewage. As they respire, the decomposers use up dissolved oxygen (O2) and the Biological Oxygen Demand (BOD) reduces.
- The flora and fauna of the rivers experience change and reduction in number due to death by suffocation.

Module# 081 Management of water pollution

- Water pollution leads to damage to human health.
- Disease carrying agents such as bacteria and viruses are carried into the surface and groundwater.
- Drinking water is affected and health hazards result. Direct damage to plants and animals nutrition also affects human health.
- Plants nutrients including nitrogen, phosphorus and other substances that support the growth of aquatic plant life could be in excess causing algal bloom and excessive weed growth.

- This makes water to have odour, taste and sometimes colour.
- Ultimately, the ecological balance of a body of water is altered. Sulphur dioxide and nitrogen oxides cause acid rain which lowers the pH value of soil and emission of carbon dioxide cause ocean POLL.

- There are many approaches that could be adopted for management of water pollution and its control. It could be through prevention, practice efforts or join a project/program; Regulation and monitoring or engaging in control measures by reducing or minimizing waste.
- Wash your car far away from any storm water drains.
- Don't throw trash, chemicals or solvents into sewer drains

- inspects your septic system every 3 – 5 years
- avoid using pesticides and fertilizers that can run off into water systems
- sweep your driveway instead of hosing it down
- always pump your waste-holding tanks on your boat
- use non-toxic cleaning materials

How do we prevent water pollution

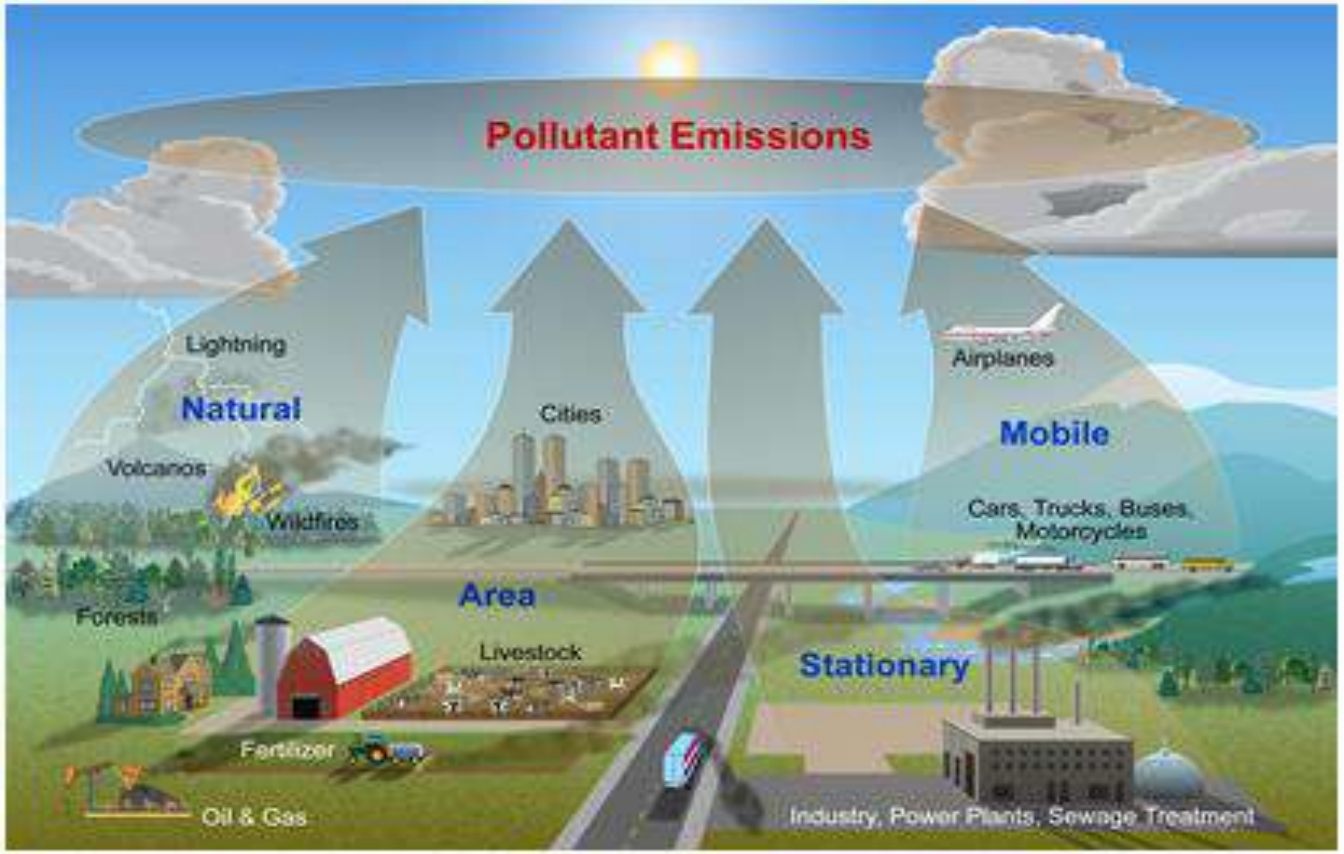


- Turn off the tap when running water is not necessary.
- Use environmentally household products.
- Avoid over dosage of pesticides and fertilisers
- Don't throw the non degradable substances to rivers, lakes or oceans.
- Pre-treatment of industrial waste water before leaving to water bodies.

Module# 82 Air pollution

- The air we breathe has a very exact chemical composition; 99% of it is made up of nitrogen, oxygen, water vapor and inert gases.
- Air pollution occurs when things that aren't normally there are added to the air. A common type of air pollution happens when people release particles into the air from burning fuels.
- This pollution looks like soot, containing millions of tiny particles, floating in the air.

- Another common type of air pollution is dangerous gases, such as sulfur dioxide, carbon monoxide, nitrogen oxides and chemical vapors.
- These can take part in further chemical reactions once they are in the atmosphere, creating acid rain and smog.
- Other sources of air pollution can come from within buildings, such as secondhand smoke.


Pollutant Emissions
Lightning
Natural
Volcanos
Wildfires
Forests
Cities
Area
Livestock
Fertilizer
Oil & Gas
Airplanes
Mobile
Cars, Trucks, Buses
Motorcycles
Stationary
Industry, Power Plants, Sewage Treatment

- Finally, air pollution can take the form of greenhouse gases, such as carbon dioxide or sulfur dioxide, which are warming the planet through the greenhouse effect.
- According to the EPA, the greenhouse effect is when gases absorb the infrared radiation that is released from the Earth, preventing the heat from escaping. This is a natural process that keeps our atmosphere warm.
- If too many gases are introduced into the atmosphere, though, more heat is trapped and this can make the planet artificially warm, according to Columbia University.

Types of Air Pollutants

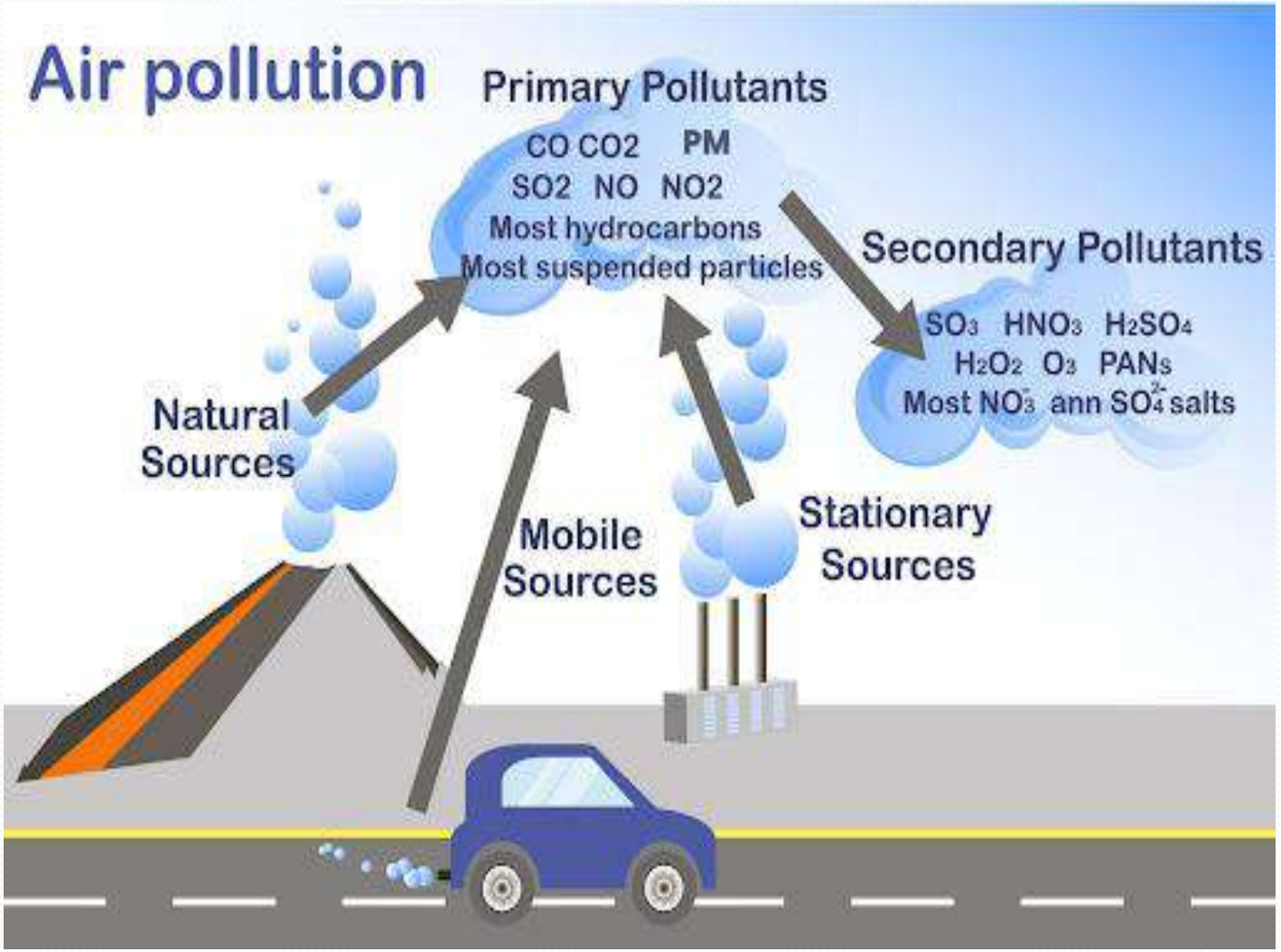
- There are two types of air pollutants:

Primary Pollutants

- The pollutants that directly cause air pollution are known as primary pollutants. Sulphur-dioxide emitted from factories is a primary pollutant.

Secondary Pollutants

- The pollutants formed by the intermingling and reaction of primary pollutants are known as secondary pollutants. Smog, formed by the intermingling of smoke and fog, is a secondary pollutant.


Air pollution
Primary Pollutants
CO CO2 PM
SO2 NO NO2
Most hydrocarbons
Most suspended particles
Secondary Pollutants
SO3 HNO3 H2SO4
H2O2 O3 PANs
Most NO3- ann SO4 2- salts
Natural Sources
Mobile Sources
Stationary Sources

Module# 83 Management of air pollution

- The use of solar, wind and geothermal energies reduce air pollution at a larger level.
- By minimising and reducing the use of fire and fire products.
- Since industrial emissions are one of the major causes of air pollution, the pollutants can be controlled or treated at the source itself to reduce its effects. For example, if the reactions of a certain raw material yield a pollutant, then the raw materials can be substituted with other less polluting materials.


SOLUTIONS
INVEST IN ENERGY-EFFICIENT POWER GENERATION.
IMPROVE DOMESTIC, INDUSTRY AND MUNICIPAL WASTE MANAGEMENT.
REDUCE AGRICULTURAL WASTE INCINERATION, FOREST FIRES AND CERTAIN AGRO-FORESTRY ACTIVITIES.
MAKE GREENER AND MORE COMPACT CITIES WITH ENERGY-EFFICIENT BUILDINGS.
PROVIDE UNIVERSAL ACCESS TO CLEAN, AFFORDABLE FUELS AND TECHNOLOGIES FOR COOKING, HEATING AND LIGHTING.
BUILD SAFE AND AFFORDABLE PUBLIC TRANSPORT SYSTEMS AND PEDESTRIAN- AND CYCLE-FRIENDLY NETWORKS.
CLEAN AIR FOR HEALTH
#AirPollution
World Health Organization

- Fuel substitution is another way of controlling air pollution. In many parts of India, petrol and diesel are being replaced by CNG – Compressed Natural Gas fueled vehicles. These are mostly adopted by vehicles that aren't fully operating with ideal emission engines.
- Although there are many practices in India, which focus on repairing the quality of air, most of them are either forgotten or not being enforced properly. There are still a lot of vehicles on roads which haven't been tested for vehicle emissions.

- Another way of controlling air pollution caused by industries is to modify and maintain existing pieces of equipment so that the emission of pollutants is minimised.
- Sometimes controlling pollutants at the source is not possible. In that case, we can have process control equipment to control the pollution.
- A very effective way of controlling air pollution is by diluting the air pollutants.
- The last and the best way of reducing the ill effects of air pollution is tree plantation. Plants and trees reduce a large number of pollutants in the air. Ideally, planting trees in areas of high pollution levels will be extremely effective.

AIR POLLUTION PREVENTION CONCEPTS



Promoting Public Transport Usage



Reducing Energy Consumption



Reduction of Forest Fires



Popularize Afforestation



Investment Into Renewable Energy



Improving Waste Management



Clean Public Transportation



Increasing Number of Parks



Available Refilling Options

Module# 84 Land Pollution

- and pollution occurs when trash, compost, and other toxins are dumped on the land, contaminating or polluting it. Land pollution is caused by human activities such as littering and waste washed ashore from boats, oil rigs, and sewage treatment plants.

- The degradation of the earth's land surfaces, both above and below ground level, is referred to as land pollution. The accumulation of solid and liquid waste products, which contaminate groundwater and soil, is the cause. The greater the permeability of the soil, the greater the risk of land contamination.

Land Pollution Causes

- Soil pollution is a form of land pollution in which the soil's upper layer is harmed. Overuse of chemical fertilizers, soil degradation caused by running water, and other pest control measures contribute to the loss of agricultural land, forest cover, and grazing pastures, among other things.




TANA

Various Causes of Land Pollution are listed below

- Agricultural Activities – As animal production grows, it becomes decoupled from crop production, causing normal nutrient cycles between plants, soil, and animals to be severely disrupted, resulting in the widespread use of synthetic herbicides, insecticides, bactericides, and fertilizers, all of which contribute to pollution.
- Mining Activities – Mining has the potential to pollute the air and water supply, damage biodiversity and ecosystems, and permanently alter natural landscapes. Mining harms the ecosystem by destroying habitats, causing soil erosion, and polluting surface water, groundwater, and soil.

- Urbanization – Intensive urbanization will exacerbate poverty by preventing local municipalities from providing services to all residents. Increased air pollution from concentrated energy usage has a direct effect on human health. Lead levels in urban air are elevated as a result of automobile emissions.
- Nuclear Waste – The soil is also contaminated by radioactive waste from nuclear research stations and nuclear power plants, as well as radioactive fallout from nuclear explosions. Since radioactive materials have a long half-life, they can survive in the soil for long periods of time.

Causes of Land Pollution

We can classify major sources in the following Categories:

- Mining, Erosion and quarrying.
- Household Garbage.
- Urbanization and Concentration of Population.
- Municipal Solid Waste.
- Industrial Waste and Hazardous Waste.
- Uncontrolled Land Treatment.
- Burning open dumps and forest fires.
- Deforestation.



Module# 85 Land Pollution Effects

- Land pollution can harm the human body in a variety of ways.
- Toxic waste and contaminants can be ingested by people. Disposal of hazardous radioactive wastes also contributes to land contamination.
- Chronic respiratory disease, lung cancer, heart disease, and even brain damage are all long-term health consequences.

Various Effects of Land Pollution are listed below

- **Climate Change** – Land contamination, such as that caused by mining, farming, and factories, may allow harmful chemicals to enter the soil and water. These chemicals have the potential to kill animals and plants, destroying the food chain. Landfills emit methane, a greenhouse gas that contributes to global warming.

- **Acid Rain** – Forests, especially those at higher elevations, are also harmed by acid rain and fog. Acid deposits deplete vital nutrients like calcium and allow aluminum to be released into the soil, making it difficult for trees to absorb water. Acids also damage the leaves and needles of trees.

- **Deterioration of fields** – A chain reaction occurs as a result of soil contamination. It alters soil biodiversity, decreases soil organic matter, and reduces soil's filtering ability. It also contaminates water contained in the soil and groundwater, resulting in nutrient imbalances in the soil.
- **Respiratory health problems** – Air pollution can irritate your airways, causing shortness of breath, coughing, wheezing, asthma attacks, and chest pain. Lung cancer, heart attacks, strokes, and, in the worst-case scenario, premature death are all risks associated with air pollution exposure.

LAND POLLUTION





Land pollution is caused through many things. Example of this is agriculture because of the fertilizers being used and landfills because the wastes could be hazardous.



The effects of land pollution could sometimes be fatal. They affect the environment and our health. They can cause cancer and pollution such as water pollution.



To solve land pollution, the things we can do is to reduce the use of trash and recycle more often.





Land Pollution





Module# 86 Management to Land Pollution

- To reduce land emissions, reduce, reuse, and recycle. It is essential to practice reforestation and afforestation. Organic fertilizers, an integrated pest control method, and crop rotation can all be used by farmers.
- One of the most important ways to help minimize landfill waste, protect natural resources, preserve wildlife, reduce noise, reduce energy use, and slow global warming is to incorporate recycling habits into your everyday life.

- Reforestation avoids river and lake silting by reducing surface erosion and preserving the fertile topsoil. It prevents the soil surface from sealing and cuts down on the amount of rainwater that runs off.
- When compared to petroleum-based plastics, biodegradable plastics produce significantly less waste. As biodegradable plastics degrade, they decompose into nontoxic, harmless components. Just 32% of the greenhouse gases released by petroleum-based plastics are produced by them.

Land Pollution Control

- Integrated Solid Waste Management
- Good agricultural practices
- Remediation of polluted soils
- Prevention of erosion and silting
- Containment of hazardous waste and waste water treatment using land treatment techniques.




SOLVENT
PAINT
AUTOMOTIVE FLUID
TOXIC MATERIALS
wikiHow

PREVENTION

The greatest prevention to land pollution is in the three 'R's' ...

Reduce Waste,

Re-use Things,

Recycle Things



Module# 87 Noise Pollution

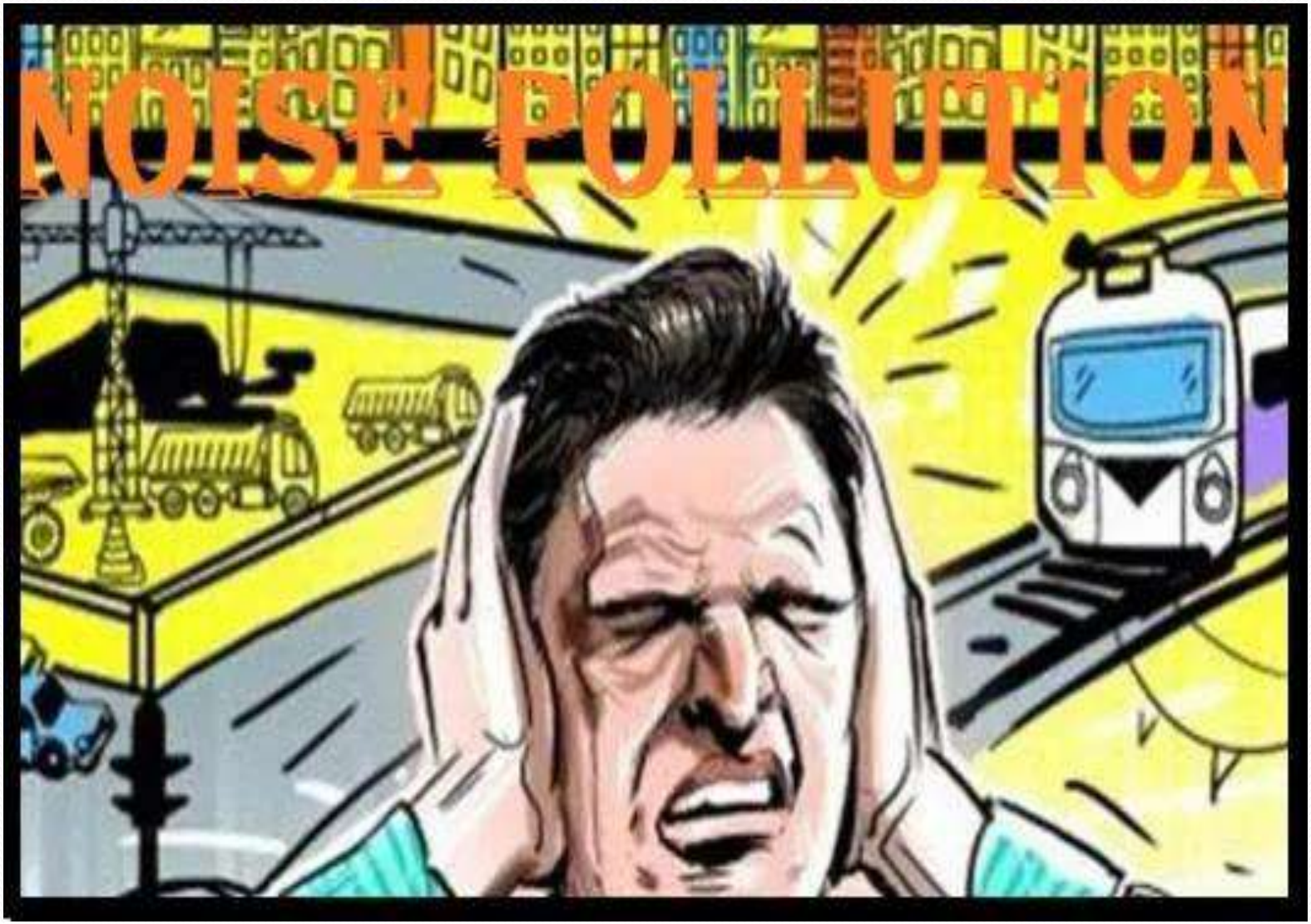
- The word noise is derived from the Latin word 'Nausea', which means sickness in which one feels the need to vomit.
- Noise is the unpleasant and undesirable sound which leads to discomfort in human beings.
- The intensity of sound is measured in decibels (dB).

- The faintest sound that the human ear can hear is 1 Db. Due to increasing noise around the civilizations, noise pollution has become a matter of concern.

- Some of its major causes are vehicles, aircraft, industrial machines, loudspeakers, crackers, etc.

- When used at high volume, some other appliances also contribute to noise pollution, like television, transistor, radio, etc.




NOISE POLLUTION


KEEP SILENCE

Module# 88 Types of Noise Pollution

- Following are the three types of pollution:

- Transport Noise
- Neighbourhood Noise
- Industrial Noise

Transport Noise

- It mainly consists of traffic noise which has increased in recent years with the increase in the number of vehicles. The increase in noise pollution leads to deafening of older people, headache, hypertension, etc.

Neighbourhood Noise

- The noise from gadgets, household utensils etc. Some of the main sources are musical instruments, transistors, loudspeakers, etc.

Industrial Noise

- It is the high-intensity sound which is caused by heavy industrial machines. According to many researches, industrial noise pollution damages the hearing ability to around 20%.

Noise Pollution Examples

- Following are the examples of noise pollution:
- Unnecessary usage of horns
- Using loudspeakers either for religious functions or for political purposes
- Unnecessary usage of fireworks
- Industrial noise
- Construction noise
- Noise from transportation such as railway and aircraft

Sources of Noise Pollution

- Following are the causes and sources of noise pollution:

- **Industrialisation**: Industrialisation has led to an increase in noise pollution as the use of heavy machinery such as generators, mills, huge exhaust fans are used, resulting in the production of unwanted noise.
- **Vehicles**: Increased number of vehicles on the roads are the second reason for noise pollution.
- **Events**: Weddings, public gatherings involve loudspeakers to play music resulting in the production of unwanted noise in the neighbourhood.
- **Construction sites**: Mining, construction of buildings, etc add to the noise pollution.

	**Urban noise**	Noise coming from cities and urbanization in general
	**Transportation noise**	Noise coming from all types of civil transportation activities including cars/trains, aircraft (helicopters, airplanes, *etc.*) and boats (or vessels/ships)
	**Industrial noise**	Noise coming from all types of industrial activities (mining, pile-driving, construction, commercial fishing, wind parks, aquaculture, *etc.*)
	**Recreational noise**	Noise coming from recreational activities (visitors to a zoo, surf, music festival, sport fishing, tourist disturbance, *etc.*)
	**Military noise**	Noise coming from military activities (water gun, sonic booms, explosion, jets and aircraft, naval sonar, *etc.*)
	**Abstract noise**	All types of abstract noise produced by computers (warning sounds, acoustic stimuli, pingers, pulses, tones, white noise, *etc.*)
Other sources of noise		All other sources of noise (aerators, aquaria filters, human voices, dog barks, *etc.*)

Module# 089 Effects of Noise Pollution on Human Health

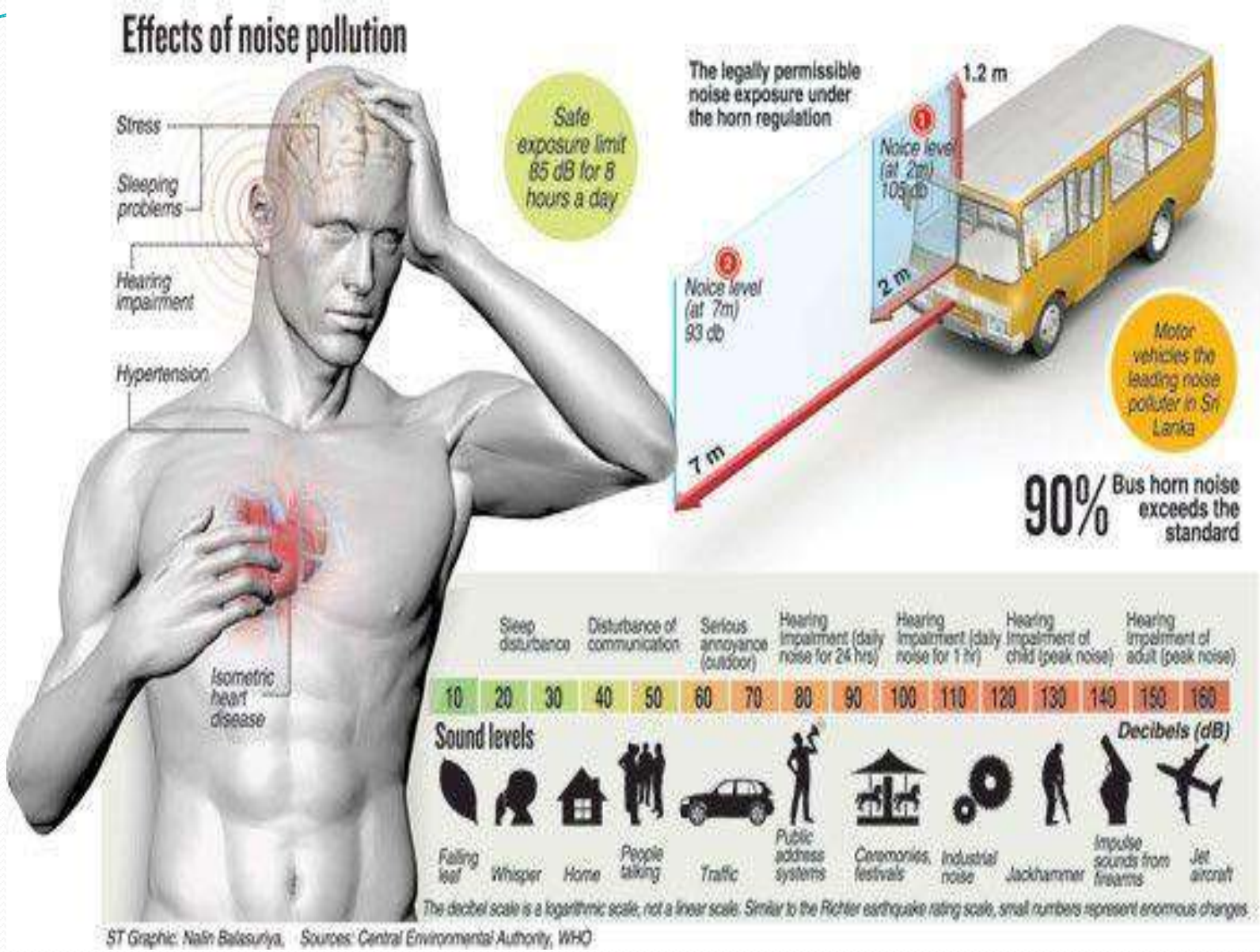
- Noise pollution can be hazardous to human health in the following ways:

- Hypertension: It is a direct result of noise pollution which is caused due to elevated blood levels for a longer duration.
- Hearing loss: Constant exposure of human ears to loud noise that are beyond the range of sound that human ears can withstand damages the eardrums, resulting in loss of hearing.
- Sleeping disorders: Lack of sleep might result in fatigue and low energy level throughout the day affecting everyday activities. Noise pollution hampers the sleep cycles leading to irritation and an uncomfortable state of mind.

- Cardiovascular issues: Heart-related problems such as blood pressure level, stress and cardiovascular diseases might come up in a normal person and a person suffering from any of these diseases might feel a sudden shoot up in the level.
- Prevention of Noise Pollution
- Some noise pollution preventive measures are provided in the points below.

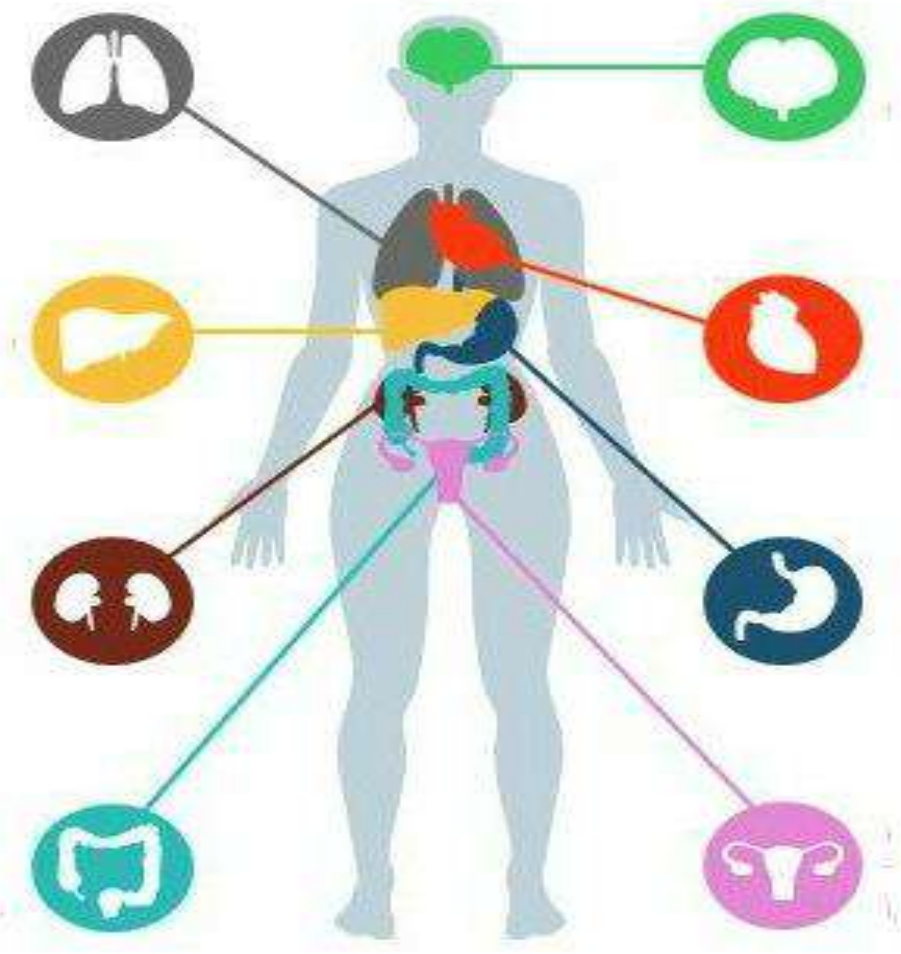
- Honking in public places like teaching institutes, hospitals, etc. should be banned.
- In commercial, hospital, and industrial buildings, adequate soundproof systems should be installed.
- Musical instruments' sound should be controlled to desirable limits.
- Dense tree cover is useful in noise pollution prevention.
- Explosives should not be used in forest, mountainous and mining areas.


Effects of Noise Pollution
Irritation
Nervousness
Increased Stress
Sleep Disturbance
Increased Blood Pressure
On Human Health



ST Graphic: Nalin Balasuriya, Sources: Central Environmental Authority, WHO

HEALTH EFFECTS OF NOISE POLLUTION



Module# 090 Control of Noise Pollution

Some of the ways to control noise pollution are as follows:

- (1) Control at Receiver's End
- (2) Suppression of Noise at Source
- (3) Acoustic Zoning
- (4) Sound Insulation at Construction Stages
- (5) Planting of Trees
- (6) Legislative Measures.

(1) Control at Receiver's End:

- For people working in noisy installations, ear-protection aids like ear-plugs, ear-muffs, noise helmets, headphones etc. must be provided to reduce occupational exposure.


WEAR EAR PLUGS

- **(2) Suppression of Noise at Source:**
- Designing, fabricating and using quieter machines to replace the noisy ones.
- Proper lubrication and better maintenance of machines.
- Installing noisy machines in sound proof chambers.
- Covering noise-producing machine parts with sound-absorbing materials to check noise production.

- Reducing the noise produced from a vibrating machine by vibration damping i.e. making a layer of damping material (rubber, neoprene, cork or plastic) beneath the machine.

- Using silencers to control noise from automobiles, ducts, exhausts etc. and convey systems with ends opening into the atmosphere.

- Using glass wool or mineral wool covered with a sheet of perforated metal for the purpose of mechanical protection.

(3) Acoustic Zoning:

- Increased distance between source and receiver by zoning of noisy industrial areas, bus terminals and railway stations, aerodromes etc. away from the residential areas would go a long way in minimising noise pollution.
- There should be silence zones near the residential areas, educational institutions and above all, near hospitals.

Module# 091 Control of Noise Pollution

(4) Sound Insulation at Construction Stages:

- Sound travels through the cracks that get left between the door and the wall. For reducing noise, this space (jamb frame gap) should be packed with sound absorbing material.
- Sound insulation can be done by constructing windows with double or triple panes of glass and filling the gaps with sound absorbing materials.
- Acoustical tiles, hair felt, perforated plywood etc. can be fixed on walls, ceilings, floors etc. to reduce noise (especially for sound proof recording rooms etc.)

(5) Planting of Trees:

- Planting green trees and shrubs along roads, hospitals, educational institutions etc. help in noise reduction to a considerable extent.

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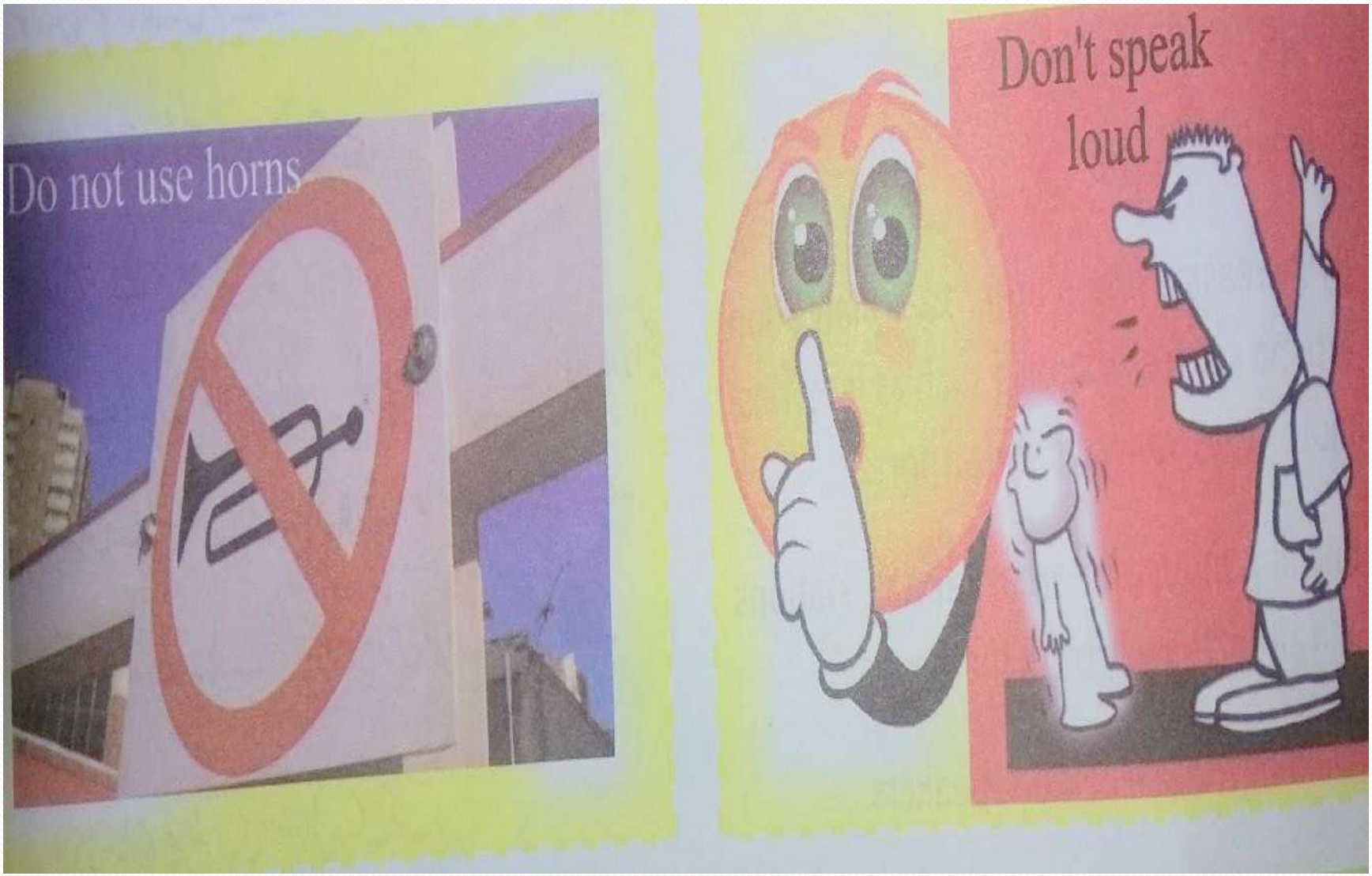
(6) Legislative Measures:

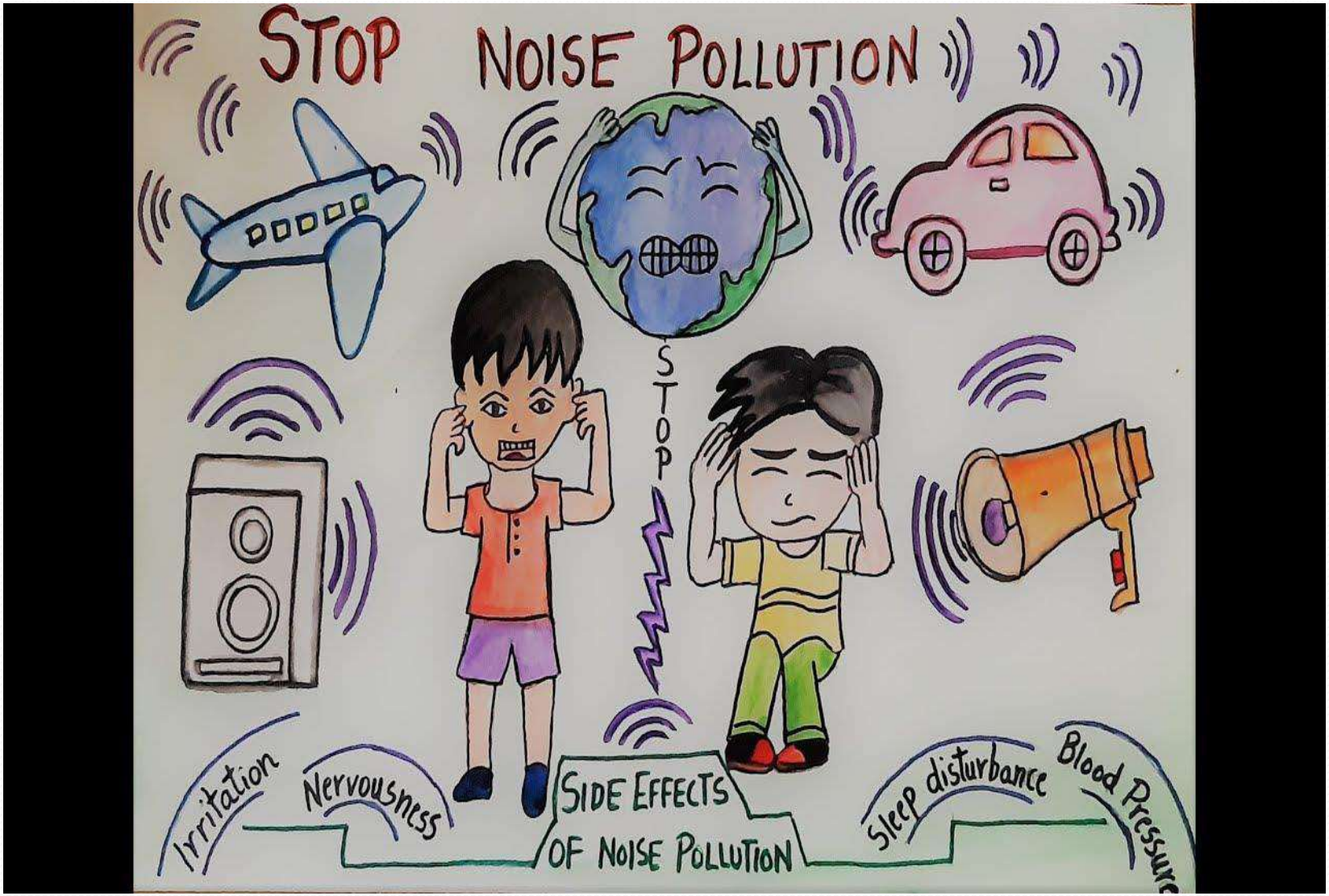
- Strict legislative measures need to be enforced to curb the menace of noise pollution. Some of these measures could be:

- (a) Minimum use of loudspeakers and amplifiers especially near silence zones.

- (b) Banning pressure horns in automobiles.

- (c) Framing a separate Noise Pollution Act.


Do not use horns
Don't speak loud

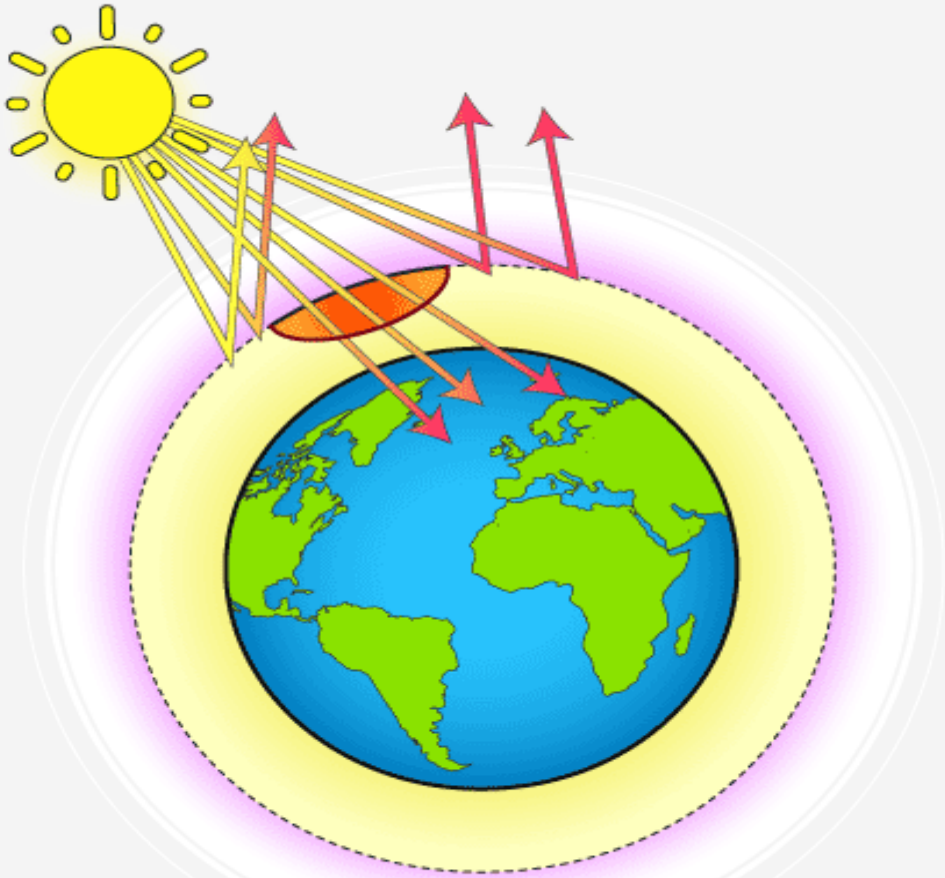

STOP NOISE POLLUTION
STOP
Irritation
Nervousness
SIDE EFFECTS
OF NOISE POLLUTION
Sleep disturbance
Blood Pressure

Module# 92 Ozone Layer

- The ozone layer is mainly found in the lower portion of the earth's atmosphere. It has the potential to absorb around 97-99% of the harmful ultraviolet radiations coming from the sun that can damage life on earth. If the ozone layer was absent, millions of people would develop skin diseases and may have weakened immune systems.

- However, scientists have discovered a hole in the ozone layer over Antarctica. This has focussed their concern on various environmental issues and steps to control them. The main reasons for the ozone hole are chlorofluorocarbons, carbon tetrachloride, methyl bromide and hydrochlorofluorocarbons.

OZONE LAYER DEPLETION



Ozone Layer Depletion

- Ozone layer depletion is the thinning of the ozone layer present in the upper atmosphere. This happens when the chlorine and bromine atoms in the atmosphere come in contact with ozone and destroy the ozone molecules. One chlorine can destroy 100,000 molecules of ozone. It is destroyed more quickly than it is created.

- Some compounds release chlorine and bromine on exposure to high ultraviolet light, which then contributes to ozone layer depletion. Such compounds are known as Ozone Depleting Substances (ODS).

- The ozone-depleting substances that contain chlorine include chlorofluorocarbon, carbon tetrachloride, hydrochlorofluorocarbons, and methyl chloroform. Whereas, the ozone-depleting substances that contain bromine are halons, methyl bromide, and hydro bromofluorocarbons.

- Chlorofluorocarbons are the most abundant ozone-depleting substance. It is only when the chlorine atom reacts with some other molecule, it does not react with ozone.

- Montreal Protocol was proposed in 1987 to stop the use, production and import of ozone-depleting substances and minimise their concentration in the atmosphere to protect the ozone layer of the earth.

Causes of Ozone Layer Depletion

- Ozone layer depletion is a major concern and is associated with a number of factors. The main causes responsible for the depletion of the ozone layer are listed below:

- **Chlorofluorocarbons**
- Chlorofluorocarbons or CFCs are the main cause of ozone layer depletion. These are released by solvents, spray aerosols, refrigerators, air-conditioners, etc.

- The molecules of chlorofluorocarbons in the stratosphere are broken down by ultraviolet radiations and release chlorine atoms. These atoms react with ozone and destroy it.

- **Unregulated Rocket Launches**
- Researches say that the unregulated launching of rockets results in much more depletion of the ozone layer than the CFCs do. If not controlled, this might result in a huge loss of the ozone layer by the year 2050.

- **Nitrogenous Compounds**
- The nitrogenous compounds such as NO2, NO, N2O are highly responsible for the depletion of the ozone layer.

- **Natural Causes**
- The ozone layer has been found to be depleted by certain natural processes such as Sun-spots and stratospheric winds. But it does not cause more than 1-2% of the ozone layer depletion.

- The volcanic eruptions are also responsible for the depletion of the ozone layer.

Effects Of Ozone Layer Depletion

- The depletion of the ozone layer has harmful effects on the environment. Let us see the major effects of ozone layer depletion on man and environment.

- Effects on Human Health
- Humans will be directly exposed to the harmful ultraviolet radiation of the sun due to the depletion of the ozone layer. This might result in serious health issues among humans, such as skin diseases, cancer, sunburns, cataract, quick ageing and weak immune system.

- Effects on Animals
- Direct exposure to ultraviolet radiations leads to skin and eye cancer in animals.

- Effects on the Environment
- Strong ultraviolet rays may lead to minimal growth, flowering and photosynthesis in plants. The forests also have to bear the harmful effects of the ultraviolet rays.

- Effects on Marine Life
- Planktons are greatly affected by the exposure to harmful ultraviolet rays. These are higher in the aquatic food chain. If the planktons are destroyed, the organisms present in the food chain are also affected.

Solutions to Ozone Layer Depletion

- The depletion of the ozone layer is a serious issue and various programmes have been launched by the government of various countries to prevent it. However, steps should be taken at the individual level as well to prevent the depletion of the ozone layer.

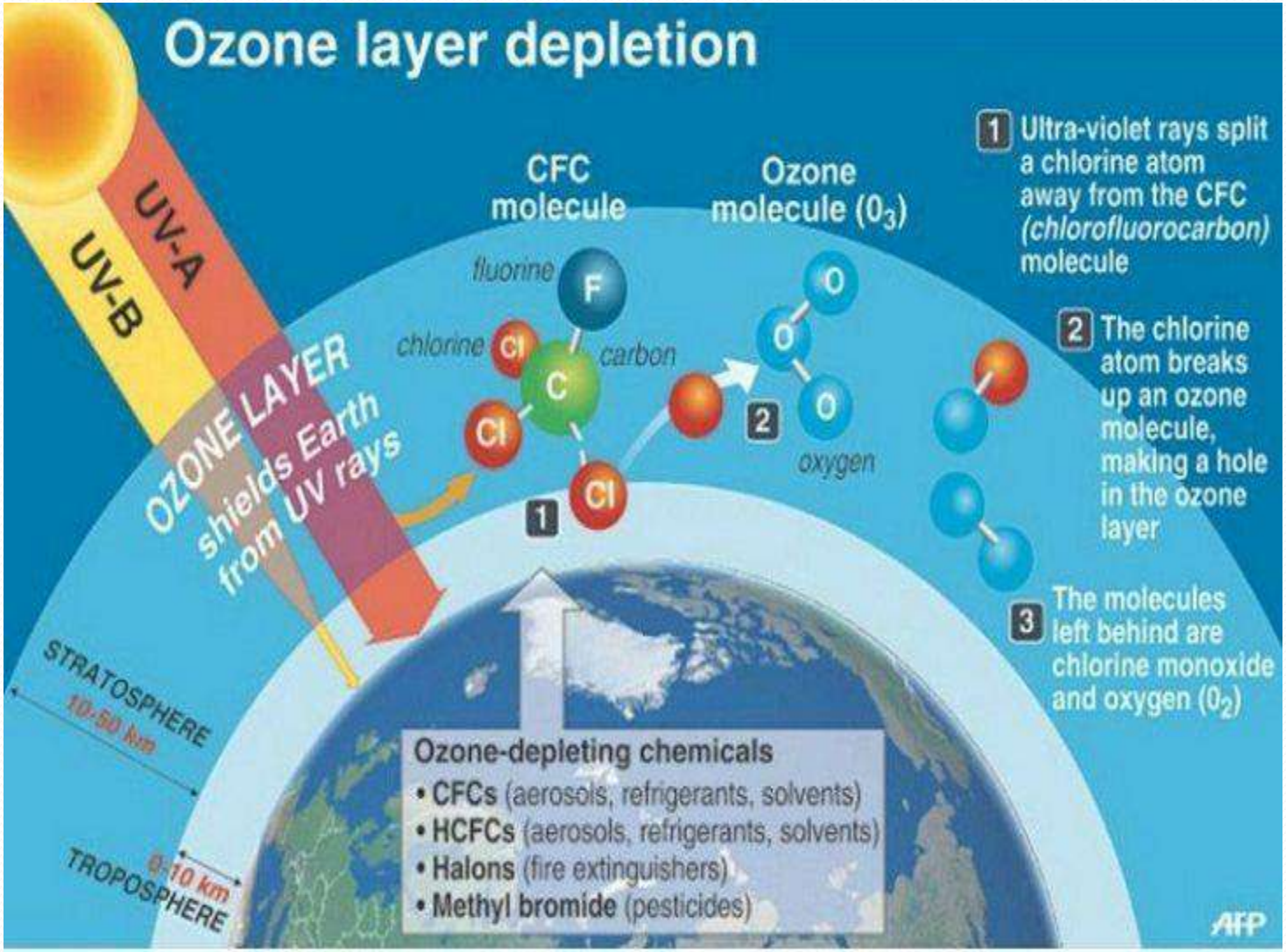
- Following are some points that would help in preventing this problem at a global level:

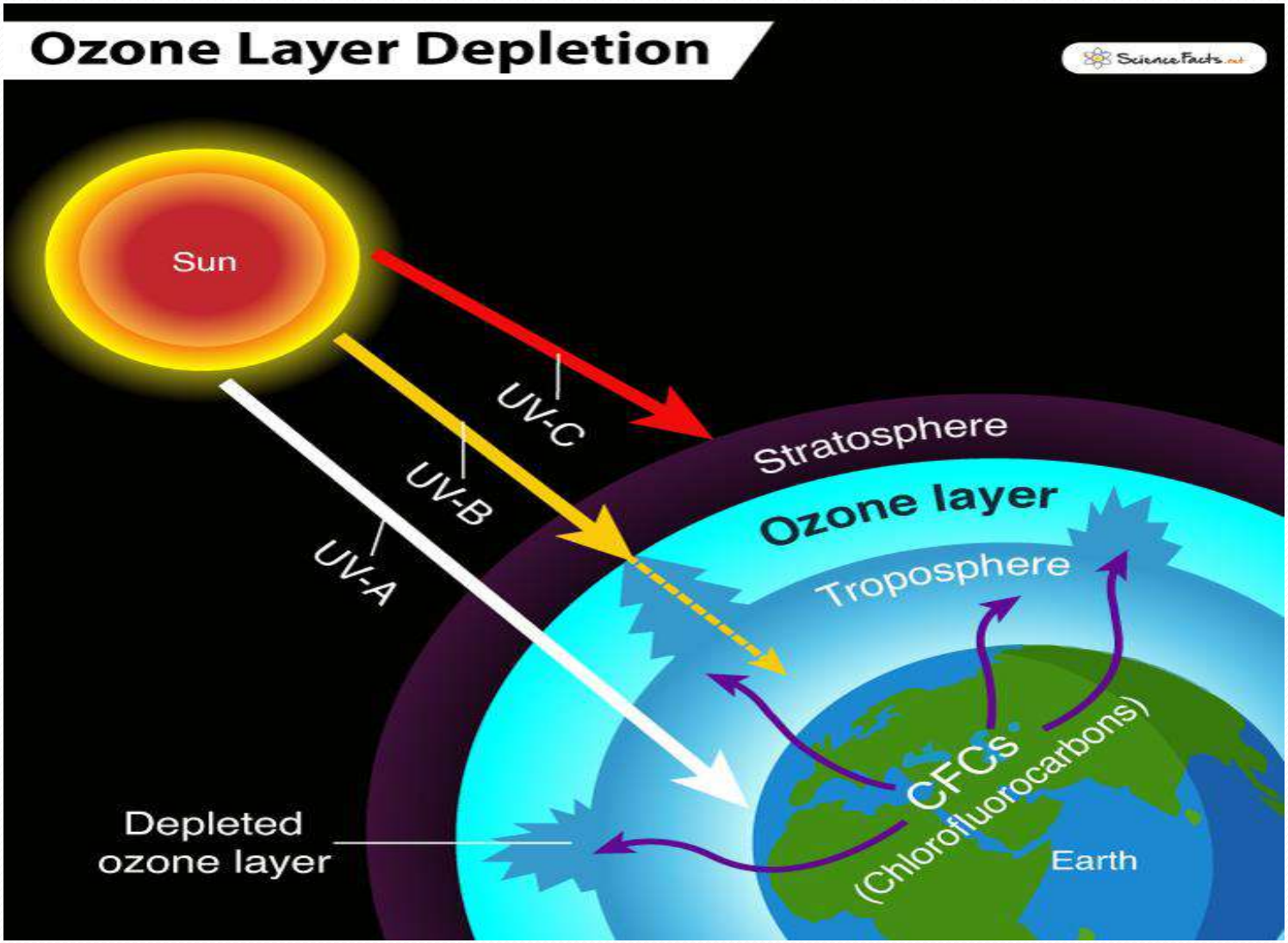
- **Avoid Using ODS**
- Reduce the use of ozone depleting substances. E.g. avoid the use of CFCs in refrigerators and air conditioners, replacing the halon based fire extinguishers, etc.

- **Minimise the Use of Vehicles**
- The vehicles emit a large amount of greenhouse gases that lead to global warming as well as ozone depletion. Therefore, the use of vehicles should be minimised as much as possible.

- **Use Eco-friendly Cleaning Products**
- Most of the cleaning products have chlorine and bromine releasing chemicals that find a way into the atmosphere and affect the ozone layer. These should be substituted with natural products to protect the environment.

- **Use of Nitrous Oxide should be Prohibited**
- The government should take actions and prohibit the use of harmful nitrous oxide that is adversely affecting the ozone layer. People should be made aware of the harmful effects of nitrous oxide and the products emitting the gas so that its use is minimised at the individual level as well.

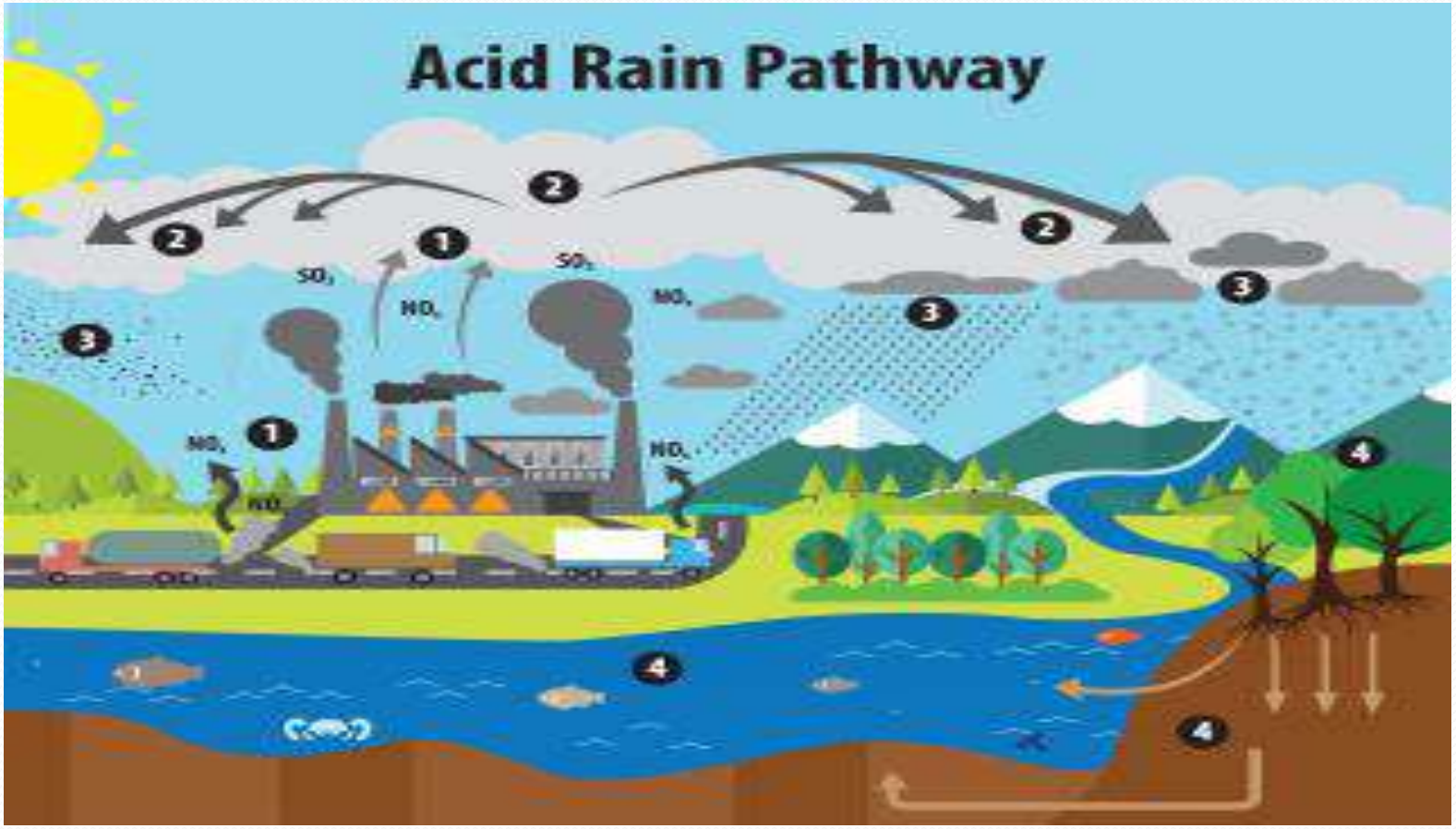

Ozone layer depletion
UV-A
UV-B
OZONE LAYER
shields Earth
from UV rays
CFC
molecule
Ozone
molecule (0_3)
fluorine
F
chlorine
Cl
C
carbon
Cl
Cl
1
2
O
O
O
oxygen
1 Ultra-violet rays split a chlorine atom away from the CFC (chlorofluorocarbon) molecule
2 The chlorine atom breaks up an ozone molecule, making a hole in the ozone layer
3 The molecules left behind are chlorine monoxide and oxygen (0_2)
STRATOSPHERE
10-50 km
0-10 km
TROPOSPHERE
Ozone-depleting chemicals
• CFCs (aerosols, refrigerants, solvents)
• HCFCs (aerosols, refrigerants, solvents)
• Halons (fire extinguishers)
• Methyl bromide (pesticides)
AFP

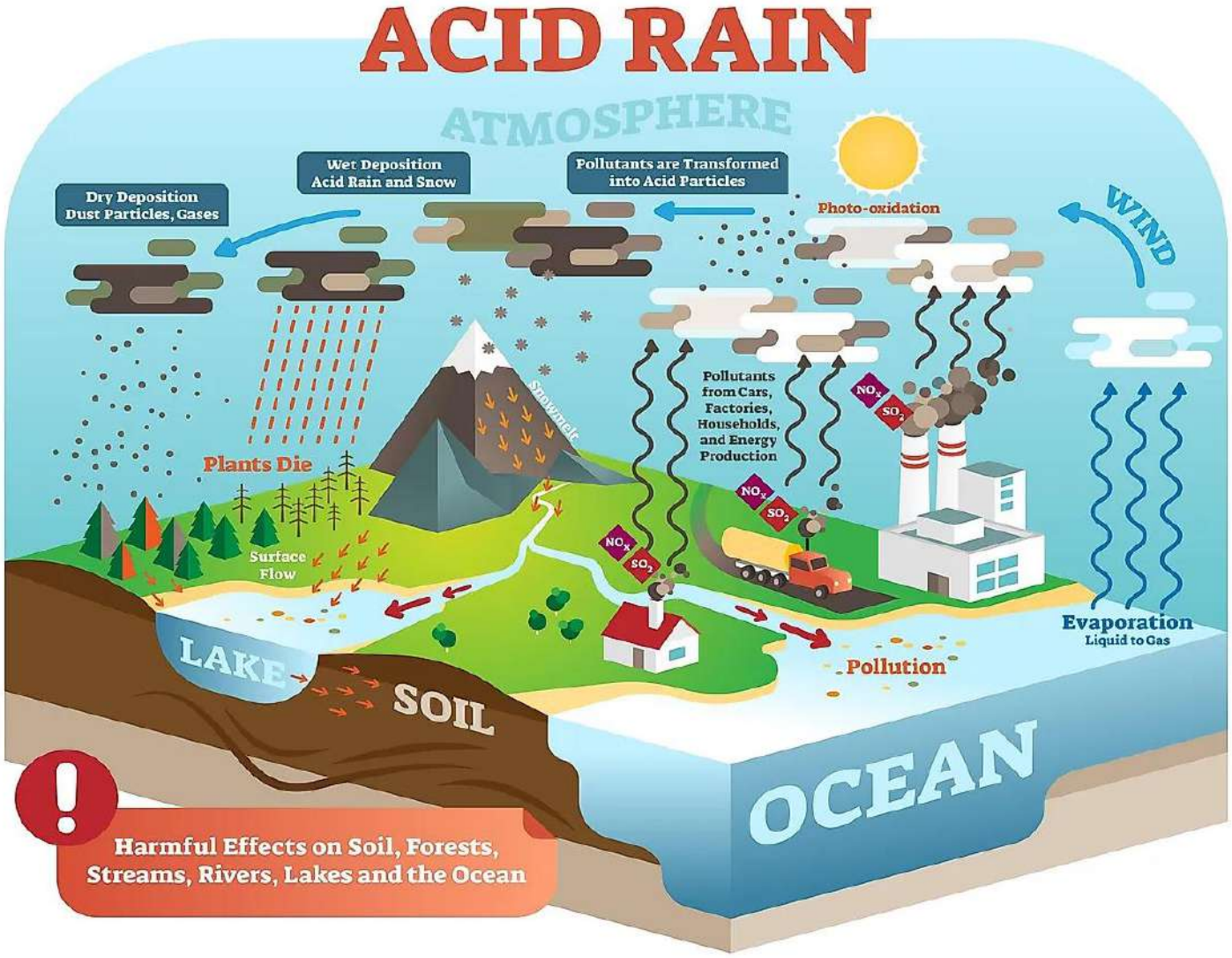

Ozone Layer Depletion
ScienceFacts.net
Sun
UV-C
UV-B
UV-A
Stratosphere
Ozone layer
Troposphere
CFCs
(Chlorofluorocarbons)
Earth
Depleted
ozone layer

Module# 093 Acid Rain

- Acid rain is made up of highly acidic water droplets due to air emissions, most specifically the disproportionate levels of sulphur and nitrogen emitted by vehicles and manufacturing processes. It is often called acid rain as this concept contains many types of acidic precipitation.

- The acidic deposition takes place in two ways: wet and dry. Wet deposition is any form of precipitation which removes acids from the atmosphere and places them on the surface of the earth. In the absence of precipitation, dry deposition of polluting particles and gases sticks to the ground through dust and smoke.


Acid Rain Pathway
SO2
NO2
SO2
NO2
NO2
NO
NO2
1
2
2
2
3
3
3
4
4
4


ACID RAIN
ATMOSPHERE
Dry Deposition
Dust Particles, Gases
Wet Deposition
Acid Rain and Snow
Pollutants are Transformed
into Acid Particles
Photo-oxidation
WIND
Pollutants
from Cars,
Factories,
Households,
and Energy
Production
Snowmelt
Plants Die
Surface
Flow
LAKE
SOIL
Pollution
Evaporation
Liquid to Gas
OCEAN
Harmful Effects on Soil, Forests,
Streams, Rivers, Lakes and the Ocean

Causes of Acid Rain

- The causes of acid rain are Sulphur and Nitrogen particles which get mixed with the wet components of rain. Sulphur and Nitrogen particles which get mixed with water are found in two ways either man-made i.e as the emissions that are given out from industries or by natural causes like lightning strike in the atmosphere releasing nitrogen oxides and volcanic eruptions releasing sulphur oxide.

- According to the Royal Society of Chemistry, which considers him the "father of acid rain," the word acid rain was invented in 1852 by Scottish chemist Robert Angus Smith. Smith decided on the word while studying rainwater chemistry near industrial towns in England and Scotland.

- The regular clean rain we experience, even though it is not clean i.e water and carbon dioxide react together to form weak carbonic acid which essentially by itself is not extremely harmful. The reaction occurring is :

- $H_2O\ (l) + CO_2\ (g) \rightleftharpoons H_2CO_3\ (aq)$

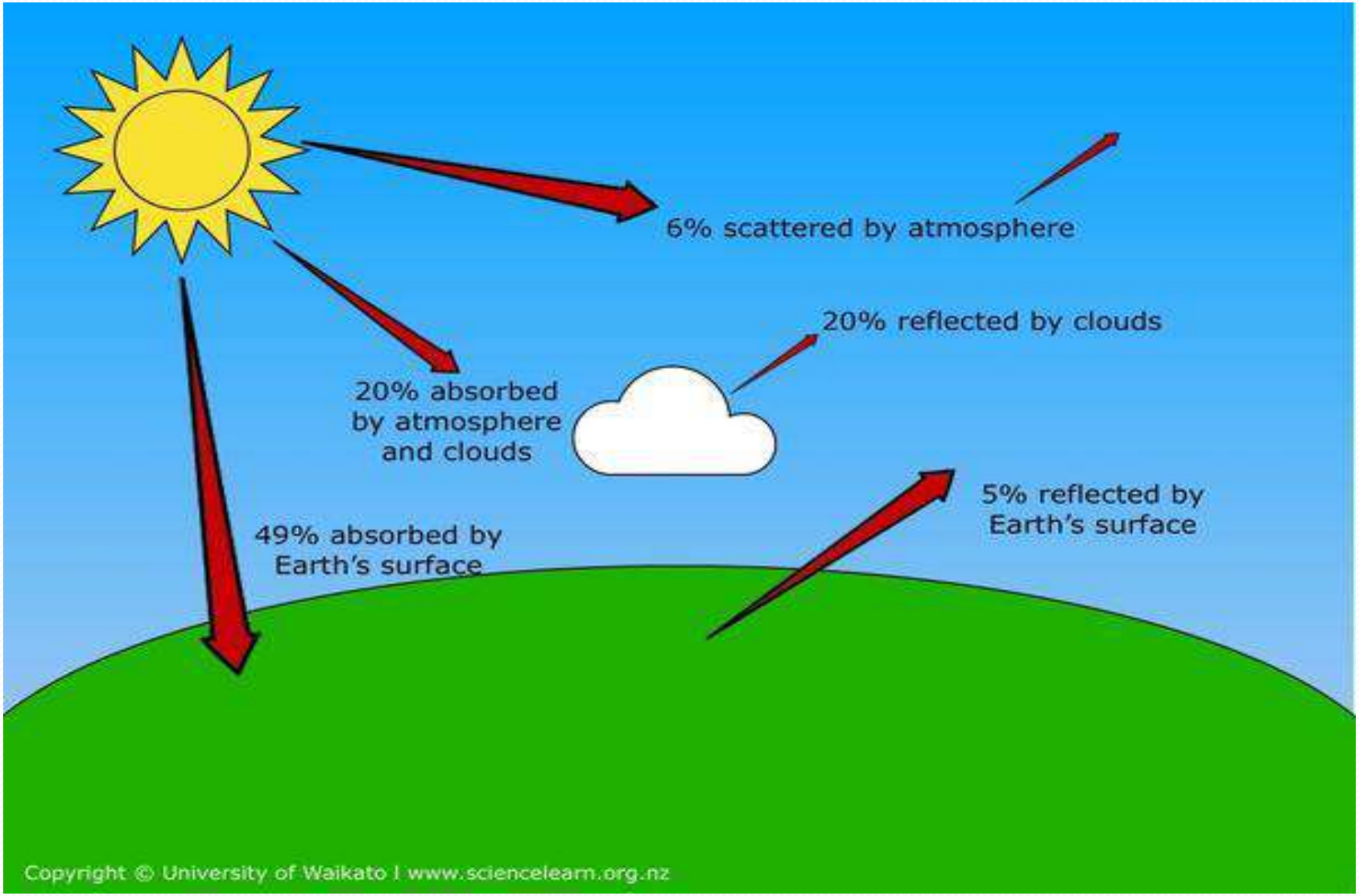
- The pH value of regular rainwater is around 5.7, giving it an acidic nature. The oxides of nitrogen and sulphur are blown away by the wind along with the dust particles. They settle on the earth's surface after coming down in the form of precipitation. Acid rain is essentially a by-product of human activities which emit oxides of nitrogen and sulphur in the atmosphere. Example – the burning of fossil fuels, unethical waste emission disposal techniques.

Prevention of Acid Rain

- The only precaution that we can take against acid rain is having a check at the emission of oxides of nitrogen and sulphur.
- Acid rain is harmful to animals, plants and the monuments.
- Being responsible citizens, one should be aware of the harmful effects they cause and of the industries which give out nitrogen and sulphur compound wastes unethically.

Module# 094 Greenhouse Effect

- The greenhouse effect is the process thanks to which Earth has a higher temperature than it would have without it.
- The gases that radiate heat also known as greenhouse gases absorb the energy radiated out by the Earth and reflect a part of it back to Earth.
- Of all the energy that the Earth receives from the Sun, a part of it around 26% is reflected back to space by the atmosphere and clouds.
- Some part of it is absorbed by the atmosphere, around 19%.


6% scattered by atmosphere
20% reflected by clouds
20% absorbed by atmosphere and clouds
5% reflected by Earth's surface
49% absorbed by Earth's surface
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- The rest hits the ground and heats the surface of the Earth.
- This absorbed energy is radiated out of the earth in the form of Infrared Waves. These IR waves warm the atmosphere above the Earth.
- The atmosphere again radiates this energy it received from the Earth both upwards and downwards.
- The energy sent downwards results in a higher equilibrium temperature than if greenhouse gases were absent.
- This greenhouse effect is essential to supporting life on Earth.

Causes of Greenhouse Effect

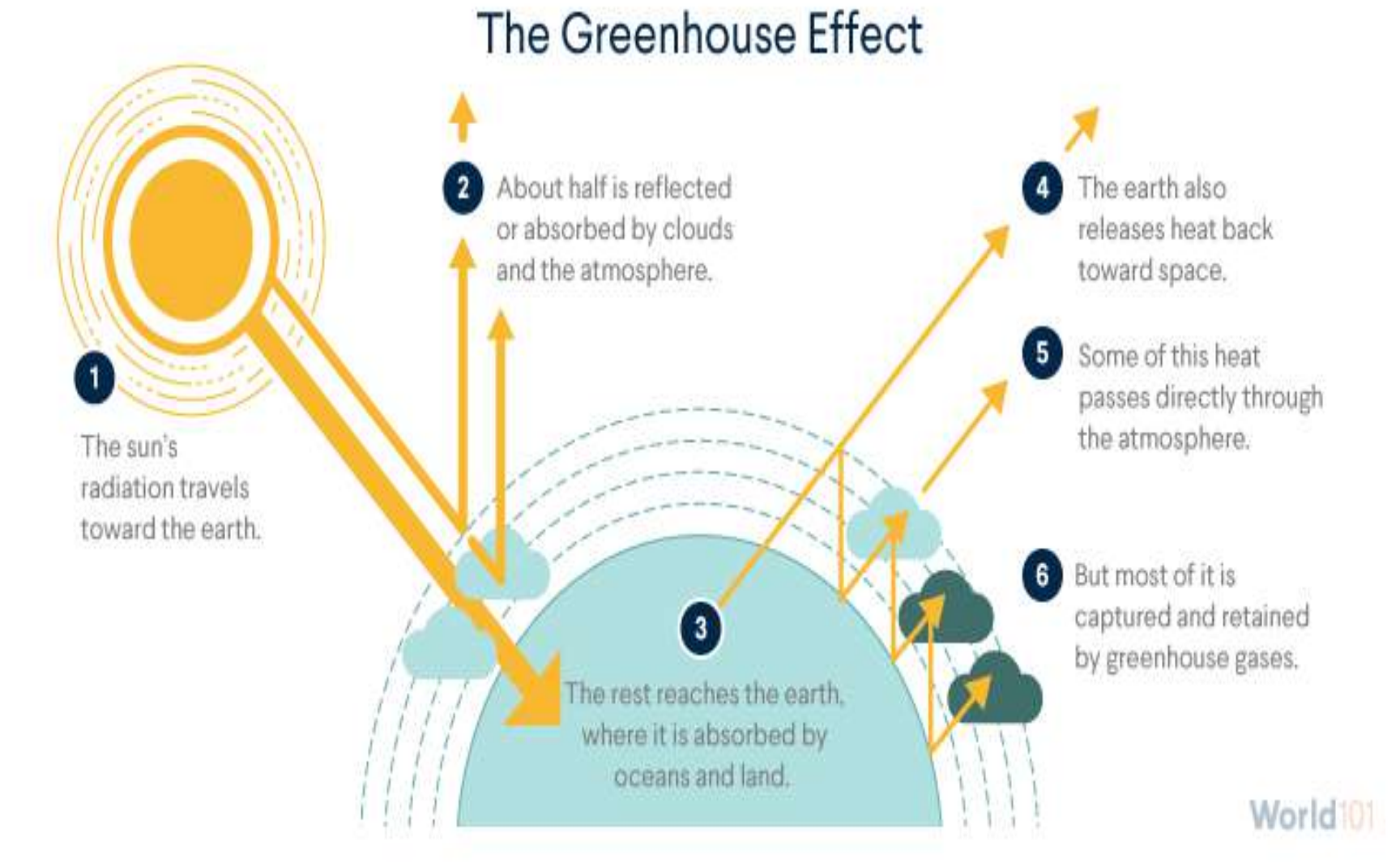
- The following are the factors that are responsible for the cause of greenhouse effect:

- Deforestation: This is considered to be one of the most responsible factors for the cause of the greenhouse effect. This is due to the reduction in the release of oxygen and absorption of carbon dioxide by the plants.
- Fossil fuel burning: Fossil fuels such as coal, oil, and natural gases are used as a means of energy which releases a huge amount of harmful gases into the environment.
- Population: As the population increases, the need for space increases which again results in deforestation.

Prevention of Greenhouse Effect

- Now that we have made a list of causes, finding alternatives to these causes becomes by following the below-mentioned preventive measures:

- Afforestation: Afforestation on a large scale area helps in decreasing the release of carbon dioxide in the atmosphere.
- Conservation of energy: Switching to renewable sources of energy such as solar energy, wind energy, etc will reduce the use of fossil fuels. This eventually reduces the release of carbon dioxide into the atmosphere.
- Policy intervention: When the government comes up with strict policies to maintain the overall air quality of the city.

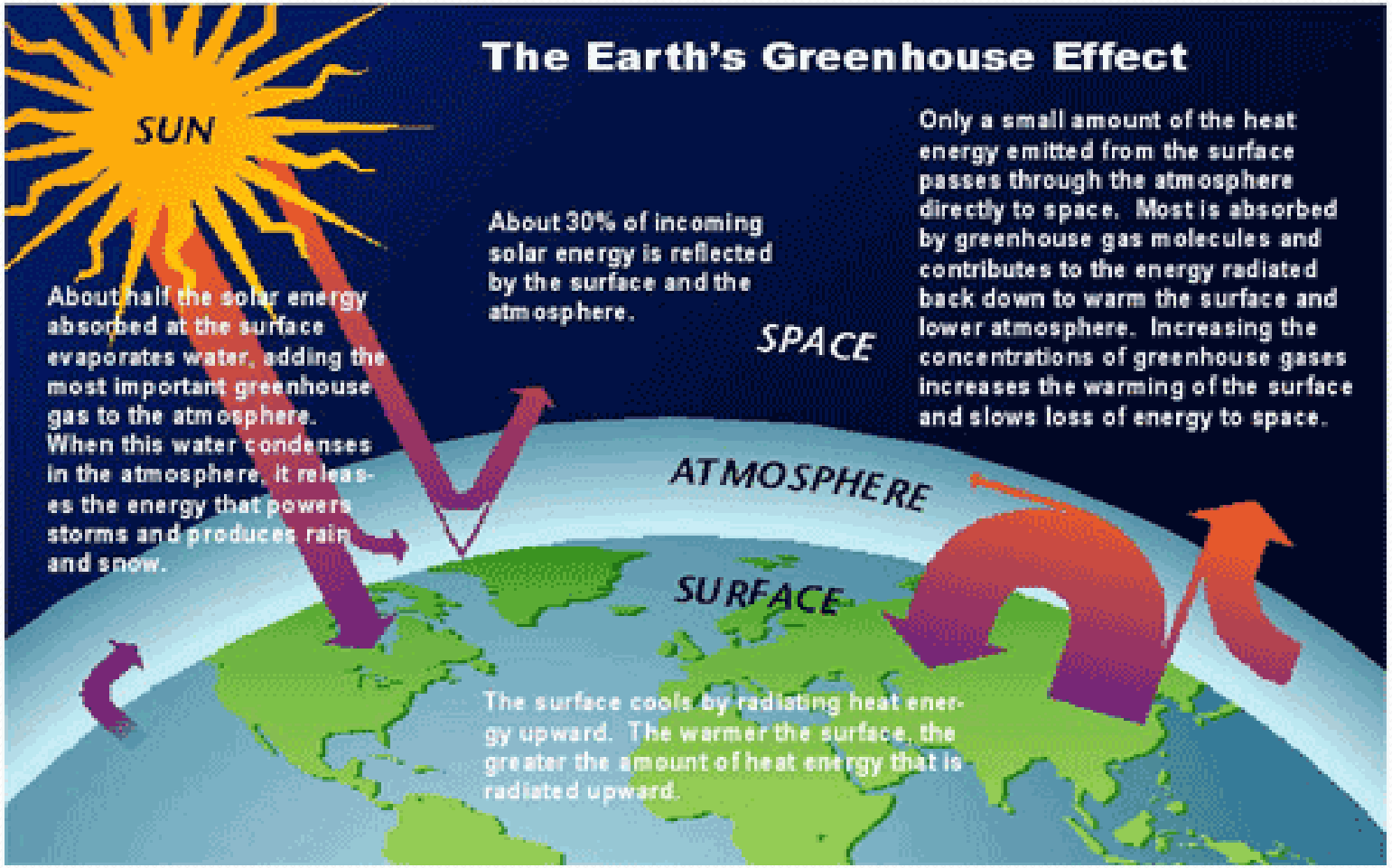

The Greenhouse Effect
1
The sun's radiation travels toward the earth.
2
About half is reflected or absorbed by clouds and the atmosphere.
3
The rest reaches the earth, where it is absorbed by oceans and land.
4
The earth also releases heat back toward space.
5
Some of this heat passes directly through the atmosphere.
6
But most of it is captured and retained by greenhouse gases.
World101

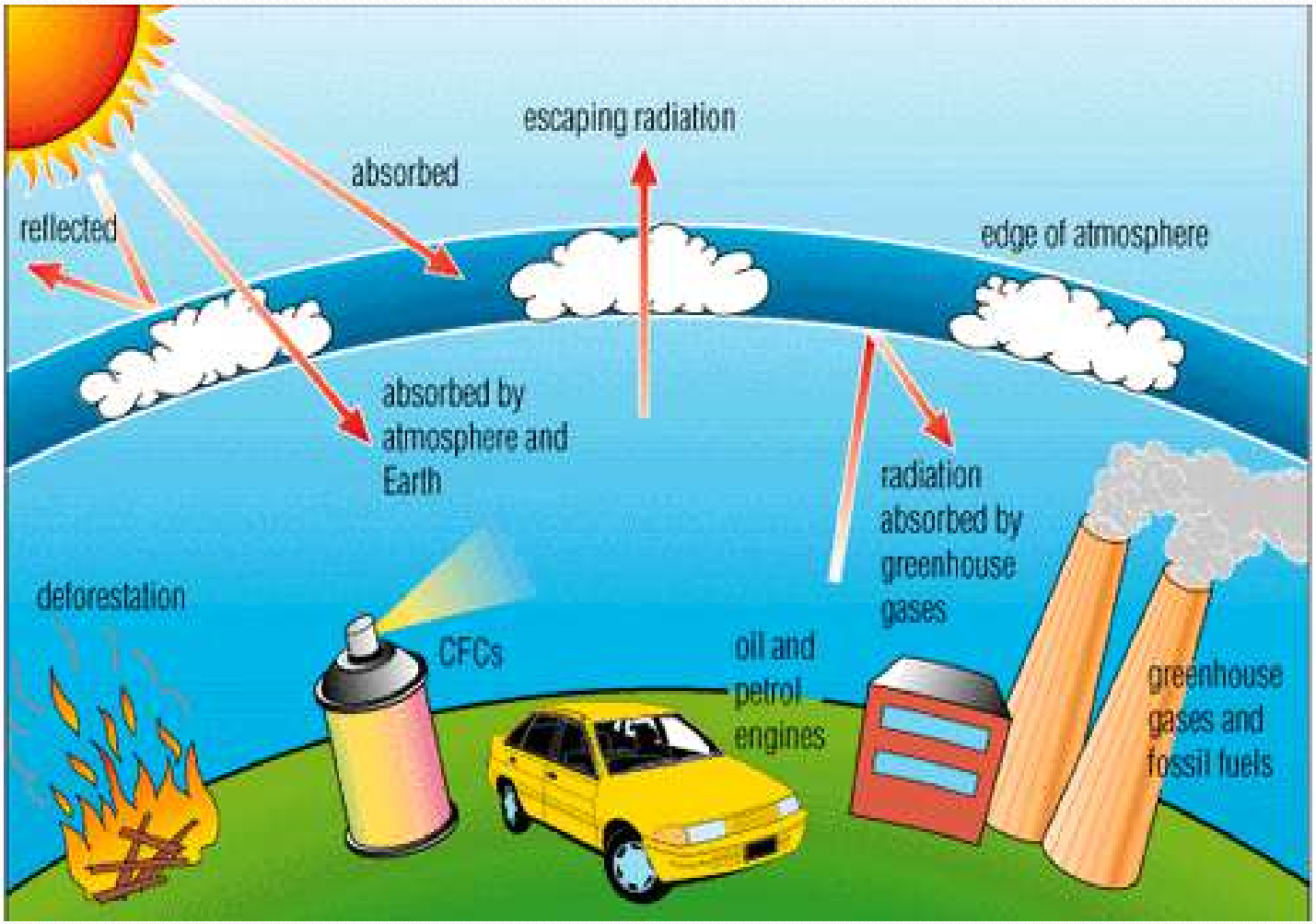
Module# 095 Greenhouse Effect and Global Warming

- Global warming refers to the increasing temperature of the Earth's climate system and its related effects. Scientific evidence has conclusively proven that the Earth's temperature is in fact rising and has risen by 0.850C.
- This has an impact that has affected different regions differently.
- The effects include rising sea levels, retreating glaciers, loss of sea ice in the poles, warming global temperatures, changing precipitation, expansion of deserts, etc.

- This raises significant threats for humans such as food security from decreasing crop yields, and submergence of a low-lying area due to the rising sea.
- To prevent irreversible damage to the delicate ecosystems on Earth, scientists have decided that global warming should be limited to a maximum of 2.00C relative to pre-industrial levels.
- The greenhouse effect plays an important role in the rising temperature. And hence to restrict global warming we need to limit the greenhouse effect and the gaseous emissions that cause the greenhouse effect.

- Fourteen of the fifteen years in the 21st century have been the hottest years on record with constant occurrences of extreme weather, cyclones, droughts, floods, etc.
- All these events are some way or the other have an association with the greenhouse effect and global warming
- **Merged BY BAQIR HUSSAIN**
- **BS ZOOLOGY 03336361244 5th Semester**


The Earth's Greenhouse Effect
SUN
About half the solar energy absorbed at the surface evaporates water, adding the most important greenhouse gas to the atmosphere. When this water condenses in the atmosphere, it releases the energy that powers storms and produces rain and snow.
About 30% of incoming solar energy is reflected by the surface and the atmosphere.
SPACE
Only a small amount of the heat energy emitted from the surface passes through the atmosphere directly to space. Most is absorbed by greenhouse gas molecules and contributes to the energy radiated back down to warm the surface and lower atmosphere. Increasing the concentrations of greenhouse gases increases the warming of the surface and slows loss of energy to space.
ATMOSPHERE
SURFACE
The surface cools by radiating heat energy upward. The warmer the surface, the greater the amount of heat energy that is radiated upward.


escaping radiation
absorbed
reflected
edge of atmosphere
absorbed by atmosphere and Earth
radiation absorbed by greenhouse gases
deforestation
CFCs
oil and petrol engines
greenhouse gases and fossil fuels



Module# 096 Kyoto Protocol

- Kyoto Protocol, in full Kyoto Protocol to the United Nations Framework Convention on Climate Change, international treaty, named for the Japanese city in which it was adopted in December 1997, that aimed to reduce the emission of gases that contribute to global warming.
- In force since 2005, the protocol called for reducing the emission of six greenhouse gases in 41 countries plus the European Union to 5.2 percent below 1990 levels during the "commitment period" 2008–12.
- It was widely hailed as the most significant environmental treaty ever negotiated, though some critics questioned its effectiveness.

Doha Amendment

- In Doha, Qatar, on 8 December 2012, the Doha Amendment to the Kyoto Protocol was adopted for a second commitment period, starting in 2013 and lasting until 2020.

- As of 28 October 2020, 147 Parties deposited their instrument of acceptance, therefore the threshold of 144 instruments of acceptance for entry into force of the Doha Amendment was achieved. The amendment entered into force on 31 December 2020.


DOHA 2012
UN CLIMATE CHANGE CONFERENCE
COP18·CMP8
COORDINATOR
PRESIDENT
SECRETARY
DOHA 2012
UN CLIMATE CHANGE CONFERENCE
COP18·CMP8
photo: jan golinski/unfccc

- The amendment includes:

- New commitments for Annex I Parties to the Kyoto Protocol who agreed to take on commitments in a second commitment period from 1 January 2013 to 31 December 2020;
- A revised list of GHG to be reported on by Parties in the second commitment period; and
- Amendments to several articles of the Kyoto Protocol which specifically referenced issues pertaining to the first commitment period and which needed to be updated for the second commitment period.

- On 21 December 2012, the amendment was circulated by the Secretary-General of the United Nations, acting in his capacity as Depositary, to all Parties to the Kyoto Protocol in accordance with Articles 20 and 21 of the Protocol.

- During the first commitment period, 37 industrialized countries and economies in transition and the European Community committed to reduce GHG emissions to an average of five percent against 1990 levels. During the second commitment period, Parties committed to reduce GHG emissions by at least 18 percent below 1990 levels in the eight-year period from 2013 to 2020; however, the composition of Parties in the second commitment period is different from the first.


Kyoto Protocol
methane
warming
ecology
emission
industry
carbon dioxide
agreement
atmosphere
gas
climate change
greenhouse gas
global warming
treaty
nitrous oxide
shutterstock.com · 1846412119

TODAY IN CHEMISTRY HISTORY

16TH FEBRUARY – THE KYOTO PROTOCOL (2005)



192 COUNTRIES

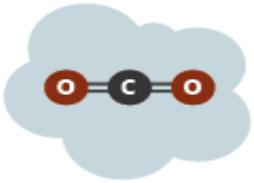
- signed and ratified
- ratified but withdrawn
- signed but not ratified

TARGET
↓5.2%
by 2012 relative to 1990

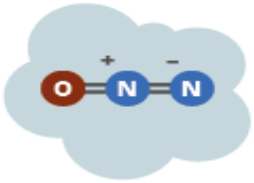




METHANE



CARBON DIOXIDE



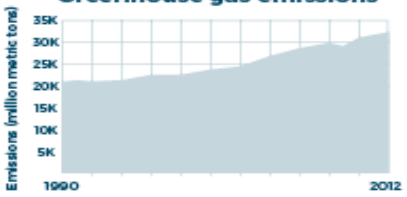
NITROUS OXIDE

GREENHOUSE GASES

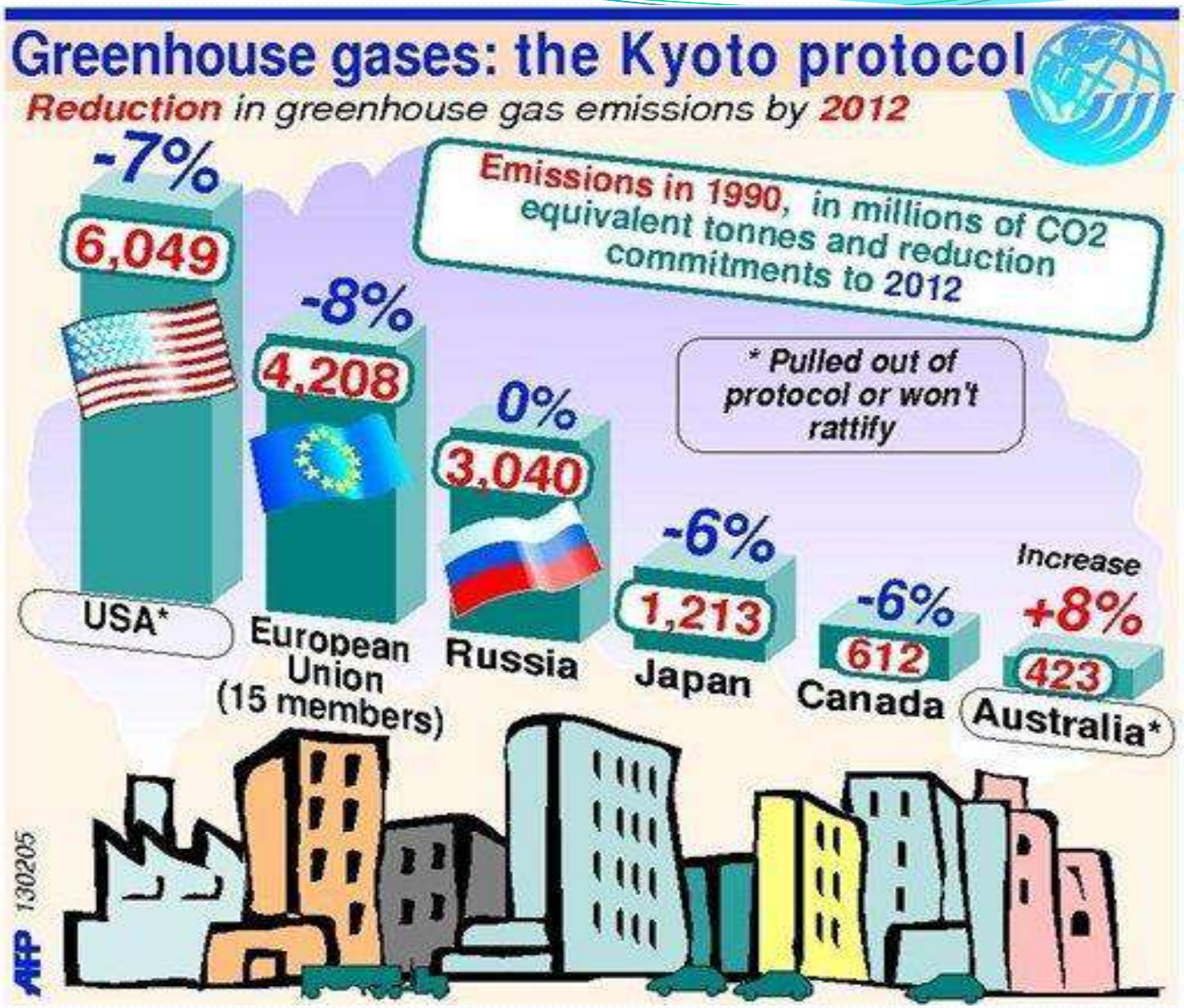
- HYDROFLUOROCARBONS
- PERFLUOROCARBONS
- SULFUR HEXAFLUORIDE

The Kyoto Protocol is an international treaty to reduce the emission of six greenhouse gases. 192 countries are party to the protocol, but only 38 had binding targets for the first period (2008–2012). These 38 countries met their target collectively – but an absence of binding targets for developing nations such as China and India meant that worldwide greenhouse gas emissions continued to rise.

Greenhouse gas emissions



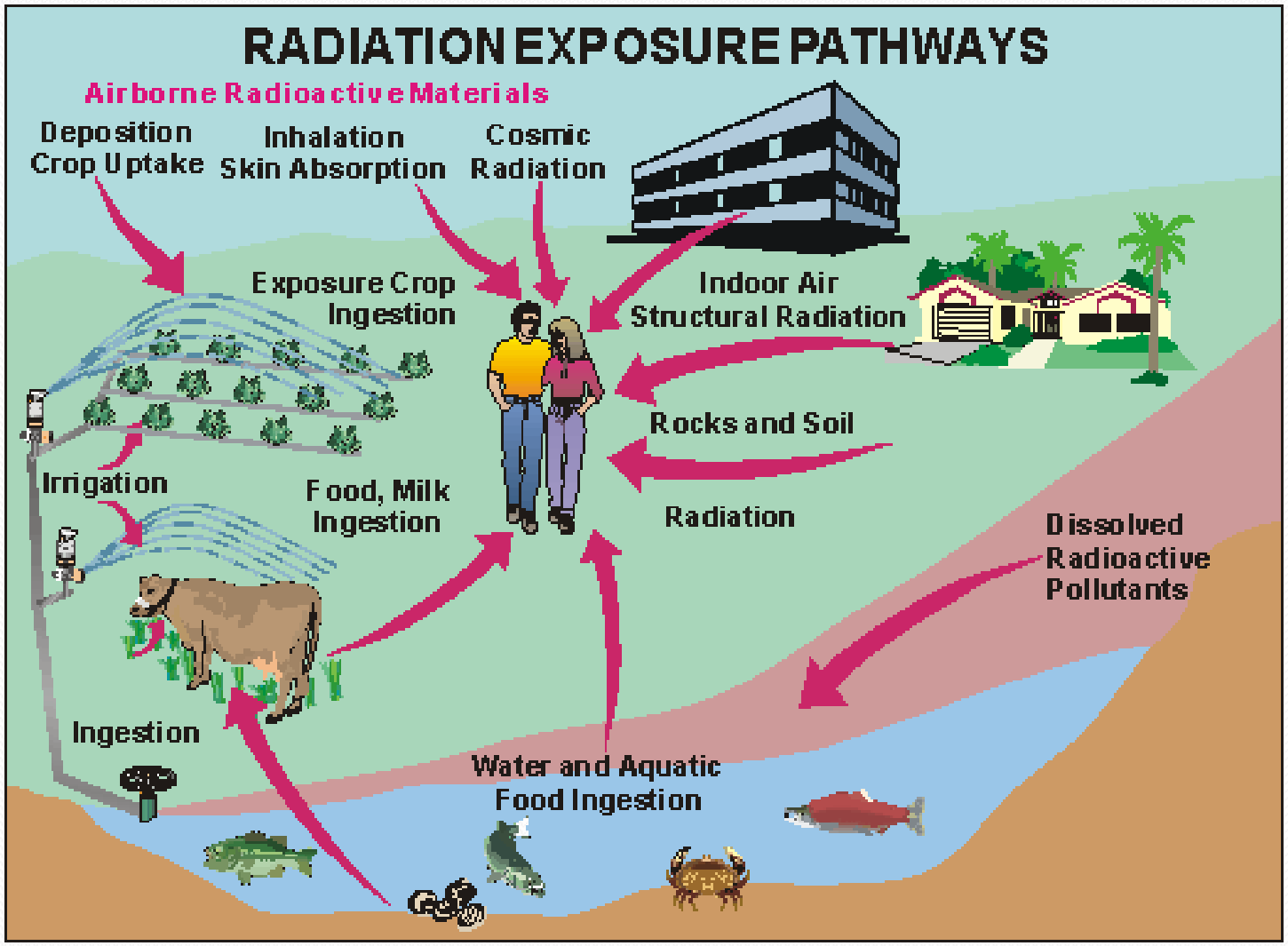
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Shared under a Creative Commons Attribution-NonCommercial-NoDerivatives licence.


Greenhouse gases: the Kyoto protocol
Reduction in greenhouse gas emissions by 2012
Emissions in 1990, in millions of CO2 equivalent tonnes and reduction commitments to 2012
* Pulled out of protocol or won't rattify
-7%
6,049
USA*
-8%
4,208
European Union (15 members)
0%
3,040
Russia
-6%
1,213
Japan
-6%
612
Canada
Increase
+8%
423
Australia*
AFP 130205

Module# 097 Radioactive contamination

- Radioactive contamination, also called radiological pollution, is the deposition of, or presence of radioactive substances on surfaces or within solids, liquids, or gases (including the human body), where their presence is unintended or undesirable (from the International Atomic Energy Agency (IAEA) definition).

- Such contamination presents a hazard because the radioactive decay of the contaminants, produces ionizing radiation (namely alpha, beta, gamma rays and free neutrons). The degree of hazard is determined by the concentration of the contaminants, the energy of the radiation being emitted, the type of radiation, and the proximity of the contamination to organs of the body. It is important to be clear that the contamination gives rise to the radiation hazard, and the terms "radiation" and "contamination" are not interchangeable.


RADIATION EXPOSURE PATHWAYS
Airborne Radioactive Materials
Deposition
Crop Uptake
Inhalation
Skin Absorption
Cosmic
Radiation
Exposure Crop
Ingestion
Indoor Air
Structural Radiation
Rocks and Soil
Irrigation
Food, Milk
Ingestion
Radiation
Dissolved
Radioactive
Pollutants
Ingestion
Water and Aquatic
Food Ingestion



- The sources of radioactive pollution can be classified into two groups: natural and man-made.
- Following an atmospheric nuclear weapon discharge or a nuclear reactor containment breach, the air, soil, people, plants, and animals in the vicinity will become contaminated by nuclear fuel and fission products.
- A spilled vial of radioactive material like uranyl nitrate may contaminate the floor and any rags used to wipe up the spill.
- Cases of widespread radioactive contamination include the Bikini Atoll, the Rocky Flats Plant in Colorado, the area near the Fukushima Daiichi nuclear disaster, the area near the Chernobyl disaster, and the area near the Mayak disaster.


Radioactive Contamination

Module# 098 Environmental law

- Environmental law is a collective term encompassing aspects of the law that provide protection to the environment.
- A related but distinct set of regulatory regimes, now strongly influenced by environmental legal principles, focus on the management of specific natural resources, such as forests, minerals, or fisheries.
- Other areas, such as environmental impact assessment, may not fit neatly into either category, but are nonetheless important components of environmental law.

- Global and regional environmental issues are increasingly the subject of international law. Debates over environmental concerns implicate core principles of international law and have been the subject of numerous international agreements and declarations.

- **Customary international law** is an important source of international environmental law. These are the norms and rules that countries follow as a matter of custom and they are so prevalent that they bind all states in the world. When a principle becomes customary law is not clear cut and many arguments are put forward by states not wishing to be bound. Examples of customary international law relevant to the environment include the duty to warn other states promptly about icons of an environmental nature and environmental damages to which another state or states may be exposed, and Principle 21 of the Stockholm Declaration ('good neighbourliness' or sic utere).

- Given that customary international law is not static but ever evolving and the continued increase of air pollution (carbon dioxide) causing climate changes, has led to discussions on whether basic customary principles of international law, such as the jus cogens (peremptory norms) and erga omnes principles could be applicable for enforcing international environmental law.[30]

- Numerous legally binding international agreements encompass a wide variety of issue-areas, from terrestrial, marine and atmospheric pollution through to wildlife and biodiversity protection.
- International environmental agreements are generally multilateral (or sometimes bilateral) treaties (a.k.a. convention, agreement, protocol, etc.). Protocols are subsidiary agreements built from a primary treaty.
- They exist in many areas of international law but are especially useful in the environmental field, where they may be used to regularly incorporate recent scientific knowledge. They also permit countries to reach an agreement on a framework that would be contentious if every detail were to be agreed upon in advance.
- The most widely known protocol in international environmental law is the **Kyoto Protocol**, which followed from the United Nations Framework Convention on Climate Change.

- While the bodies that proposed, argued, agreed upon, and ultimately adopted existing international agreements vary according to each agreement, certain conferences, including 1972's United Nations Conference on the Human Environment, 1983's World Commission on Environment and Development, 1992's United Nations Conference on Environment and Development, and 2002's World Summit on Sustainable Development have been particularly important.
- Multilateral environmental agreements sometimes create an International Organization, Institution or Body responsible for implementing the agreement.
- Major examples are the **Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) and the International Union** for Conservation of Nature (IUCN).

- International environmental law also includes the opinions of international courts and tribunals. While there are few and they have limited authority, the decisions carry much weight with legal commentators and are quite influential on the development of international environmental law.
- One of the biggest challenges in international decisions is to determine an adequate compensation for environmental damages.
- The courts include the International Court of Justice (ICJ), the International Tribunal for the Law of the Sea (ITLOS), the European Court of Justice, European Court of Human Rights[32] and other regional treaty tribunals.

PAKISTAN ENVIRONMENTAL PROTECTION ACT 1997

The PEPA' 1997 provides

- The framework for implementation of NCS
- Establishment of Provincial Sustainable development Funds
- Protection and Conservation of species
- Conservation of renewable resources
- Establishment of Environmental Tribunals and appointment of Environmental Magistrates
- Initial Environmental Examination (IEE)
- Environmental Impact Assessment (EIA)

Module# 99 Desertification

- Desertification, also called desertization, the process by which natural or human causes reduce the biological productivity of drylands (arid and semiarid lands).
- Declines in productivity may be the result of climate change, deforestation, overgrazing, poverty, political instability, unsustainable irrigation practices, or combinations of these factors.
- The concept does not refer to the physical expansion of existing deserts but rather to the various processes that threaten all dryland ecosystems, including deserts as well as grasslands and scrublands.



Causes and consequences of desertification

- A woman in Nigeria pouring grains of millet into a bowl in the village of Sadongori Kolita, near Maradi. Low rainfall and locusts caused a food crisis during the summer of 2005.
- In general, desertification is caused by variations in climate and by unsustainable land-management practices in dryland environments. By their very nature, arid and semiarid ecosystems are characterized by sparse or variable rainfall. Thus, climatic changes such as those that result in extended droughts can rapidly reduce the biological productivity of those ecosystems.
- Such changes may be temporary, lasting only a season, or they may persist over many years and decades. On the other hand, plants and animals are quick to take advantage of wetter periods, and productivity can rapidly increase during these times.

- Since dryland environments are used for a variety of human purposes (such as agriculture, animal grazing, and fuelwood collection), the various activities undertaken in them can exacerbate the problem of desertification and bring about lasting changes to dryland ecosystems. In 1977, at the United Nations Conference on Desertification (UNCOD) in Nairobi, Kenya, representatives and delegates first contemplated the worldwide effects of desertification.
- The conference explored the causes and contributing factors and also possible local and regional solutions to the phenomenon.
- In addition, the delegates considered the varied consequences of desertification, such as crop failures or decreased yields in rain-fed farmland, the loss of perennial plant cover and thus loss of forage for livestock, reduced woody biomass and thus scarcity of fuelwood and building materials, a decrease in potable water stocks from reductions in surface water and groundwater flow, increased sand dune intrusion onto croplands and settlements, increased flooding due to rising sedimentation in rivers and lakes, and amplified air and water pollution from dust and sedimentation.

Desertification

Causes

- Deforestation
- Farming
- Excessive use of fertilizers and pesticides
- Animal grazing
- Global warming
- Overpopulation
- Changes in land use
- Mining
- Excessive consumption
- Waste production and disposal
- Soil pollution
- Acid rain
- Natural causes

Effects

- Decrease in crop yields
- Poverty
- Hunger
- Starvation
- Weak immune system
- Destruction of habitats
- Biodiversity loss
- Endangerment and extinction of species
- Floods
- Water pollution
- Global warming
- Migration

Solutions

- Stricter regulations regarding farming practices
- Reduction in fertilizer and pesticides
- Reforestation
- Population control
- Technological advancements
- Confinement of mining practices
- Avoidance of pollution
- Adjustments in consumption
- Avoid waste production
- Education
- Convince others

Solutions for Desertification

- Rising awareness of the problem
- Planting indigenous trees and shrubs
- Developing sustainable agricultural practices
- Controlled grazing
- Crop rotation (increase fertility & reduce soil erosion)
- Proper land management
- Soil and water management
- Introduce proper technologies
- Manage dry land resources
- Tree planting schemes to reduce soil erosion.

Module# 100 Deforestation

- Deforestation refers to the decrease in forest areas across the world that are lost for other uses such as agricultural croplands, urbanization, or mining activities. Greatly accelerated by human activities since 1960, deforestation has been negatively affecting natural ecosystems, biodiversity, and the climate

Causes of Deforestation



Overpopulation Demands



Logging



Timber Production



Agricultural Expansions

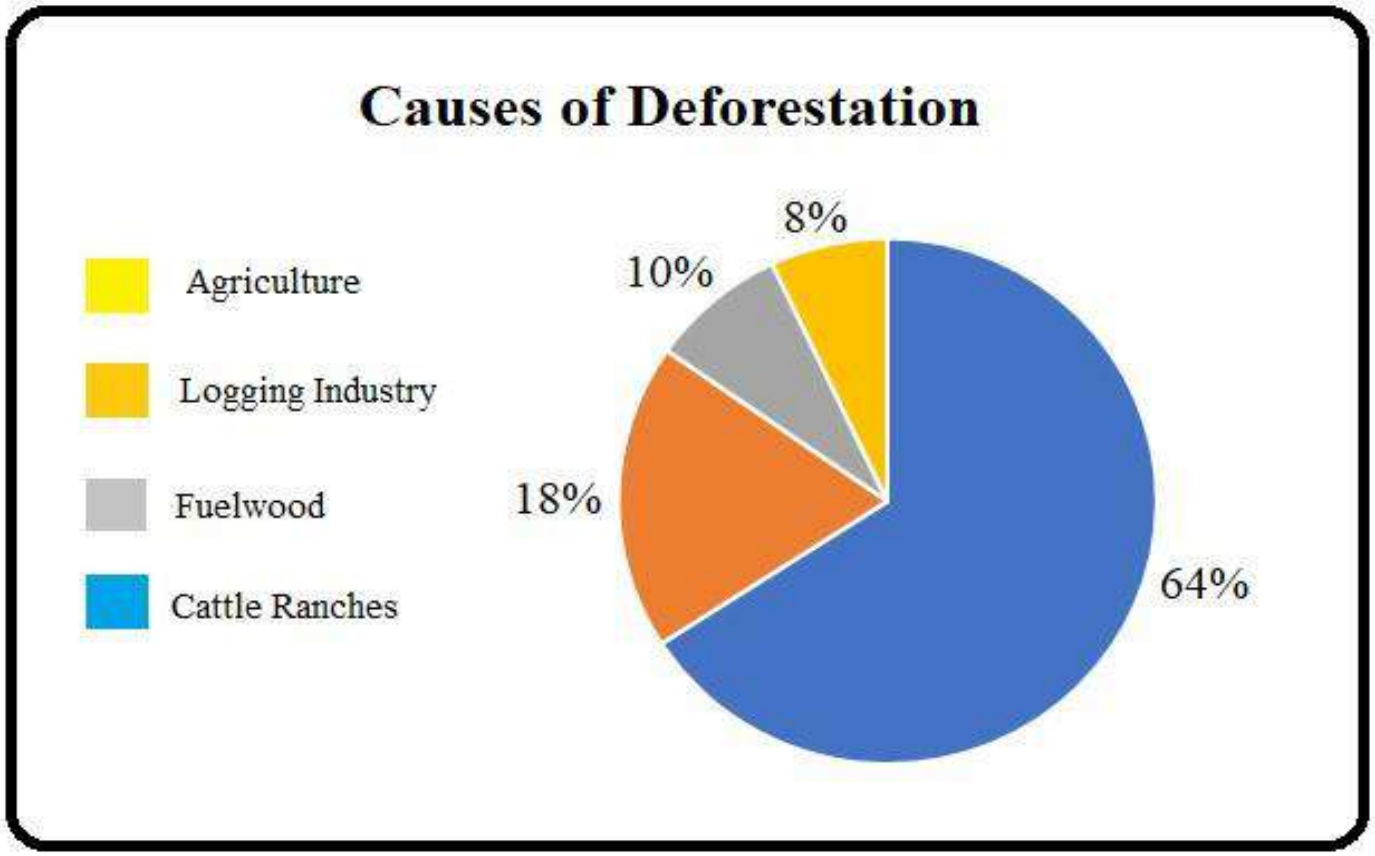


Forest Fire

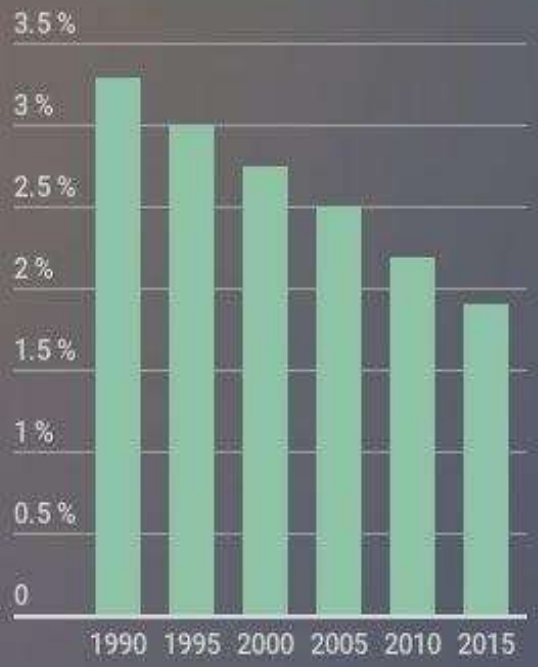


Mining

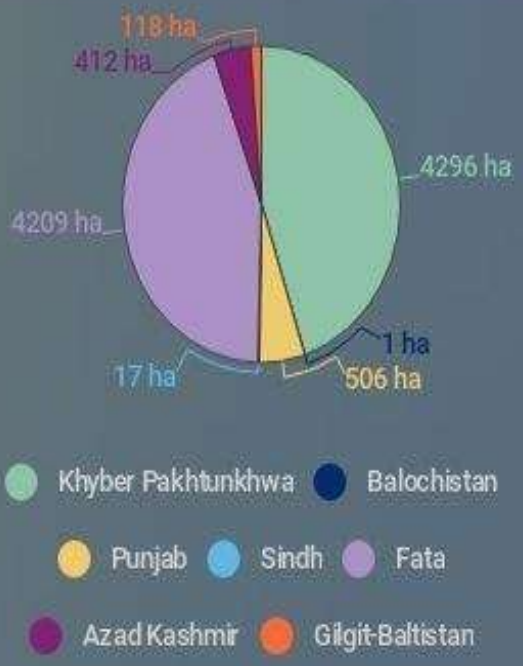
www.earthreminder.com


Causes of Deforestation
Agriculture
Logging Industry
Fuelwood
Cattle Ranches
8%
10%
18%
64%

Deforestation in Pakistan



Decline in the area under natural forests in Pakistan between 1990 and 2015



The tree cover loss in Pakistan's provinces between 2001 and 2016

Preventive Measures of Deforestation



Educational Campaigns



Reforestation



Rules and Regulations



Renewable Forest Resources



Reduce Paper Consumption



Eco-friendly products

www.earthreminder.com

Module# 101 Exotic and invasive species

- Exotic and invasive species are two types of non-native species. Non-native species can be found in a second ecosystem apart from the ecosystem they evolved from. Non-native species spread by natural process or human activities.
- Most non-native or indigenous species are harmless. They are called exotic species. When an exotic species becomes harmful to the ecosystem, it is called an invasive species. The main difference between exotic and invasive species is that exotic species is harmless to the ecosystem whereas invasive species is harmful to the ecosystem.
- Invasive species can be harmful the native habitats, natural areas such as lakes, forests, and rivers, agriculture, economy, and humans

EXOTIC SPECIES VERSUS INVASIVE SPECIES

EXOTIC SPECIES	INVASIVE SPECIES
An exotic species refers to a plant, animal or microorganism species which is introduced into an area outside of its native range	An invasive species refers to an exotic species whose introduction causes environmental and economic harm to the ecosystem
Not harmful to ecosystems	Harmful to ecosystems
May require a lot of resources for the growth	Have a fast growth, rapid reproduction ability, and a high dispersal ability
May consist of natural competitors or enemies	Do not consist of natural competitors or enemies
Do not have an effect on native species	May completely replace native species
Examples: Venus flytrap, purple beautyberry, voodoo lily, sensitive plant, cows, pigs, chicken, pigeons, mute swans	Examples: Kudzu, English ivy, feral cats, feral horses, lionfish, Asian carp, Japanese beetle, and cane toads

Visit www.pediaa.com







Exotic species

Species living outside its native distributional range







INVASIVE ALIEN SPECIES

- An alien species become 'invasive' when they are **intentionally or unintentionally** introduced outside their normal ranges, where they **outcompete native species.**
- Invasive alien species can be **animals, plants or other organisms.**
- They have **High Phenotypic plasticity** refers to the capacity to adapt physiologically to new requirements.

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- One of the **biggest causes of biodiversity loss and species extinctions.**
- A **global threat to food security and livelihoods**.
- **Climate change reduces the resilience of habitats** to biological invasions.

Characteristics of invasive species

- **Rapid reproduction and expansion**
- **Strong dispersing ability**
- **Capacity to thrive** on a variety of food forms and in a variety of environmental settings

- Example: **Lantana, Prosopis juliflora**

Characteristics of Invasive Species

